



RIVISION CAPSULE

by

GATE WALLAH

***ELELCTRICAL ENGINEERING
AND
ELECTRONIC & COMMUNICATION
ENGINEERING***



Revision Capsule by GATEWALLAH

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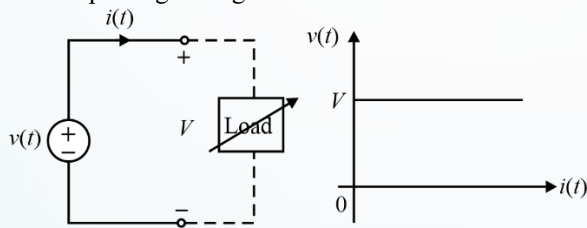
NETWORK THEORY

BASICS

Ideal Sources

• Ideal Voltage Source

Voltage always remains constant for any value of current passing through it.

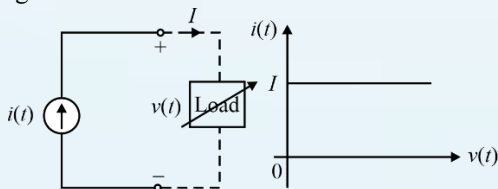


Note:

- The current through any voltage source will depend upon.
 - The current source which is connected in series with it.
 - The load resistance which is connected in parallel with it.
- Can't be SC
- $R_S = 0 \Omega$

Ideal Current Source

- Current always remains constant for any value of voltage across it



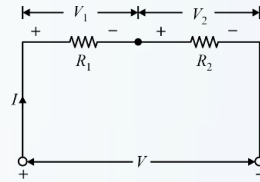
Note:

- The voltage across current source will depend upon
 - Voltage source which is connected in parallel with it.
 - The load resistance which is connected in series with it.
- Can't be OC
- $R_S = \infty$

Voltage Division Equations

$$V_1 = \frac{R_1}{R_1 + R_2} V$$

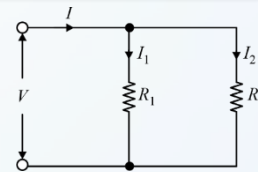
$$V_2 = \frac{R_2}{R_1 + R_2} V$$



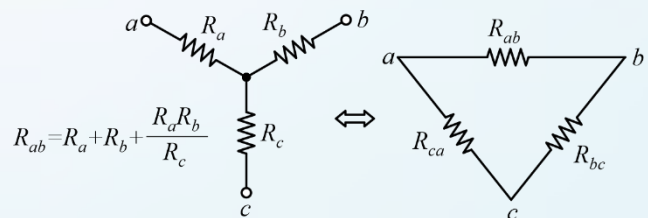
Current Division Equations

$$I_1 = \frac{R_2 I}{R_1 + R_2};$$

$$I_2 = \frac{R_1 I}{R_1 + R_2}$$



Star to Delta Transformation



$$R_{ab} = R_a + R_b + \frac{R_a R_b}{R_c}, \quad R_{bc} = R_b + R_c + \frac{R_b R_c}{R_a}, \quad R_{ca} = R_c + R_a + \frac{R_c R_a}{R_b}$$

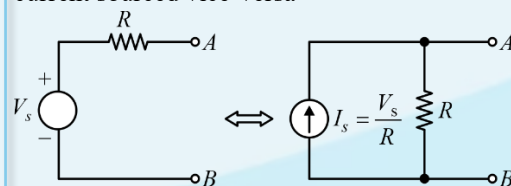
Delta to Star Transformation

$$R_a = \frac{R_{ca} R_{ab}}{R_{ab} + R_{bc} + R_{ca}}, \quad R_b = \frac{R_{ab} R_{bc}}{R_{ab} + R_{bc} + R_{ca}}$$

$$R_c = \frac{R_{bc} R_{ca}}{R_{ab} + R_{bc} + R_{ca}}$$

Source Transformation

Transformation of a practical voltage source to a practical current source vice-versa



Network Theorems

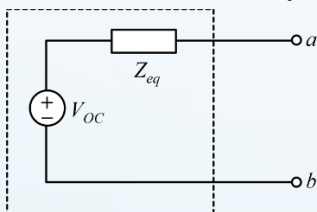
Super Position Theorem

The response in any element of a linear, bilateral RLC network containing more than one independent voltage or current source is the algebraic sum of responses produce by the independent source when each of them acting along with

- All other independent voltage sources are short circuited (S.C.).
- All other independent current sources are open circuited (O.C.).
- All dependent voltage and current sources remain as they are and therefore, they are neither S.C. nor O.C.

Thevenin's Theorem

A linear active RLC network which contains one or more independent or dependent voltage or current sources can be replaced by a single voltage source V_{OC} in series with equivalent impedance Z_{eq} .

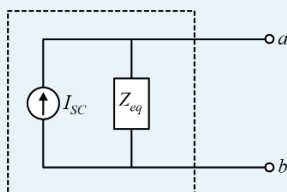


where, V_{OC} = Open circuit voltage between a and b (when $I = 0$).

Z_{eq} = Equivalent impedance between a and b , when.

Norton's Theorem

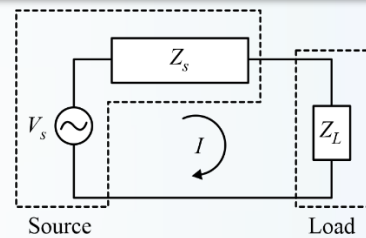
A linear, active RLC network which contains one or more independent or dependent voltage or current sources can be replaced by a single current source I_{SC} in shunt with equivalent impedance Z_{eq} .



Where, I_{SC} = Short circuit current between a and b (when $V = 0$).

Z_{eq} = Same is that of Thevenin's theorem

Maximum Power Transfer Theorem



$Z_L = Z_s^*$ For maximum power transfer

Case 1: If $Z_s = R_s + jX_s$ and $Z_L = R_L + jX_L$
then $R_L = R_s$ and $X_L = -X_s$

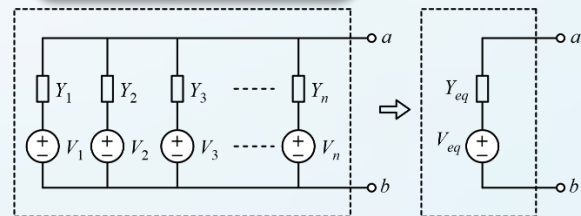
Case 2: If $Z_s = R_s + jX_s$ and $Z_L = R_L$
then $R_L = \sqrt{R_s^2 + X_s^2}$

Case 3: If $Z_L = R_L$ and $Z_s = R_s$
then $R_L = R_s$

Tellegan's Theorem

- In any network, the sum of instantaneous power consumed by various elements of the branches is always equal to zero.
- Total power given out by different voltage sources is equal to total power consumed by various passive elements in various branches of the network.

Milliman's Theorem

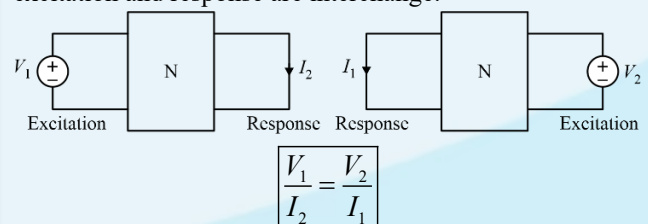


$$V_{eq} = \frac{\sum_{i=1}^n V_i Y_i}{\sum_{i=1}^n Y_i}$$

$$Y_{eq} = \sum_{i=1}^n Y_i$$

Reciprocity Theorem

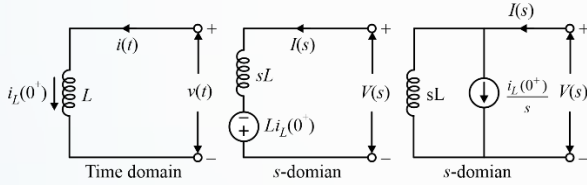
In a linear bilateral single source network, the ratio of excitation to the response is constant then the position of excitation and response are interchange.



Transient

Inductor

Energy storing element



In time domain

$$v(t) = L \frac{di(t)}{dt}$$

$$i(t) = \frac{1}{L} \int_{-\infty}^t v(t) dt$$

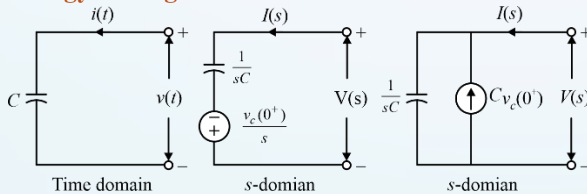
In s-domain

$$V(s) = sL \cdot I(s) - Li_L(0^+) \text{ (with initial condition)}$$

$$I(s) = \left[\frac{V(s)}{sL} + \frac{i_L(0^+)}{s} \right]$$

Capacitor

Energy storing element



In time domain

$$i(t) = C \frac{dv(t)}{dt}$$

$$v(t) = \frac{1}{C} \int_{-\infty}^t i(t) dt$$

In s-domain

$$I(s) = sCV(s) - Cv_c(0^+) \text{ (with initial condition)}$$

$$V(s) = \frac{I(s)}{sC} + \frac{v_c(0^+)}{s}$$

Transient Response

RL Circuit

The transient current through the inductor L at any time t

$$i_L(t) = i_{\infty} - [i_{\infty} - i_0] e^{-R_{eq}t/L}$$

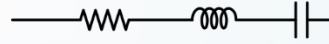
RC Circuit

The transient voltage across Capacitor C at any time t .

$$v_C(t) = v_{\infty} - [v_{\infty} - v_0] e^{-t/R_{eq}C}$$

Series Resonance

- Acceptor circuit because offers min impedance hence, current is max.



- $f < f_0 \rightarrow X_C > X_L \rightarrow$ leading PF
- $f > f_0 \rightarrow X_L > X_C \rightarrow$ lag PF
- $f = f_0 \rightarrow X_L = X_C \rightarrow$ UPF
- L & C Combo $\rightarrow$ SC

$$Q = \frac{1}{R} \sqrt{\frac{L}{C}}$$

$$s^2 + \frac{R}{L}s + \frac{1}{LC} = 0$$

$$w_0 = \frac{1}{\sqrt{LC}} = \sqrt{w_1 \cdot w_2}$$

$$\varepsilon = \frac{1}{2Q}$$

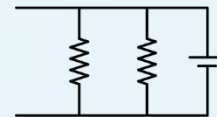
$$B.W. = w_2 - w_1 = \frac{w_0}{2} = \frac{w_0}{Q} = \frac{R}{L}$$

$$w_1 \text{ or } w_2 = \pm \frac{R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

$$w_1 \text{ or } w_2 \approx w_0 \mp \frac{BW}{2}$$

Parallel Resonance

- Rejector circuit because offers max impedance hence, current is min.



- $w < w_0, b_L > b_C \rightarrow$ Lag PF
- $w > w_0, b_C > b_L \rightarrow$ Lead PF
- $w = w_0, b_C = b_L \rightarrow$ UPF
- L & C Combo $\rightarrow$ OC

- $Q = R\sqrt{\frac{C}{L}}$
- $S^2 + \frac{1}{RC}S + \frac{1}{LC}$
- $w_0 = \frac{1}{\sqrt{LC}} = \sqrt{w_1 \cdot w_2}$
- $\varepsilon = \frac{1}{2Q}$
- $B.W. = \frac{1}{RC}$
- $W_1 \text{ (or) } W_2 = +\frac{1}{2RC} \pm \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}}$
- $W_1 \text{ (or) } W_2 \cong w_0 \mp \frac{Bw}{2}$

Quality Factor

- $Q = \frac{|V_L|}{|V|}$ (or) $Q = \frac{|V_C|}{|V|}$ and
- $Q = \frac{|I_L|}{|I|}$ (or) $Q = \frac{|I_C|}{|I|}$
- $Q = \frac{\text{Reactive Power}}{\text{Real Power}}$

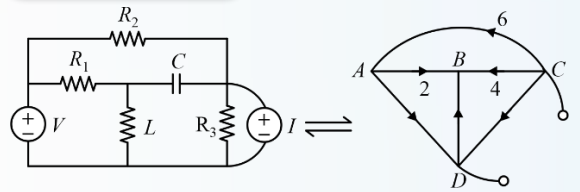
Tank Circuit

- $w_0 = \frac{1}{\sqrt{LC}} \sqrt{1 - \frac{R^2 C}{L}}$
- $Z = \frac{1}{R_L C}$ dynamic R

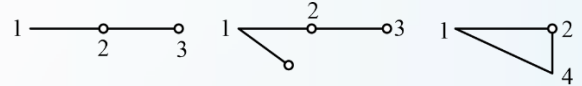
TWO PORT NETWORK

Parameter	Equation	Modelling	Sym	Reci
	$\begin{pmatrix} V_1 \\ V_2 \end{pmatrix} = \begin{pmatrix} Z_a + Z_b & Z_b \\ Z_b & Z_b + Z_c \end{pmatrix} \begin{pmatrix} I_1 \\ I_2 \end{pmatrix}$		$Z_{11} = Z_{22}$	$Z_{12} = Z_{21}$
	$\begin{pmatrix} I_1 \\ I_2 \end{pmatrix} = \begin{pmatrix} Y_a + Y_b & Y_b \\ Y_b & Y_b + Y_c \end{pmatrix} \begin{pmatrix} V_1 \\ V_2 \end{pmatrix}$		$Y_{11} = Y_{22}$	$Y_{12} = Y_{21}$

Graph Theory



- **Path:** Traversal plan one node to another.
- **Subgraph:** Part of a graph, may/may not be loop



$$\text{No. of branches } {}^nC_2 = \frac{n(n-1)}{2}$$

- **Tree:** Connected & Spanning of subgraph contains all the nodes of main graph but doesn't form closed path.
- **Twigs:** Tree branches are called twigs.
- **Links/Chords:** After forming a tree, removed branches are called links/chords.

$$\text{No. of links } l = b - (n - 1) = b - n + 1$$

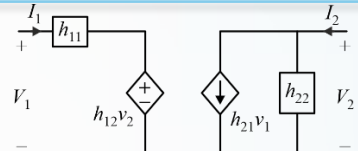
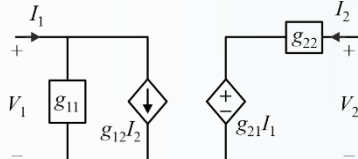
$$\text{No. of twigs} = \text{branches in tree} = n - 1$$

Reduced Incidence Matrix

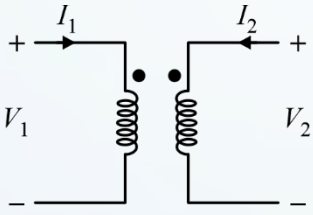
$$A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & -1 & -1 & -1 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & 1 & 0 \end{bmatrix}$$

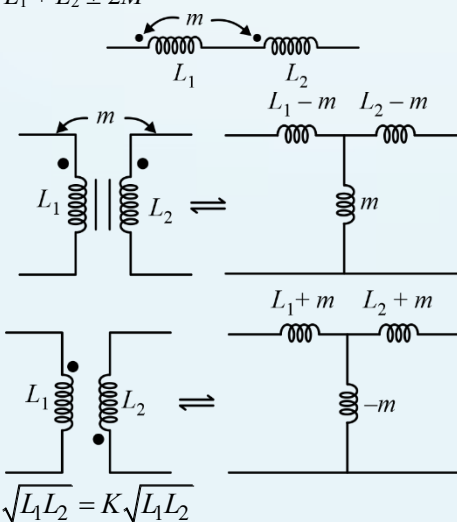
$$A_r = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & -1 & -1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- No. of possible Trees = $|[A_r][A_r]^T|$
- **Tie-Set:** Contains one link & rest as twig.
- **Cut-Set:** Contains one twig rest as link.

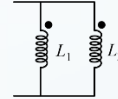
h	$\begin{pmatrix} V_1 \\ I_2 \end{pmatrix} = h \begin{pmatrix} I_1 \\ V_2 \end{pmatrix}$		$\Delta h = 1$	$h_{21} = -h_{12}$
g	$\begin{pmatrix} I_1 \\ V_2 \end{pmatrix} = g \begin{pmatrix} V_1 \\ I_2 \end{pmatrix}$		$\Delta g = 1$	$g_{21} = -g_{12}$
T	$\begin{pmatrix} V_1 \\ I_1 \end{pmatrix} = T \begin{pmatrix} V_2 \\ -I_2 \end{pmatrix}$		$A = D$	$\Delta T = 1$

Transformer Parameters

- $\frac{V_1}{V_2} = n = -\frac{I_2}{I_1}$
- 
- $h = \begin{bmatrix} 0 & n \\ -n & 0 \end{bmatrix}; T = \begin{bmatrix} n & 0 \\ 0 & 1/n \end{bmatrix}$
- Shortcut:
- $h = \begin{bmatrix} \frac{1}{Y_{11}} & \frac{Z_{12}}{Z_{22}} \\ \frac{Y_{21}}{Y_{11}} & \frac{1}{Z_{22}} \end{bmatrix}; T = \begin{bmatrix} \frac{Z_{11}}{Z_{21}} & -\frac{1}{Y_{21}} \\ \frac{1}{Z_{21}} & -\frac{Y_{11}}{Y_{21}} \end{bmatrix}$
- $L_{eq} = L_1 + L_2 \pm 2M$

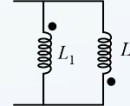


Parallel Aiding:



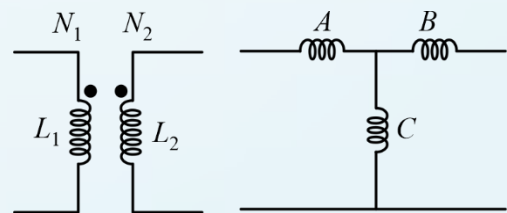
$$L_{eq} = \frac{L_1 L_2 - M^2}{L_1 + L_2 - 2M}$$

Parallel Opposing:



$$L_{eq} = \frac{L_1 L_2 - M^2}{L_1 + L_2 + 2M}$$

Transformers



$$A = L_1 - \frac{M}{K} \quad \text{Where } k = \left(\frac{N_2}{N_1} \right)$$

$$C = \frac{M}{K}$$

$$B = \frac{L_2}{K^2} - \frac{M}{K}$$

SIGNALS AND SYSTEMS

Basics of Signal System

Elementary signal classification

Name	Continuous	Discrete	Name	Continuous	Discrete
Unit step function	$u(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}$	$u(t) = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$	Signum signal	$\text{sgn}(t) = \begin{cases} 1, & t > 0 \\ -1, & t < 0 \end{cases}$	$\text{sgn}(n) = \begin{cases} 1, & n > 0 \\ -1, & n < 0 \end{cases}$
Ramp signal	$r(t) = \begin{cases} t, & t \geq 0 \\ 0, & t < 0 \end{cases}$	$r[n] = nu(n) = \begin{cases} n, & n \geq 0 \\ 0, & n < 0 \end{cases}$	Sinusoidal signal	$x(t) = \sin(2\pi f_0 t + \theta)$	$x(n) = \sin(2\pi f_0 n + \theta)$
Impulse function	$\delta(t) = 0, t \neq 0$	$\delta(n) = \begin{cases} 1, & n = 0 \\ 0, & \text{otherwise} \end{cases}$	sinc function	$\text{sinc}(\omega_0 t) = \frac{\sin(\pi \omega_0 t)}{\pi \omega_0 t}$	$\text{sinc}[\omega_0 n] = \frac{\sin(\pi \omega_0 n)}{\pi \omega_0 n}$
Rectangular pulse function	$\Pi\left(\frac{t}{\tau}\right) = \begin{cases} 1, & t \leq \tau/2 \\ 0, & t > \tau/2 \end{cases}$	$\Pi\left[\frac{n}{2N}\right] = \begin{cases} 1, & n \leq N \\ 0, & n > N \end{cases}$	Triangular pulse	$\Delta\left(\frac{t}{\tau}\right) = \begin{cases} 1 - \frac{ t }{\tau}, & t \leq \tau \\ 0, & t > \tau \end{cases}$	$\Delta\left(\frac{n}{N}\right) = \begin{cases} 1 - \frac{ n }{N}, & n \leq N \\ 0, & \text{elsewhere} \end{cases}$

Important Properties of Signals

Name	Properties
Impulse properties	$\int_{-\infty}^{\infty} \delta(t) dt = 1$ $\delta(at) = \frac{1}{ a } \delta(t)$ $\delta(at+b) = \frac{1}{ a } \delta\left(t + \frac{b}{a}\right)$ $\int_{-\infty}^{\infty} x(t) \delta(t-\lambda) dt = x(\lambda)$ $x(t) \delta(t-\lambda) = x(\lambda) \delta(t-\lambda)$
Signal in terms of unit step and vice versa.	$u(t) = \int_{-\infty}^t \delta(\tau) d\tau$ $\delta(t) = \frac{d}{dt} u(t)$ $r(t) = tu(t)$ $u(t) = \frac{d}{dt} r(t)$ $\text{sgn}(t) = u(t) - u(-t)$ $\text{sgn}(t) = 2u(t) - 1$ $\Pi\left(\frac{t}{\tau}\right) = u\left(t + \frac{\tau}{2}\right) - u\left(t - \frac{\tau}{2}\right)$
Derivative of impulse (doublet)	$\frac{d}{dt} \delta(t) = \delta'(t) = \begin{cases} \text{undefined}, & t = 0 \\ 0, & \text{otherwise} \end{cases}$ $\delta'(at) = \frac{1}{a a } \delta'(t)$ $\int_{-\infty}^{\infty} x(t) \delta'(t-\lambda) dt = -\frac{d}{dt} x(t) \Big _{t=\lambda}$ $x(t) \delta'(t-\lambda) = x(\lambda) \delta'(t-\lambda) - x'(\lambda) \delta(t-\lambda)$

Property	Continuous	Discrete
Energy	$E_x = \int_{-\infty}^{\infty} x(t) ^2 dt < \infty$	$E_x = \sum_{n=-\infty}^{\infty} x[n] ^2$
Power	$P_x = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t) ^2 dt < \infty$	$P_x = \lim_{N \rightarrow \infty} \frac{1}{2N} \sum_{n=-N}^N x[n] ^2 < \infty$
Periodic	$x(t-T_0) = x(t), T_0 = \text{period}$	$x[n-N_0] = x[n], N_0 = \text{period}$
Even	$x(-t) = x(t)$	$x[-n] = x[n]$
Odd	$x(-t) = -x(t)$	$x[-n] = -x[n]$

LTI System

Types of System

System Type	Description / Example
Linear System	Satisfies additivity and homogeneity Example: $y(t) = 2x(t)$, $y(t) = \int x(t) dt$
Non-Linear System	Does not satisfy additivity or homogeneity
Time-Invariant	Time shift in input causes the same shift in output Example: $y(t) = x(t^2)$
Time-Varying	Time shift in input causes a different output shift
Causal	Output depends on past and present input Example: $y(t) = x(t)$

Non-Causal	Output depends on future input
Static (Memoryless)	Output depends only on the present input Example: $y(t) = 2x(t)$
Dynamic	Output depends on past/future input (has memory)
Stable	Bounded input gives bounded output Example: $y(t) = e^{-t}$
Unstable	Bounded input does not guarantee bounded output
Invertible	Has an inverse system Example: $y(t) = \int x(t)dt \Leftrightarrow x(t) = d/dt y(t)$
Non-Invertible	Does not have a unique inverse

Convolution

Continuous Time	Discrete time
$y(t) = x_1(t) \otimes x_2(t)$ $= \int_{-\infty}^{\infty} x_1(\tau) x_2(t - \tau) d\tau$ $= \int_{-\infty}^{\infty} x_1(t - \tau) x_2(\tau) d\tau$	$y(n) = x_1(n) \otimes x_2(n)$ $= \sum_{k=-\infty}^{\infty} x_1[k] x_2[n - k]$ $= \sum_{k=-\infty}^{\infty} x_1[n - k] x_2[k]$
$x(t - a) \otimes \delta(t - b) = x(t - a - b)$	$x(n - a) \otimes \delta(n - b) = x(n - a - b)$
$\delta(t) \otimes \delta(t) = \delta(t)$	$u(n) \otimes u(n) = (n + 1)u(n)$
$\delta(t - a) \otimes \delta(t - b) = \delta(t - a - b)$	$\delta(n - n_1) \otimes \delta(n - n_2) = \delta(n - n_1 - n_2)$

Fourier Series

Trigonometric Fourier Series

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

Where, ω_0 = Fundamental frequency; a_0, a_n, b_n = Trigonometric Fourier series coefficient; $n\omega_0$ = n^{th} harmonic of ω_0

$$a_0 = \frac{1}{T} \int_0^T f(t) dt; a_n = \frac{2}{T} \int_0^T f(t) \cos n\omega_0 t dt; b_n = \frac{2}{T} \int_0^T f(t) \sin n\omega_0 t dt$$

Exponential Fourier series

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t} \quad \text{Where } c_0 = a_0 = \frac{1}{T} \int_0^T f(t) dt$$

$$c_n = \frac{1}{T} \int_0^T f(t) e^{-jn\omega_0 t} dt$$

Relation between Exponential and Trigonometric Fourier Series

$$a_n = c_n + c_{-n}; c_n = \frac{a_n}{2} - j \frac{b_n}{2}$$

$$b_n = j(c_n - c_{-n}); c_{-n} = \frac{a_n}{2} + j \frac{b_n}{2}$$

Effect of symmetry of Fourier coefficients

Odd symmetry	Even symmetry	Half-wave symmetry
For signals with odd symmetry, the Fourier coefficients a_0 and a_n are zero. $x(t) = -x(-t)$	For signals with even symmetry, the Fourier coefficient b_n is zero. $x(t) = x(-t)$	For signals with half wave symmetry. The Fourier series will consist of odd harmonic terms of the sine and cosine signals. $x(t \pm T/2) = -x(t)$

Properties of exponential form of Fourier series coefficients

Property	Continuous Time periodic signal	Fourier series coefficient
Time shifting	$x(t \pm t_0)$	$c_n e^{\pm jn\omega_0 t_0}$
Conjugation	$x^*(t)$	c_{-n}^*
Time reversal	$x(-t)$	c_{-n}
Time scaling	$x(\alpha t); \alpha > 0$ [x(t)] is periodic with period T/α	c_n (no change in Fourier coefficient)

Multiplication	$x(t)y(t)$	$\sum_{m=-\infty}^{\infty} c_m d_{n-m}$
Differentiation	$\frac{d}{dt} x(t)$	$j n \omega_0 c_n$
Integration	$\int_{-\infty}^t x(t) dt$ (Finite valued and periodic only if $a_0 = 0$)	$\frac{1}{jn\omega_0} c_n$

Periodic convolution	$\int_t x(\tau)y(t-\tau)d\tau$	$TC_n d_n$
Symmetry of real signals	$x(t)$ is real	$c_n = c_{-n}^*$ $ c_n = c_{-n} $; $\angle c_n = -\angle c_{-n}$ $\text{Re}\{c_n\} = \text{Re}\{c_{-n}\}$ $\text{Im}\{c_n\} = -\text{Im}\{c_{-n}\}$
Real and even	$x(t)$ real and even	c_n are real and even
Real and odd	$x(t)$ real and odd	c_n are imaginary and odd

Properties of Discrete Time Fourier Series

Property	Discrete Time Periodic Signal	Fourier Series Coefficient
Time shifting	$x(n \pm n_0)$	$c_k e^{\pm jk\left(\frac{2\pi}{N}\right)n_0}$
Conjugation	$x^*[n]$	c_{-k}^*
Time reversal	$x[-n]$	c_{-k}
Time scaling	$x(m)[n] = \begin{cases} x\left[\frac{n}{m}\right]; & \text{if } n \text{ is a multiple of } M \\ 0; & \text{otherwise} \end{cases}$ It n is a multiple of M otherwise	$\frac{1}{m} c_k$ (period mN)
Multiplication	$x[n]y[n]$	$\sum_{L=0}^{N-1} c_L b_{K-L}$
First difference	$x[n] - x[n-1]$	$\left(1 - e^{-j\left(\frac{2\pi}{N}\right)k}\right) c_k$

Running sum	$\sum_{n=-\infty}^n x[k]$	$\frac{1}{\left(1 - e^{-j\left(\frac{2\pi}{N}\right)k}\right)} c_k$
Convolution	$\sum_{r=0}^{N-1} x[r]y[n-r]$	$Nc_k d_k$

Note:

- Parseval's Power Theorem for Continuous time signals is given as:

$$P = \frac{1}{T} \int_0^T |x(t)|^2 dt = \sum_{n=-\infty}^{\infty} |c_n|^2 = a_0^2 + \frac{1}{2} \sum_{n=1}^{\infty} (a_n^2 + b_n^2)$$

- Parseval's Power Theorem for Discrete time signals is given as:

$$P = \frac{1}{N} \sum_{n=0}^{N-1} |x[n]|^2 = \sum_{k=0}^{N-1} |c_k|^2$$

Fourier Transform

Basic Definitions

Continuous time Fourier Transform:

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt$$

Continuous time Inverse Fourier Transform:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega)e^{j\omega t} d\omega$$

Discrete time Fourier Transform:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

Inverse Discrete time Fourier Transform:

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n} d\omega$$

Properties of Fourier Transformation

Property	Continuous time Fourier Transform	Discrete time Fourier Transform
Time Shifting	$x(t \pm t_0) \leftrightarrow e^{\pm j\omega t_0} X(\omega)$	$x[n \pm n_0] \leftrightarrow e^{\pm j\omega n_0} X(e^{j\omega})$
Frequency Shifting	$e^{\pm j\omega_0 t} x(t) \leftrightarrow X(\omega \mp \omega_0)$	$e^{\pm j\omega_0 n} x[n] \leftrightarrow X(e^{j(\omega \mp \omega_0)})$
Conjugation	$x^*(t) \leftrightarrow X^*(-j\omega)$	$x^*[n] \leftrightarrow X^*(e^{-j\omega})$
Time Reversal	$x(-t) \leftrightarrow X(-\omega)$	$x[-n] \leftrightarrow X(e^{-j\omega})$
Time and Frequency Scaling	$x(at) \leftrightarrow \frac{1}{ a } X\left(\frac{\omega}{a}\right)$	$x_{(k)}[n] = \begin{cases} x\left[\frac{n}{k}\right], & \text{if } n \text{ is multiple of } k \\ 0, & \text{if } n \text{ is not multiple of } k \end{cases} \leftrightarrow X(e^{jk\omega})$
Convolution	$x(t) * y(t) \leftrightarrow [X(\omega)Y(\omega)]$	$x[n] * y[n] \leftrightarrow X(e^{j\omega})Y(e^{j\omega})$
Multiplication	$x(t) \cdot y(t) \leftrightarrow \frac{1}{2\pi} [X(\omega) * Y(\omega)]$	$x[n]y[n] \leftrightarrow \frac{1}{2\pi} [X(e^{j\omega}) * Y(e^{j\omega})]$

Differentiation in Time	$\frac{d}{dt}x(t) \leftrightarrow j\omega X(\omega)$	$x[n] - x[n-1] \leftrightarrow (1 - e^{-j\omega})X(e^{j\omega})$
Integration/Accumulation	$\int_{-\infty}^t x(t)dt \leftrightarrow \frac{1}{j\omega}X(\omega) + \pi X(0)\delta(\omega)$	$\sum_{k=-\infty}^n x[k] \leftrightarrow \frac{1}{1 - e^{-j\omega}}X(e^{j\omega})$
Differentiation in Frequency	$tx(t) \leftrightarrow j \frac{d}{d\omega}X(\omega)$	$nx[n] \leftrightarrow j \frac{d}{d\omega}X(e^{j\omega})$
Duality	$x(t) \leftrightarrow X(\omega)$ $X(t) \leftrightarrow 2\pi x(-\omega)$	---
Parseval's Theorem	$\int_{-\infty}^{+\infty} x^2(t)dt = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(\omega) ^2 d\omega$	$\sum_{n=-\infty}^{+\infty} x[n] ^2 = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) ^2 d\omega$

Basic Fourier Transform pairs

Continuous Time Pairs	Discrete Time Pairs
$e^{j\omega_0 t} \leftrightarrow 2\pi\delta(\omega - \omega_0)$	$e^{j\omega_0 n} \leftrightarrow 2\pi \sum_{l=-\infty}^{+\infty} \delta(\omega - \omega_0 - 2\pi l)$
$x(t) = \begin{cases} 1, & t < T_1 \\ 0, & t > T_1 \end{cases} \leftrightarrow \frac{2\sin \omega T_1}{\omega}$	$x[n] = \begin{cases} 1, & n < N_1 \\ 0, & n > N_1 \end{cases} \leftrightarrow \frac{\sin[\omega(N_1 + 1/2)]}{\sin(\omega/2)}$
$\frac{\sin WT}{\pi t} \leftrightarrow \begin{cases} 1, & \omega < W \\ 0, & \omega > W \end{cases}$	$\frac{W}{\pi} \text{sinc}\left(\frac{Wn}{\pi}\right), 0 < W < \pi \leftrightarrow \begin{cases} 1, & 0 \leq \omega \leq W \\ 0, & W < \omega \leq \pi \end{cases}$
$u(t) \leftrightarrow \frac{1}{j\omega} + \pi\delta(\omega)$	$u[n] \leftrightarrow \frac{1}{1 - e^{-j\omega}} + \sum_{k=-\infty}^{+\infty} \pi\delta(\omega - 2\pi k)$
$e^{-at}u(t), \text{Re}\{a\} > 0 \leftrightarrow \frac{1}{a + j\omega}$	$a^n u[n], a < 1 \leftrightarrow \frac{1}{1 - ae^{-j\omega}}$

Fourier Transform based on signal characteristics

Signal	Fourier Transform
Real	Conjugate symmetric
Imaginary	Conjugate anti-symmetric
Real + even	Real + even
Real + odd	Imaginary + odd
Imaginary + odd	Real + odd
Imaginary + even	Imaginary + even
Continuous + Periodic	Discrete + Aperiodic
Continuous + Aperiodic	Continuous + Aperiodic
Discrete + Periodic	Discrete + Periodic
Discrete + Aperiodic	Continuous + Periodic

Laplace Transform and Z-Transform

Basic Definitions

Bilateral Laplace Transform: $X(s) = \int_{-\infty}^{\infty} x(t)e^{-st} dt$

Unilateral Laplace Transform: $X(s) = \int_0^{\infty} x(t)e^{-st} dt$

Inverse Laplace Transform: $x(t) = \int_{-\infty}^{\infty} X(s)e^{st} ds$

Bilateral Z-Transform: $X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$

Unilateral Z-Transform: $X(z) = \sum_{n=0}^{\infty} x[n]z^{-n}$

Properties of Laplace and Z-Transformation

Property	Laplace Transform		Z- Transform	
	Pairs	ROC	Pairs	ROC
Time Shifting	$x(t \pm t_0) \leftrightarrow e^{\pm st_0} X(s)$	R	$x[n - n_0] \leftrightarrow z^{-n_0} X(z)$	R
Shifting in the s -Domain/Z/domain	$e^{s_0 t} x(t) \leftrightarrow X(s - s_0)$	Shifted version of R	$a^n x[n] \leftrightarrow X(a^{-1} z)$	Scaled version of R
Time scaling/reversal	$x(at) \leftrightarrow \frac{1}{ a } X\left(\frac{s}{a}\right)$	Scaled ROC	$x(-n) \leftrightarrow X(z^{-1})$	Inverted R
Conjugation	$x^*(-t) \leftrightarrow X^*(s^*)$	R	$x^*[n] \leftrightarrow X^*(z^*)$	R
Convolution	$x_1(t) * x_2(t) \leftrightarrow X_1(s)X_2(s)$	$R_1 \cap R_2$	$x_1[n] * x_2[n] \leftrightarrow X_1(z)X_2(z)$	Intersection of R_1 and R_2 .
Differentiation in the Time Domain	$\frac{d}{dt} x(t) \leftrightarrow sX(s)$	At least R	-----	-----
Differentiation in the s -Domain/Z-Domain	$-tx(t) \leftrightarrow \frac{d}{ds} X(s)$	R	$nx[n] \leftrightarrow -z \frac{dX(z)}{dz}$	R
Integration in the Time Domain/Accumulation	$\int_{-\infty}^t x(\tau) d\tau \leftrightarrow \frac{1}{s} X(s)$	$R \cap \{\text{Re}\{s\} > 0\}$	$\sum_{k=-\infty}^n x[k] \leftrightarrow \frac{1}{1-z^{-1}} X(z)$	Intersection of R and $ z > 1$
Final Value Theorem	$x(\infty) = \lim_{s \rightarrow 0} sX(s)$		$x(\infty) = \lim_{z \rightarrow 1} (1 - z^{-1})X(z)$	
Initial Value theorem	$x(0) = \lim_{s \rightarrow \infty} sX(s)$		$x(0) = \lim_{z \rightarrow \infty} X(z)$	

Common Laplace and Z-Transform Pairs

Laplace Transform Pairs	Z-Transform Pairs
$u(t) \leftrightarrow \frac{1}{s} \quad \text{Re}\{s\} > 0$	$u[n] \leftrightarrow \frac{1}{1-z^{-1}} \quad z > 1$
$u(-t) \leftrightarrow -\frac{1}{s} \quad \text{Re}\{s\} < 0$	$-u[-n-1] \leftrightarrow \frac{1}{1-z^{-1}} \quad z < 1$
$e^{-at} u(t) \leftrightarrow \frac{1}{s+a} \quad \text{Re}\{s\} > -a$	$\alpha^n u[n] \leftrightarrow \frac{1}{1-\alpha z^{-1}} \quad z > \alpha$
$-e^{-at} u(-t) \leftrightarrow \frac{1}{s+a} \quad \text{Re}\{s\} < -a$	$-\alpha^n u[-n-1] \leftrightarrow \frac{1}{1-\alpha z^{-1}} \quad z < \alpha$
$[\cos \omega_0 t] u(t) \leftrightarrow \frac{s}{s^2 + \omega_0^2} \quad \text{Re}\{s\} > 0$	$[\cos \omega_0 n] u[n] \leftrightarrow \frac{1 - [\cos \omega_0] z^{-1}}{1 - [2 \cos \omega_0] z^{-1} + z^{-2}} \quad z > 1$
$[\sin \omega_0 t] u(t) \leftrightarrow \frac{\omega_0}{s^2 + \omega_0^2} \quad \text{Re}\{s\} > 0$	$[\sin \omega_0 n] u[n] \leftrightarrow \frac{[\sin \omega_0] z^{-1}}{1 - [2 \cos \omega_0] z^{-1} + z^{-2}} \quad z > 1$

Causality and Stability Condition

Property	Laplace Transform (Continuous-Time)	Z-Transform (Discrete-Time)
Causality Condition	ROC is right-sided and $\text{Re}(s) > \sigma_{\max}$ (right of the rightmost pole).	ROC is outside the outermost pole, i.e., $ z > r_{\max}$ (right-sided sequence).
Stability Condition	ROC includes the $j\omega$ -axis (i.e., $\text{Re}(s) = 0$ is in ROC).	ROC includes the unit circle (i.e., $ z = 1$ is in ROC).
Causal + Stable	All poles must lie in the left-half s-plane ($\text{Re}(s) < 0$).	All poles must lie inside the unit circle ($ z < 1$).

DFT and FFT

Discrete Fourier Transformation

Consider a sampled and periodic signal $x[n]$ its DFT is given as

$$\text{DFT}\{x[n]\} = X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}$$

And the inverse DFT can be obtained as

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\frac{2\pi}{N}kn} : \Omega = \frac{2\pi}{N}$$

Where 'N' is period of the signal $x[n]$

Properties of DFT

If $x[n] \xrightarrow{\text{DFT}} X[k]$ then

Property	DFT
Linearity	$\alpha x_1[n] + \beta x_2[n] \xrightarrow{\text{DFT}} \alpha X_1[k] + \beta X_2[k]$
Time shifting	$x[n - n_0] \xrightarrow{\text{DFT}} e^{-j\frac{2\pi}{N}kn_0} X[k]$
Shift in k-domain	$x[n] e^{j\frac{2\pi}{N}k_0n} \xrightarrow{\text{DFT}} X[k - k_0]$
Conjugate symmetry	$x^*[n] \xrightarrow{\text{DFT}} X^*[-k]$
Time reversal	$x[-n] \xrightarrow{\text{DFT}} X[-k]$
Linear convolution	$x_1[n] * x_2[n] \xrightarrow{\text{DFT}} X_1[k] X_2[k]$ with period, $N_1 + N_2 - 1$
Periodic convolution	$x_1[n](N)x_2[n] \xrightarrow{\text{DFT}} \frac{1}{N} [X_1[k] * X_2[k]]$
Parseval's relation	$E = \sum_{n=0}^{N-1} x[n] ^2 = \frac{1}{N} \sum_{n=0}^{N-1} X[k] ^2 =$

Fast Fourier Transformation (FFT)

The N-point DFT is given as

$$X[k] = \sum_{n=0}^{N-1} x[n] \left(e^{-j\frac{2\pi}{N}} \right)^{kn}$$

- Number of addition = $\text{Nog}_2 N$
- Number of multiplication = $\frac{N}{2} \log_2 N$

Miscellaneous

Hilbert transforms

- It provides a phase shift of -90° to all signals
- Signal and its Hilbert transform have same energy spectral density or power spectral density and auto-correlation.
- Impulse response of Hilbert transform is given as:

$$h(t) = \frac{1}{\pi t}, \quad H(j\omega) = -j \text{sgn}(\omega)$$

Sampling

- Any band-limited signal is uniquely determined by its samples if the sampling frequency is greater than twice the maximum frequency i.e. $\omega_s > 2\omega_m$.
- Band pass sampling: For a band pass signal with the spectral components from ω_L to ω_H , the sampling frequency is given by:

$$\omega_s \geq \frac{2\omega_H}{k}, \text{ where } k = \frac{\omega_H}{\omega_H - \omega_L} \quad \{K \text{ must be an integer}\}$$

FIR Filter V/S IIR Filter

Property	FIR Filter	IIR Filter
Impulse Response	Impulse response is of finite duration.	Impulse response is of infinite duration.
System Type	Non-recursive (no feedback path).	Recursive (feedback path from output to input).
Phase Response	Always designed as a linear phase system.	Cannot be designed as a linear phase system.
Stability	Stability is guaranteed.	Stability cannot be guaranteed.
Processing Speed	Slower due to more additions and multiplications for the same frequency response as IIR.	Faster due to fewer additions and multiplications.

□□□



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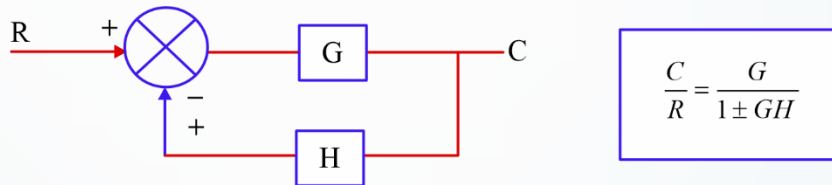
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CONTROL SYSTEM

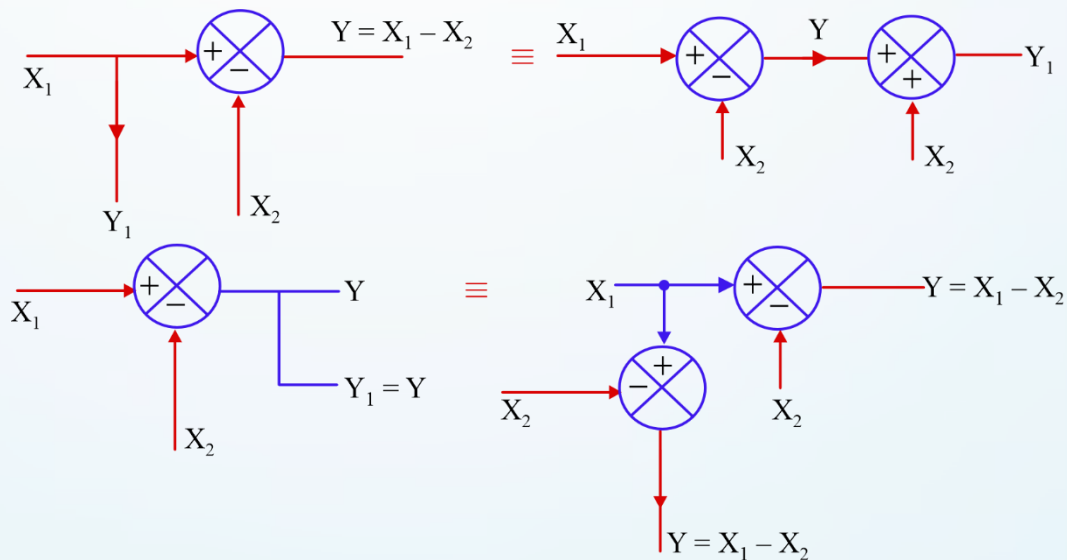
BLOCK DIAGRAM REPRESENTATION AND SIGNAL FLOW GRAPH DIODE CIRCUITS & APPLICATIONS

Feedback System:



Block diagram Reduction:

- Take off point and summer



- Macon Gain formula** – used to calculate for the transfer function

Limitations

(1) $\frac{\text{Output Node}}{\text{Input Node}}$

(2) $\frac{\text{Intermediate Node}}{\text{Input Node}}$

$$\frac{Y}{X} = \frac{\sum_{k=1}^n p_k \Delta_k}{\Delta}$$

n = no of forward path from X to Y

$\Delta = 1 - (\text{sum of all loop gains}) + (\text{sum of product of two non-touching loop gain}) + (\text{sum of product of 3 non-touching loop gains})$

Δ_K = It is dependent on forward path

$\Delta_K = 1 - [\text{sum of all loop gains not touching } K^{th} \text{ forward path}] + (\text{sum of product of 2 non-touching loop gains not touching } K^{th} \text{ Forward path}) + \dots$

TIME RESPONSE ANALYSIS

First Order System:

- Time constant = $\frac{1}{|\text{Real part of dominant pole}|}$
 - 3 dB band width of RC circuit = $\frac{1}{RC} \text{ rad/sec} = \frac{1}{\text{Time constant}}$
- $$T(s) = \frac{K/T}{S + \frac{1}{T}} \rightarrow \text{Time constant} = T, \text{ 3dB BW} = 1/T \text{ rad/sec}$$

Rise Time

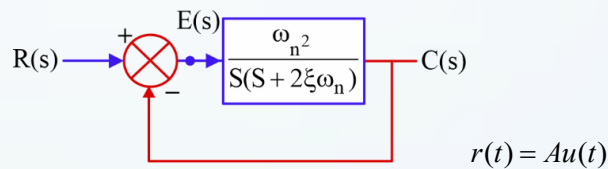
Time taken by step response of first order CLS to reach from 10% to 90% of its final value.

$$t_s = 4T \quad 2\% \text{ error band}$$

$$t_r = 2.2T, \quad t_d = 0.693T$$

Second Order System:

Step Response of under damp system



➤ Error signal $E(S) = R(S) - C(S)$

$$C(S) = \frac{A\omega_n^2}{S(S^2 + 2\xi\omega_n S + \omega_n^2)} \quad C(t) = A \left\{ 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}} \sin(\omega_d t + \theta) \right\} \rightarrow u(t)$$

$$\omega_d = \omega_n \sqrt{1-\xi^2} \quad \cos \theta = \xi, \quad \tan \phi = \frac{\sqrt{1-\xi^2}}{\xi}$$

$$e(t) = A \left[\frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}} \sin(\omega_d t + \theta) \right] u(t)$$

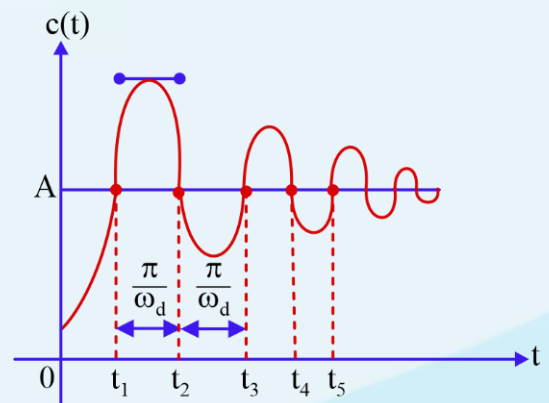
$$(1) \quad t_k [\text{Time Constant When } C(t) = A] \quad t = \frac{n\pi - \phi}{\omega_d}$$

$$(2) \quad \text{peak time} \quad t_p = \frac{n\pi}{\omega_d}$$

$$(3) \quad t_{po(K)} = \frac{(2K-1)\pi}{\omega_d}, \quad K = 1, 2, K^{\text{th}} \text{ peak overshoot time}$$

$$(4) \quad t_{pu(K)} = \frac{2K\pi}{\omega_d} \quad K = 1, 2, \quad K^{\text{th}} \text{ peak undershoot time}$$

$$(5) \quad \text{At peak overshoot time } \frac{dc(t)}{dt} = 0, \text{ same for peak undershoot also}$$



(6) At 1st peak overshoot time value, output is maximum $c(t = t_{po(1)}) = [c(t)]_{\max}$

(7) Time gap b/w two successive $\rightarrow p.o | p.u$ is $2\pi / \omega_d$

(8) Maximum peak percentage undershoot

$$\%M_{pu} = \frac{C(t)|_{ss} - C(t)|_{\min}}{C(t)|_{ss}} \times 100\% = e^{-2\pi\xi\sqrt{1-\xi^2}} \times 100\%$$

(9) Decay ratio $= \frac{M_{po(2)}}{M_{po(1)}} = \frac{M_{pu(2)}}{M_{pu(1)}} = e^{-2\pi\xi\sqrt{1-\xi^2}}$

(10) $\%M_{p0} = m\% = \%M_{po(1)}$ s.1 $p = \frac{m}{100}$ s.2 $\xi = \sqrt{\frac{(\ln p)^2}{\pi^2 + (\ln p)^2}}$

Note : If $r(t) \rightarrow Au(t - t_0)$ then all-time formulas will be replaced by $(t - t_0)$

Impulse Response of 2nd order s/s

(1) Under damped system $0 < \xi < 1$

$$C(t) = \frac{AK\omega_n}{\sqrt{1-\xi^2}} e^{-\xi\omega_n t} \sin(\omega_d t) u(t)$$

(2) Undamped system $\xi = 0$

$$C(t) = AK\omega_n \sin \omega_n t$$

$$t_r = \frac{\pi - \phi}{\omega_d}, t_p \text{ (peak time)} = \frac{n\pi}{\omega_d}, t_s = \frac{4}{\xi\omega_n}, \%M_p e^{\frac{-\pi\xi}{\sqrt{1-\xi^2} \times 100}}$$

(3) Critically damped System $\xi = 1$

$$C(t) = AK\omega_n^2 t e^{-\omega_n t} u(t)$$

(4) Overdamped System $1 < \xi < \infty$

$$C(t) = a_0 e^{-p_1 t} + a_1 e^{-p_2 t}$$

STEADY STATE ERROR AND ROUTH STABILITY TIME RESPONSE ANALYSIS

System Error

(1) Must tend to 0 as $t \rightarrow \infty$

(2) $e_{sys}(t) = r(t) - c(t)$ for $r(t) = Au(t)$

(3) $e_{sys}(t) = \int_{-\infty}^t r(t) - \int_{-\infty}^t c(t) dt$ for $r(t) = A\delta(t)$

(4) $e_{sys}(t) = \frac{dr(t)}{dt} - \frac{dc(t)}{dt}$ for $r(t) = At$ $u(t)$

Amplitude Error

$$e(t) = r(t) - c(t) \rightarrow \text{output}$$

$$e_{ss} = \lim_{s \rightarrow 0} \frac{SR(S)}{1 + G(S)}$$

Steps :

(1) Identify feedback (unity)

$$(2) e_{ss} = \lim_{s \rightarrow 0} \frac{SR(S)}{1 + G(S)}$$

(3) Pole location of $\frac{SR(S)}{1 + G(S)}$ is strictly on L.H.P then perform calculation.

Steady state Error for different inputs.

Input	Check	e_{ss}
$Au(t)$	Poles of $\frac{A}{1+G(S)}$	$e_{ss} = \frac{A}{1+K_p}$
$Atu(t)$	Poles of $\frac{A}{S(1+G(S))}$	$e_{ss} = \frac{A}{K_v}$
$\frac{At^2}{2}u(t)$	Poles of $\frac{A}{S^2(1+G(s))}$	$e_{ss} = \frac{A}{K_a}$

$K_p = \lim_{s \rightarrow 0} G(s)$ = Positional Error constant / coefficient

$K_v = \lim_{s \rightarrow 0} sG(s)$ = velocity Error constant / coefficient

$K_a = \lim_{s \rightarrow 0} s^2 G(s)$ Acceleration Error constant / coefficient

Key Point:

- (1) If any row of RH table multiply or divided by + constant result remains same
- (2) 1st column elements have same sign $\rightarrow$ No roots in
- (3) Any row becomes zero then roots will be at image location
- (4) If all elements in 1st column $\rightarrow$ same sign + No row becomes zero
Then all roots in LHP
- (5) The no of sign changes in 1st column = No of roots lying in RHP
- (6) If any power of s is missing then
 - (i) 1 or more than 1 root may exist in RHP
 $D(s) = s^2 - 1$
 Roots = $s = \pm 1$
 - (ii) Non repeated roots on $j\omega$ axis may exist
 $D(s) = s^2 + 1 \Rightarrow s = \pm j$
 - (iii) Repeated roots may exist on $j\omega$ axis
 - (iv) There may be complex roots
 - (v) If only even power of s exists then root will be at image location.
 - (vi) If only odd power of s exists then few roots will be at origin and remaining roots at image location
 - (vii) In RH table $\rightarrow$ odd ROZ $\rightarrow$ some roots will be at image location

$\downarrow$

Sign of elements in 1st $\rightarrow$ Image location will be imaginary axis

Column is same

(viii) In RH table $\rightarrow$ No odd ROZ $\rightarrow$ No root at image location

$\downarrow$

No root at imaginary location

Odd Row becomes Zero Twice

2 Auxillary equation

- (1) $A_1(s)$: Higher order AE $A_2(s)$: Lower order AE
- (2) Roots of $A_2(s)$ will automatically be covered by $A_1(s)$

- (3) Roots of $A_1(s)$ will be image location

- (4) $\frac{D(s)}{A_1(s)} = P(s) \rightarrow$ Remaining roots

$$\text{No. of roots of } D(s) \text{ on Imaginary axis} = \left(\text{Highest order AE} \right) - 2 \times \left(\text{No. of sign changes below highest order AE} \right) - C$$

No. of roots of $D(s)$ is RHP = No of sign charge in 1st column (2)

No. of roots of $D(s)$ in LHP = order of $D(s)$ - (i) - (ii)

ROOT LOCUS

Rule:1 Angle of asymptotes

Formula:
$$\theta_n = \frac{(2n+1)180^\circ}{(p-z)}$$

$$\theta_n = \frac{(2n+1)180^\circ}{(z-p)}$$
 $Z = \text{finite open loop zeros}$

Rules 2: Centroid

formula
$$\sigma = \frac{\sum p - \sum z}{p - z}$$
 σ : Real numbers

Rule 3: Break point

- (1) Where 2 or more then 2 poles of C.L.S coincides simultaneously. If is part of R.L.

Types:

(1) Break away point

- 2 or more then 2 poles of C.L.S coincides.
- After B.A.P R.L. Breaks into some parts and it can not remains on real axis. It moves into different parts in complex conjugate location
- BAP means shifting on R.L from real axis to complex conjugate Location,
- At BAP k achieves max value for which root remains on real axis if $K \uparrow$ then R.L. moves on complex conjugate location
 $0 \leq K \leq K_{BAP}$: Root locus is on real axis.
 $K > K_{BAP}$: Root locus on complex conjugate location.

(2) Break In point

- 2 or more then 2 poles of C.L.S coincides.
- After the BIP R.L Breaks into some parts and if dan not remain on complex conjugate location. If move on into different parts of real axis.
- BIP means shifting of root locus from complex conjugate location to real axis.
- At BIP K achieves min value for which root locus is on real axis. If $K \uparrow$ if remains on real axis $0 < K \leq K_{BIP} \rightarrow$ complex conjugate location
 $K \geq K_{BIP} \rightarrow$ Real axis
 S.1 from the CE $1 \pm G(s) H(s) = 0$

Rule 4: Angle of departure

- For complex OLP, given originating direction to Branch of DRL

$$\theta_d = 180^\circ - [\phi_p - \phi_z]$$

ϕ_p = Angle sustained by remaining OLP towards desired pole

ϕ_z = Angle sustained by remaining all OLZ towards desired pole

Rule 5: Angle of arrival

- For complex OLZ, it gives terminating direction.

$$\theta_a = 180^\circ - [\phi_z - \phi_p]$$

ϕ_z = Angle sustained by remaining OLZ toward desired OLZ

d_p = Angle sustained by remaining all OLP toward desired OLZ.

FREQUENCY RESPONSE ANALYSIS

Introduction

Test inputs: Sinusoidal input

$$x(t) = A \cos \omega_o t \rightarrow \text{const frequency} = \omega_o \quad S = j\omega$$

$$A \cos \omega_o t \longrightarrow \boxed{H(S)} \longrightarrow y(t) = A |H(j\omega_o)| \cos(\omega_o t + \angle H(j\omega_o))$$

$$A \sin \omega_o t \longrightarrow \boxed{H(S)} \longrightarrow y(t) = A |H(j\omega_o)| \sin(\omega_o t + \angle H(j\omega_o))$$

Few observations

$$A \cos(\omega_o t + \phi) \longrightarrow \boxed{H(S)} \longrightarrow y(t)$$

$$\boxed{y(t) = A |H(j\omega_o)| \cos(\omega_o t + \phi + \angle H(j\omega_o))} \rightarrow [y(t)]_{ss}$$

↓

Resonant frequency $\omega_r = \omega_n \sqrt{1 - 2\xi^2}$

Resonant Peak $M_r = \frac{K}{2\xi\sqrt{1-\xi^2}}$

$$\omega_c = \omega_n \sqrt{(1-2\xi^2) + \sqrt{(1-2\xi^2)^2 + 1}}, \quad \omega_c = \omega_n \sqrt{(1-2\xi^2) + \sqrt{4\xi^4 - 4\xi^2 + 2}}$$

$\rightarrow \omega_c \downarrow$ as $\xi \uparrow$, ω_n constant

$\rightarrow \omega_c \propto \omega_n$ When ξ is constant

Steady state error from Bode plot –

Initial Line slope	Information	e_{ss}
0dB / decade	$Amp = 20 \log_{10} K_p$	$\frac{A}{1 + K_p}$
-20dB/decade	0dB axis Intersection = K_v	A/K_v
-40dB/decade	0 dB axis Intersection = $\sqrt{K_a}$	A/K_a

Nyquist Stability and Plot

- Contour (closed curve in S - plane) or (specified region in s plane) encircles encircles (contains) m poles of $Q(S)$ strictly inside if.



$Q(S)$ plot in $Q(S)$ plane encircles origin $(0,0)$ m times.

- Contour in s -plane passes through one or more pole of $\theta(S)$

$Q(S)$ plot in $Q(S)$ plane remains open curved. Hence poles do not contribute to encirclement of origin.

- Contour in s -plane encircle m zeros of $Q(S)$ strictly inside it

↓ Same direction

$Q(S)$ plot in $Q(S)$ plane encircles the origin m times.

- Contour in s plane has m zero on the boundary of the contour

↓ same sense

- $Q(S)$ plot in $Q(S)$ plane is closed.
- $Q(S)$ plot crosses the origin m times.
- Such zeros do not contribute to encirclement of origin

Rule of Mapping: let TF is $G(S)H(S) \Rightarrow N = P_c - Z_c$
 N = No of encirclement of $(0,0)$ by $G(S)H(S)$ plot in $G(S)H(S)$ plane.

N — $\begin{cases} \rightarrow +ve & G(S)H(S) \text{ plot in ACW direction} \\ \rightarrow -ve & G(S)H(S) \text{ plot in CW direction} \end{cases}$

P_c = No of poles of $G(S)H(S)$ lying inside contour in S plane.

OR

No of poles of $G(S)H(S)$ lying in right-side plane $P_c = P_r$

Z_c = No of zero of $G(S)H(S)$ lying inside contour in S plane.

OR

No of zeros of $G(S)H(S)$ lying in RHP $Z_c = Z_r$

$$N = P_+ - Z_+$$

For OLS to be stable $P_c = 0$

G.M and P.M from Nyquist

G.M (1) $\omega_{pc} = ? \angle T(j\omega) = -180^\circ$ [Nyq plot must cross $-ve$ real axis $\rightarrow \omega_{pc}$ exist]

$$(2) G.M = \frac{1}{|T(j\omega_{pc})|} = \frac{1}{|\text{length on negative real axis till } \omega_{pc}|}$$

Phase Margin –

$$(i) \omega_{gc} \rightarrow |T(j\omega_{gc})| = 1$$

Nyquist plot intersects unity radius circles then ω_{gc} exist .

$$(ii) PM = \angle T(j\omega_{gc}) + 180^\circ$$

$\angle T(j\omega_{gc}) \rightarrow$ from $+ve$ real axis $\begin{cases} \rightarrow CW = -ve \\ \rightarrow ACW = +ve \end{cases}$

For OLS : Min phase

		G.M (in dB)	PM in degree	C.L.S
$K = K_1$	$\omega_{pc} > \omega_{gc}$	+ve (dB)	$+ve^\circ$	Stable
$K = K_2$	$\omega_{pc} > \omega_{gc}$	0 (dB)	0°	M.S
$K = K_3$	$\omega_{pc} > \omega_{gc}$	- ve (dB)	$-ve^\circ$	Unstable

STATE SPACE ANALYSIS

Introduction

Important:

$$[X(S)] = \underbrace{[SI - A]^{-1}[X(0^-)]}_{\text{Solution of state variable due to non zero I.C}} + \underbrace{[SI - A]^{-1}[B][U(S)]}_{\text{Solution of state Variable due to input.}}$$

State Transition Matrix – (S.T.M)

(a) S.T.M in S-domain = $[SI - A]^{-1} \quad n \times n$

(b) S.T.M in time domain = $[\phi(t)]$ or $[e^{At}]_{n \times n}$
 $[\phi(t)]_{n \times n} \xleftrightarrow{\text{L.T}} [SI - A]^{-1}$

Properties of STM

$$[e^{At}] = [\phi(t)]$$

(1) $[e^{A_0}] = [\phi(0)] = [I]_{n \times n}$

(2) $\left[\left(\frac{de^{At}}{dt} \right)_{t=0} \right] = (A)_{n \times n}$

(3) $[\phi(-t)] = [\phi^{-1}(t)]$

(4) $[\phi(t_1 + t_2)] = [\phi_1(t)\phi_2(t)]$

(5) $[\phi^K(t)] = [\phi(Kt)]$

Kalman Test

• Controllability

(1) $[Q_C] = [B : AB : A^2B : \dots : A^{n-1}B]$ Square
Rectangular
 (2) Q_C : Rectangular, $\rho(Q_C) = \rho(A) \rightarrow$ Controllable
 $\rho(Q_C) < \rho(A) \rightarrow$ Uncontrollable

• Observability

(1) $[Q_0] = [C^T : A^T C^T : (A^T)^2 C^T : \dots : (A^T)^{n-1} C^T]$
 (2) Q_0 : Square
 $|Q_0| = 0$ Non observable
 $|Q_0| \neq 0$ observable
 (3) Q_C : Rectangular, $\rho(Q_0) = \rho(A) \rightarrow$ Observable
 $\rho(Q_C) < \rho(A) \rightarrow$ Not observable

CONTROLLER AND COMPENSATOR

Phase Lead compensator

$$T(S) = \frac{\alpha(1+ST)}{1+S\alpha T} \quad \frac{1}{\alpha} > 1 \quad \alpha = \frac{R_2}{R_1 + R_2} < 1$$

Zero Dominant Compensator

Phase $\phi = \tan^{-1}(\omega T) - \tan^{-1}(\alpha \omega T)$

Max. value of phase, $\frac{d}{d\omega} \phi(\omega) = 0$

Max $\leftarrow \omega = \frac{1}{T\sqrt{\alpha}} \quad \tan \phi_{\max} = \frac{1-\alpha}{2\sqrt{\alpha}}$

$$\sin \phi_{\max} = \frac{1-\alpha}{1+\alpha}$$

- Behave as HPF
- Decrease gain of system
- Increase steady state error
- Increase $\omega_{gc}, BW \uparrow$
- Increase P.M, improve relative stability

- Increase $\xi \nearrow$ Mr decreases
 $\searrow$ % Mp decreases
- ω_n increases, t_s decreases
- Improve or reduces the transient region
- Increase the speed of system

Phase lag Compensator

$$T(S) = \left(\frac{1+ST}{1+\beta ST} \right)$$

$$S = \frac{-1}{\beta T}, \quad \frac{-1}{T}, \quad \beta = \frac{R_1 + R_2}{R_1} > 1, \quad T = R_1 C_1$$

- Pole dominant
- $\phi = \tan^{-1} \omega T - \tan^{-1} \beta \omega T$
- $\omega_m = \frac{1}{T\sqrt{\beta}}$
- $\sin \phi_m = \left(\frac{1-\beta}{1+\beta} \right) \beta > 1$
- LPF
- Gain remains constant

- $\beta \begin{cases} = 1 & e_{ss} \text{ same} \\ \geq 1 & e_{ss} \text{ reduces} \end{cases}$
- Reduces $\omega_{gc} \rightarrow$ B.W reduced
- Reduces P.M $\rightarrow$ Relative stability decreases
- $\xi \downarrow \rightarrow M_p \uparrow$ and $M_r \uparrow$
- $\xi \downarrow, \omega_n \downarrow \rightarrow t_s \uparrow$
- Increases transient region, speed of operation decreases.

Proportional Controller Effects

- (1) e_{ss} reduces if $K_p > 1$
- (2) $\xi \omega_n = \text{constant}, t_s = \text{constant}$, stability same
- (3) $\xi \downarrow, \%M_p \uparrow, \omega_d \uparrow, t_r \downarrow$

Derivative Controller

- Derivative Controller reduces type of system by 1
- $e_{ss} \uparrow$ for same input
- Transient region reduced

Proportional Derivative Controller (P-D)

$$\frac{Y}{X} = K_p + K_d S \xrightarrow{K_p=1} 1 + K_d S$$

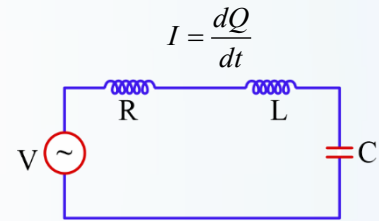
- It reduces $\%M_p, t_s, t_r$
- It improves relative stability and transient region

PID Controller

- Improves stability and decreases e_{ss}
- Increases type and decreases e_{ss}

Mathematical Modelling

Force Voltage Analogy

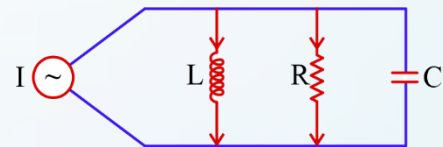


$$V = \frac{Ld^2Q}{dt^2} + \frac{RdQ}{dt} + \frac{1}{C}Q$$

$$F = \frac{Md^2x}{dt^2} + \frac{Bdx}{dt} + K_x$$

$$\begin{cases} F \rightarrow V \\ M \rightarrow L \\ B \rightarrow R \end{cases} \quad \begin{cases} x \rightarrow Q \\ K \rightarrow \frac{1}{C} \end{cases} \quad v \longleftrightarrow I$$

Force Current Analogy



$$V = \frac{dQ}{dt}$$

$$I = C \frac{d^2Q}{dt^2} + \frac{1}{R} \frac{dQ}{dt} + \frac{1}{L}Q$$

$$F \rightarrow I$$

$$M \rightarrow C$$

$$B \rightarrow 1/R$$

$$K \rightarrow 1/L$$

$$x \rightarrow Q$$

$$\text{Voltage} \rightarrow \text{Velocity}$$



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DIODE CIRCUITS & APPLICATIONS

Diode

- A two terminal semiconductor device with PN junction is called a diode
- A PN junction diode is a two terminal device formed by doping with acceptor and dopant impurities at different regions.

VI Characteristics

- In ideal condition, a diode works as a short circuit at during forward bias and open circuit during reverse bias, i.e. like an ideal switch
- In practical condition a diode works as low resistance, and high resistance at reverse bias, i.e. like a practical switch.

Application of Diode in Clipper circuit

- The Circuit clips a portion of the input signal
- On the basis of which part the circuit clips, the circuit is named as Positive or Negative clipper
- A series or shunt clipper is named according to the placement of diode in the circuit.

Positive Clipper: Clips positive portion of the input signal.

Negative Clipper: Clips negative portion of the input signal.

Application of Diode in Clamper Circuit

- A clamper circuit clamps or adds a DC shift to the input signal.
- Based on the polarity of shift, a Positive or Negative clamper is named.

Positive Clamper Circuit: Adds a positive DC shift to the input signal.

Negative Clamper Circuit: Adds a negative DC shift to the input signal.

Note: For Proper functioning of clamper circuit, time constant of circuit should be much greater than time period of input signal ($R_L C \gg T$)

Application of Diode in Peak Detector

A peak detector detects the peak of the input signal.

- **Positive Peak Detector:** Detects the positive peak of input signal.
- **Negative Peak Detector:** Detects the negative peak of input signal

Application of Diode in Voltage Doubler

- Voltage doubler gives the output as double of the input signal.
- A level shifter or clamper followed by a peak detector gives a Voltage Doubler

Application of Diode in Voltage Multiplier

Using multiple stages of a set of a clamper and peak detector gives voltage multiplier, also called as Cockcroft-Walton Voltage Multiplier.

Zener Diode VI Characteristics

- A "reverse biased" diode blocks current in the reverse direction, but will suffer from premature breakdown or damage if the reverse voltage applied across it is too high.
- Zener Diode are basically the same as the standard PN junction diode but are specially designed to have a low predetermined Reverse Breakdown Voltage that takes advantage of this high reverse voltage.

Zener Diode in Forward Bias:

Works same as a normal PN diode i.e short circuit ideally and a low resistance practically.

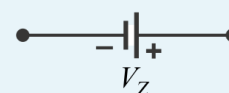
Zener Diode in Reverse Bias:

1. $V < V_Z$, works as a normal PN diode in reverse bias i.e open circuit ideally and a high resistance practically.
2. $V > V_Z$, works as a voltage regulator i.e becomes a source of constant voltage for the connected load.

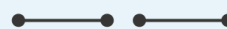
Ideal Zener Diode



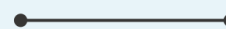
Zener Diode



Reverse Biased
(with Breakdown)

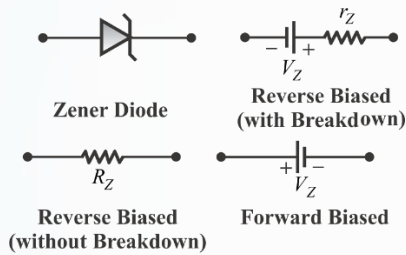


Reverse Biased
(without Breakdown)



Forward Biased

Practical Zener Diode



Operating Conditions for Zener diode to maintain break down characteristics:

1. Current through Zener diode, $I_{Z(\text{knee})} \leq I \leq I_{Z(\text{max})}$
2. The magnitude of open circuit reverse voltage across the Zener diode should be greater than or equal to V_Z .
 - I_Z (knee) is the minimum current required for the Zener diode to work as a voltage regulator
 - $I_Z(\text{max})$ is the maximum current the Zener diode can operate without damaging the device. It is specified by manufacturer.

Zener Diode as Voltage Regulator

Zener Diodes are used to produce a stabilised voltage output with low ripple under varying load current conditions.

Note: Zener diodes can also be connected together in series along with normal silicon diodes to produce a variety of different reference voltage output values

BJT BIASING AND REGION OF OPERATION

Operating Region of BJT

Normal Active Region

In this region, J_E operating in forward bias while J_C operating in reverse bias. It is used as amplifier.

Saturation Region

Both J_E and J_C are operating in forward bias and is used as switch (ON).

Cut-OFF Region

Both J_E and J_C are operating in reverse bias and is used as switch (OFF).

Reverse Active Region

In this region, J_E in reverse bias and J_C in forward bias. It is used as attenuator.

Some Standard values for *nnp* transistor.

Note : If nothing is mentioned then we will assume transistor is silicon type.

Different Methods used to Identify Operating Region of BJT

Method-I : Assume transistor in Saturation Region

1. $I_C \neq \beta I_B$
2. $V_{CE} = V_{CE(\text{sat})}$
3. $I_{B(\text{min})} = \frac{I_C(\text{sat})}{\beta_{dc}} \quad (\beta_{dc} = h_{fe})$
4. If $I_B \geq I_{B(\text{min})}$ then transistor will work in saturation region otherwise in active region.

Method-II : Assume transistor in active region.

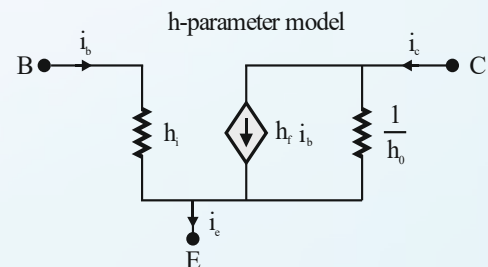
1. $I_C = \beta_{dc} I_B$
2. $I_E = I_C + I_B = (1 + \beta_{dc}) I_B$
3. For active region =

$$\begin{cases} V_{CB} > 0 & \text{for npn transistor} \\ V_{CB} < 0 & \text{for pnp transistor} \end{cases}$$

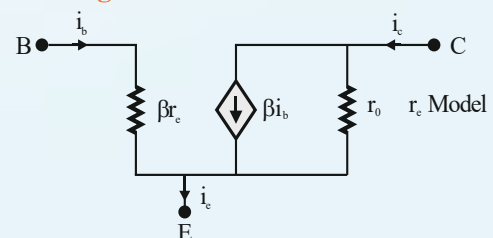
LOW FREQUENCY BJT AMPLIFIER

Amplifiers

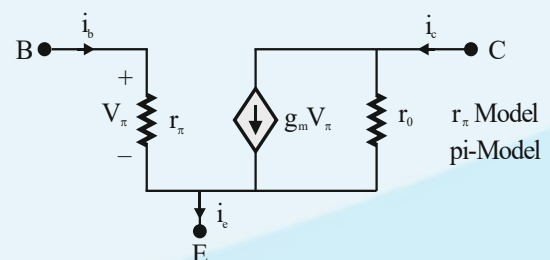
1. Approximated h-model



2. r_e modelling



3. r_π modelling



Relation between small signal modelling parameters

1. $h_i = \beta r_e = r_\pi$ 2. $h_f = \beta$
3. $\frac{1}{h_o} = r_o = V_A / I_{CQ} = \infty$ if no early effect

Note: Practically $r_o = 0$, and $1/h_o = \infty$, if not mentioned explicitly.

Procedure of AC analysis

Step 1: Do the mid frequency analysis

- (a) $C_{c1}, C_{c2}, C_E \rightarrow$ short circuit
 $C_T, C_D, C_{sh} \rightarrow$ Open circuit
- (b) all DC independent voltage source $\rightarrow$ Short circuit
 All independent current source = open circuit

Step 2: Replace BJT with small signal equivalent.

Internal Performance Parameter of an Amplifier

1. Current Gain (A_i) = $-I_2 / I_1$
2. Input Resistance (R_i) = V_1 / I_1
3. Voltage Gain (A_v) = V_2 / V_1
4. Output Resistance (R_o) = $1 / \text{Output Admittance } (1/y_o)$
 $= I_2 / V_2 \mid V_S = 0$
1. $R_{L1} \neq f(R_{o1}, R_{sa}) \quad R_{L1} = f(R_{i2})$
2. $R_{s2} \neq f(R_{L1}, R_{i2}) \quad R_{s2} = f(R_{o1})$
3. $R_i [\text{cascade}] = R_i [1^{\text{st}} \text{ stage}]$
4. o/p $R_o (\text{cascade}) = R_{oN} (\text{last stage})$
5. $A_v = A_{v1} \times A_{v2} \times A_{v3} \times \dots \times A_{vN}$
 (for proper impedance matching)
6. $A_i = A_{i1} \times A_{i2} \times A_{i3} \times \dots \times A_{iN}$
 (for proper impedance matching)

Loading Effect

1. The decrease in the gain of first stage due to low value of input impedance of second stage is called Loading effect
2. Loading effect occurs in BJT amplifiers, and not in JFET and MOSFET amplifiers because they have a very high input impedance.

Cascode Amplifier

- A combination of CE followed by CB is referred as Cascode amplifier
- CE acts as input stage and CB acts as output stage.
- Cascode amplifier can amplify both voltage and current
- Also known as direct coupled amplifier because output of CE configuration is directly connected to input of CB configuration.

Cascode Amplifier Parameters

1. Transconductance (g_m) = $A_I [\text{CB}] \times g_m [\text{CE}]$
2. Input Impedance (R_i) = $R_i [\text{CE}]$
3. Output Impedance (R_o) = $R_o [\text{CB}]$
4. Current Gain (A_I) = $A_I [\text{CB}] \times A_I [\text{CE}] > A_I [\text{CE}]$
5. Voltage Gain (A_V) = $A_V [\text{CB}] \times A_V [\text{CE}] > -A_V [\text{CB}]$

Darlington Pair

- Series combination of two CC configurations.
- Also referred as direct coupled amplifier

Darlington Amplifier Parameters

1. Transconductance (g_m) = $A_i [2^{\text{nd}} \text{ stage}] \times g_m [1^{\text{st}} \text{ stage}]$
2. Input Impedance (R_i) = $R_i [1^{\text{st}} \text{ stage}]$
3. Output Impedance (R_o) = $R_o [2^{\text{nd}} \text{ stage}]$
4. Current Gain (A_i) = $A_i [1^{\text{st}} \text{ stage}] \times A_i [2^{\text{nd}} \text{ stage}]$
5. Voltage Gain (A_v) = 1

MOSFET AMPLIFIER WITH BIASING

MOSFET

- **Source :** terminal through which majority charge carrier enters into the SC bar.
- **Drain :** terminal through which majority charge carriers leaves the SC bar.
- **Channel:** path between source & drain by which majority carrier travel from source to drain
- **Gate :** Terminal to control the flow of charge carrier from source to drain

Channel Current

- The current flowing from source to drain via the channel. It is a function of aspect ratio of MOSFET and applied voltages V_{GS} , and V_{DS} .

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} \left[(v_{GS} - v_{TH}) v_{DS} - \frac{v_{DS}^2}{2} \right]$$

Trans Conductance :

It is a figure of merit indicates that how well a transistor convert the voltage to the current.

$$g_m = \frac{\partial I_D}{\partial V_{GS}} \bigg|_{V_{DS} = \text{const.}}$$

In cut-off Region: MOSFET works as an OFF switch.

$$V_{GS} < V_T$$

$$I_D = 0$$

In Triode Region: MOSFET works as a resistor

$$V_{GS} > V_T$$

$$V_{DS} > (V_{GS} - V_T)$$

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} \left[(v_{GS} - v_{TH}) v_{DS} - \frac{v_{DS}^2}{2} \right]$$

$$\frac{\partial I_D}{\partial V_{GS}} \bigg|_{(V_{DS} = \text{const.})} = g_m = \mu_n C_{ox} \frac{W}{L} v_{DS}$$

In Saturation region :

MOSFET works as on ON switch

$$V_{GS} \gg V_T$$

$$v_{DS} = v_{GS} - v_{TH}$$

$$I_{D \text{ sat}} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} [v_{GS} - v_{TH}]^2$$

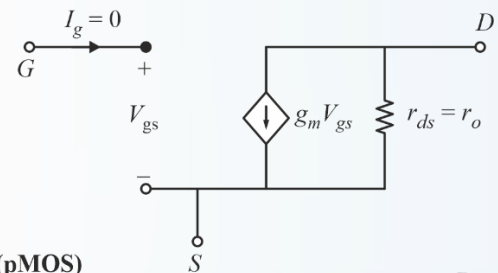
and $g_m = \mu_n C_{ox} \frac{W}{L} [v_{GS} - v_{TH}]$

$$g_m = \frac{2I_{D \text{ sat}}}{v_{GS} - v_{TH}}$$

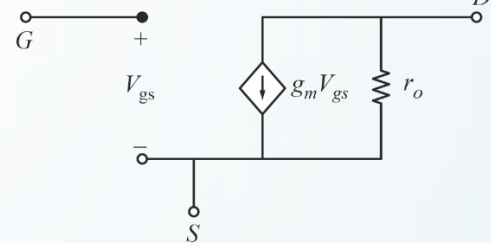
$$g_m = \sqrt{2\mu_n C_{ox} \left(\frac{W}{L} \right) I_D}$$

Small Signal (or) AC Analysis

(nMOS)



(pMOS)



If no channel length modulation ($\lambda = 0$).

$$V_A = \frac{1}{\lambda} = \infty ; r_{ds} = r_o = \frac{V_A}{I_D} = \infty$$

(a) RC Coupled amplifier with Coupling Capacitor Current Mirror

- It copies the ref current flowing at the input of the system to the output.
- It is also known as practical current source.
- It is designed by using IC technology.

MOSFET Amplifiers

Common Source MOSFET Amplifier without R_S and its AC Equivalent circuit

AC parameter	CS without R_S	CS with R_S	Common drain	Common Gate
Input resistance	$R_i' = \infty$ [$\because I_g = 0$]	$R_i' = \infty$ [$\because I_g = 0$]	$R_i' = \infty$	$R_i' \approx \frac{1}{g_m}$
Output resistance	$R_o' = r_o$ [$\lambda \neq 0$] $R_o' = \infty$ [$\lambda = 0$] [$r_o = \frac{1}{\lambda I_D}$]	$R_o' = r_o + (1 + \mu)R_S$ $R_o' = \infty$ [if $\lambda = 0 \Rightarrow r_o = \infty$]	$R_o' \approx \frac{1}{g_m}$	$R_o' = r_o + R_S(1 + \mu)$ $R_o' = \infty$ [if $\lambda = 0 \Rightarrow r_o = \infty$]
Voltage gain	$A_v' = -g_m R_L''$ ($R_L'' = r_o \parallel R_D \parallel R_L$)	$A_v' = \frac{-\mu R_L'}{R_L' + r_o + R_S(1 + \mu)}$ ($R_L' = R_D \parallel R_L$)	$A_v' = \frac{g_m R_L'}{1 + g_m R_L'}$ ($R_L' = R_S \parallel R_L$)	$A_v' = g_m R_L''$ ($R_L'' = r_o \parallel R_D \parallel R_L$)
Current gain	Does not exist [$I_g = 0$ A]	Does not exist [$I_g = 0$ A]	Does not exist [$I_g = 0$ A]	$A_I' = 1.0$

FEEDBACK AMPLIFIER

Introduction

Feedback : It is a process of taking sample from output and mix with input.

1. Overall Gain:

$$A_F = \frac{A}{1 + A\beta} \quad \text{gain reduced by } (1 + A\beta)$$

$$\text{if } A\beta \gg 1 \Rightarrow \boxed{A_f = \frac{1}{\beta}}$$

2. Bandwidth: Gain $\times$ B.W. = Constant

$$\Rightarrow (B.W.)_f = (BW)(1 + A\beta)$$

so, bandwidth increased by $(1 + A\beta)$

noise and distortion reduced by $(1 + A\beta)$

5. Gain Sensitivity:

$$S_g = \frac{\partial A_f / A_f}{\partial A / A} = \frac{A}{A_f} \times \frac{\partial A_f}{\partial A} = \frac{1}{1 + A\beta}$$

6. Desensitivity

$$D = \frac{1}{S_g} = (1 + A\beta)$$

(1) Series-Shunt Feedback: (Voltage Amplifiers)

Observations:

$$\text{Gain} = \frac{A_v}{1 + A_v\beta}, \quad \text{Bandwidth} = BW[1 + A_v\beta]$$

$$(R_{in})_f = R_i[1 + A_v\beta]$$

$$(R_o)_f = \frac{R_o}{1 + A_v\beta}$$

(2) Shunt-Shunt Feedback: (Trans-resistance Amplifiers)

Observations:

$$1. \quad \text{Gain } (R_m)_f = \frac{R_{ms}}{1 + R_{ms}\beta} \text{ decreased.}$$

$$2. \quad (R_i)_f = \frac{R_i}{1 + R_{ms}\beta} \text{ decreased}$$

$$3. \quad (R_o)_f = \frac{R_o}{1 + R_{ms}\beta} \text{ decreased.}$$

Observations:

$$1. \quad \frac{I_o}{V_s} = \frac{G_{ms}}{1 + G_{ms}\beta} \quad 2. (R_{in})_f = R_{in}(1 + G_m\beta)$$

$$3. \quad (R_o)_f = R_o(1 + G_m\beta)$$

4. Shunt-Series Feedback: (Current Amplifier)

Observations:

$$1. \quad \frac{I_o}{I_s} = \frac{A_I}{1 + A_I\beta} \quad 2. (R_{in})_f = \frac{R_{in}}{1 + A_I\beta}$$

$$3. \quad (R_o)_f = R_o[1 + A_I\beta]$$

OPERATIONAL AMPLIFIERS

Op-amp

A difference amplifier (amplifies the difference between two inputs)

$$\text{where, } V_d = V_1 - V_2$$

Op-amp Characteristics/Parameters

1. Open Loop Voltage Gain $\left(A = \frac{V_o}{V_i}\right)$: The internal gain of op-amp without any feedback.
Ideally = ∞ ; Practically = very high
2. Gain Bandwidth Product (GBW): Unity gain bandwidth product i.e. $GBW = 1$ refers to be behavior that the gain reduces at the same rate as frequency increases.
3. Input Resistance $\left(R_i = \frac{V_i}{I_i}\right)$: It is the internal input impedance of the op-amp.
Ideally = ∞ ; practically = very high
4. Output Resistance $\left(R_o = \frac{V_o}{I_o}\right)$: Ideally = 0; practically = very small
5. CMRR (Common Mode Rejection Ratio): Metric used to quantify the ability of the device to reject common-mode signals.
Ideally CMRR is infinite, Practically very high.

$$CMRR = \left| \frac{A_d}{A_c} \right|$$

$$CMRR \text{ in dB} = 20 \log \left| \frac{A_d}{A_c} \right| = (A_d)_{dB} - (A_c)_{dB}$$

6. Slew Rate: The maximum rate of change in output voltage per unit of time

$$SR = \left| \frac{dV_o}{dt} \right|_{\max} = \left| \frac{dV_o}{dV_i} \right|_{\max} \times \left| \frac{dV_i}{dt} \right|_{\max} = |A_{CL}| \times \left| \frac{dV_i}{dt} \right|_{\max}$$

Non-Idealities in Op-amp

- Offset voltage (V_{OS}): If the two transistors are not perfectly matched, an offset will show up as a non-zero DC offset at the output.
- Bias current (I_{bias}): The transistor inputs actually do draw some current. The bias current is defined to be the average of the currents of the two inputs.
- Offset current (I_{OS}): The difference between the input bias currents.

Schmitt Trigger

$$V_{UT} = \frac{V_{sat}R_2}{R_1 + R_2}, \quad V_{LT} = \frac{-V_{sat}R_2}{R_1 + R_2}$$

$$V_H = V_{UT} - V_{LT}$$

$$V_H = \frac{2V_{sat}R_2}{R_1 + R_2}$$

DIGITAL ELECTRONICS

Logic Gate & Minimization

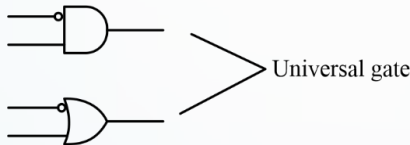
1. Logic Gate

Basic Gate – OR, AND, NOT

Arithmetic Gate – Ex-OR, EX-NOR

Universal Gate – NAND, NOR

Note :



2. Distribution Theorem:

$$A(B + C) = AB + AC$$

$$A + BC = (A + B)(A + C)$$

Consensus Theorem:

$$AB + \bar{B}C + AC = AB + \bar{B}C$$

$$(A + B)(\bar{B} + C)(A + C) = (A + B)(\bar{B} + C)$$

Transposition Theorem:

$$AB + \bar{B}C = (A + \bar{B})(B + C)$$

De-Morgan's Theorem

$$\overline{A \cdot B} = \bar{A} + \bar{B}, \overline{A + B} = \bar{A} \cdot \bar{B}$$

3. Number of Boolean Functions

Number of Boolean function (from n-variable) = 2^{2^n}

Number of self-dual function = $2^{2^{n-1}}$

Self-dual Function:

1. Neutral Function

Number of minterm = Number of Maxterm

2. Two mutually exclusive term should not present

0	1	2	3
7	6	5	4

Mutually exclusive term

4. XOR & XNOR

$$A \oplus B = \bar{A}B + A\bar{B} = (\bar{A} + \bar{B})(A + B)$$

$$A \odot B = \bar{A}\bar{B} + AB = (A + \bar{B})(\bar{A} + B)$$

$$\overline{A \oplus B} = \bar{A} \oplus B = A \oplus \bar{B} = A \odot B$$

$$A \oplus B \oplus AB = (A + B)$$

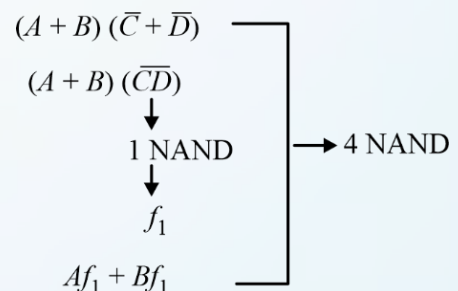
$$A \oplus B \oplus (A + B) = AB$$

5. Universal Gate Implementation

	NAND	NOR
NOT	1	1
AND	2	3
OR	3	2
XOR	4	5
XNOR	5	4

Note:

$A + B \rightarrow 3 \text{ NAND}$	$AB \rightarrow 3 \text{ NOR}$
$A + BC \rightarrow 3 \text{ NAND}$	$A(B + C) \rightarrow 3 \text{ NOR}$
$AD + BC \rightarrow 3 \text{ NAND}$	$(A + D)(B + C) \rightarrow 3 \text{ NOR}$



- SOP and POS expression of a Boolean function are not complement of each-other. In-fact it represents two different expression of same Boolean function.
- We can get more than one minimized expression from k-map.

PI: In k-map, the biggest group that we can make using each minterm is called prime implicant.

EPI: The biggest group, that contains at least one minterm that is uniquely grouped is called essential prime implicant.

Combinational Circuit

Half adder and Half subtractor.

$$\rightarrow S = D = A \oplus B$$

$$\rightarrow \text{Carry} = AB$$

$$\rightarrow \text{Borrow} = \bar{A}B(A - B) = \bar{A}\bar{B}(B - A)$$

$$\rightarrow 5 \text{ NAND/NOR (H.A. and H.S.)}$$

$$\rightarrow 3 (2 : 1 \text{ MUX}) (\text{H.A. and H.S.})$$

Full Adder and Full Subtractor

$$S = D = A \oplus B \oplus C_{in} = \Sigma m(1, 2, 4, 7)$$

$$\text{Carry} = AB + C_{in}(A \oplus B)$$

$$= AB + BC_{in} + AC_{in}$$

= Boolean function of 3-input majority gate

$$= \Sigma m(3, 5, 6, 7)$$

$$\text{Borrow} = \bar{A}B + C_{in}(\overline{A \oplus B})(A - B - C_{in})$$

$$= A\bar{B} + C_{in}(\overline{A \oplus B})(B - A - C_{in})$$

→ 9 NAND/NOR (F.A. and F.S.)

→ 7 (2 : 1 MUX) (F.A. and F.S.)

MUX

MUX contains AND gate followed by an OR-gate.

It is called “many to one circuit” or “universal logic circuit” or “data selector circuit”.

MUX contains 2^n input lines and 1 output line where ‘n’ is the number of select line.

Using one $2^n : 1$ MUX

→ All ‘n’ variable logic functions can be implemented.

→ Some ‘n + 1’ variable logic function can be implemented.

Using one $2^n : 1$ MUX and 1 NOT gate All ‘n’ and ‘n + 1’ variable logic functions can be implemented.

One (2 : 1 MUX) is required to implement NOT, AND, OR.

Two (2 : 1 MUX) are required to implement NAND, NOR, XOR, XNOR.

DEMUX

- One to many circuit or data distributor circuit.
- Number of select lines is $\log_2 n$ (where n is number of output line.)

Decoder

- It is used to convert binary to other codes.
- A “Decoder” with an enable input can function as a “Demultiplexer”.

Comparator

If A and B contain equal number of bits n.

$$\text{Number of minterm } f(A = B) = 2^n$$

Number of minterm $f(A > B) = f(A < B)$

$$= \frac{2^{2n} - 2^n}{2}$$

If A contains n_1 bit and B contains n_2 bit ($n_1 > n_2$).

Number of minterm $f(A = B) = 2^{n_2}$.

Number of minterm $f(B > A) = \Sigma(2^{n_2} - 1)$

Number of minterm

$$f(A > B) = 2^{n_1+n_2} - 2^{n_2} - \Sigma(2^{n_2} - 1)$$

Sequential Circuit

Flip-Flops (Stores “one bit of information”)

(1) S-R-FF

$$(i) Q_{n+1} = S + \bar{R}Q_n$$

(ii) Invalid state Present ($S = R = 1$)

(2) J-K-FF

$$(i) Q_{n+1} = J\bar{Q}_n + \bar{K}Q_n$$

(ii) Race around condition occur ($J = K = 1$) in level triggered to avoid race-around condition.

(a) Master-slave FF is used

(b) Edge triggered FF is used

(c) $t_{pw} < t_{pd}(FF) < T$

(3) D-FF

$$(i) Q_{n+1} = D$$

(4) T-FF

$$(i) Q_{n+1} = T \oplus Q_n$$

Registers

	Input(CLK)	Output(CLK)
SISO	n	n - 1
SIPO	n	0
PISO	1	n - 1
PIPO	1	0

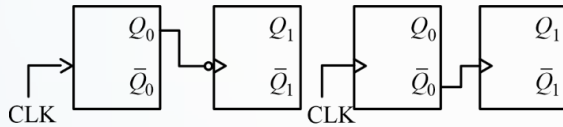
Counter

Asynchronous (ripple counter)

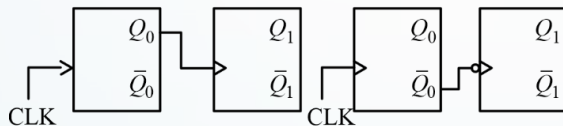
$$T_{CLK} \geq nt_{pd}(FF)$$

All FFs operate in toggle mode.

Clock is directly connected to LSB FF.



UP-Counter



Down Counter

Synchronous Counter

- (i) Ring Counter (non-self starting counter)
→ n -bit ring counter MOD Number = n
- (ii) Johnson Counter (Self starting counter)
→ n -bit Johnson Counter
MOD Number = $2n$

ADC-DAC

- Dual slope ADC
 - (i) Slowest ADC
 - (ii) Most Accurate therefore used in digital voltmeter
- Counter type
 - (i) Maximum Conversion time = $(2^n - 1)T_{CLK}$
- SAR type
 - (i) Maximum conversion time = $n T_{CLK}$.
 - (ii) Conversion time is independent of input analog voltage.

- Flash type ADC
 - (i) $2^n - 1$ comparator
 - (ii) $2n$ resistor, one $2^n : n$ priority encoder.
 - (iii) Fastest ADC
- Weighted Register type DAC ($n \sim$ bit)
 - (i) Resolution = $\frac{V_r}{2^{n-1}}$
 - (ii) LSB resistance = (2^{n-1}) MSB resistance
- $R - 2R$ ladder type DAC (n - bit)
 - (i) Resolution = $\frac{V_r}{2^n}$
 - (ii) $(n - 1) R$ - resistance
 - (iii) $(n + 1) 2R$ - resistance.

Number System

Range of 1's complement number

$$-(2^{n-1} - 1) \text{ to } +(2^{n-1} - 1)$$

$$\text{Total number} = 2^n - 1$$

Range of 2's complement number

$$-2^{n-1} \text{ to } +(2^{n-1} - 1)$$

$$\text{Total number} = 2^n$$

Logic Family

ECL → Fastest Logic

CMOS – Highest Fan out, least power dissipation

HTL → Highest Noise Margin

P^2L → Best FOM





**GATE
WALLAH**

UPSC ESE 2024 RESULTS

32 Students Secured Under AIR 100

**ROHIT
DHONDGE**



**RANK
1**

CIVIL ENGINEERING
IGP

**HARSHIT
PANDEY**



**RANK
2**

CIVIL ENGINEERING
IGP

**RAJESH
KASANIYA**



**RANK
2**

MECHANICAL ENGINEERING
IGP

**SATYAM
CHANDRAKANT
KHAIRNAR**



**RANK
2**

ELECTRICAL ENGINEERING
IGP

**SANCHIT
GOEL**



**RANK
6**

CIVIL ENGINEERING
IGP

**MAYANK
JAIMAN**



**RANK
8**

ELECTRICAL ENGINEERING
IGP

**ROHIT
KUMAR**



**RANK
8**

CIVIL ENGINEERING
IGP

**ANKIT
MEENA**



**RANK
9**

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**KETAN
KUMAR
SINHA**



**RANK
11**

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**BOLLA
USHNEESH
NANDAN**



**RANK
12**

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**DHAWAL
TAYAL**



**RANK
14**

CIVIL ENGINEERING
IGP

**MOHAMMAD
SHAQUIB**



**RANK
15**

CIVIL ENGINEERING
IGP

**ANKIT
ANAND**



**RANK
16**

CIVIL ENGINEERING
IGP

**MAHAVEER
SINGH**



**RANK
17**

ELECTRONIC & COMMUNICATION ENGINEERING
IGP

**GADDIPATI
YASWANTH
BABU**



**RANK
18**

CIVIL ENGINEERING
IGP

**AKASH
TANWAR**



**RANK
19**

CIVIL ENGINEERING
IGP

& More...

ELECTROMAGNETIC FIELD THEORY

Co-ordinate Systems

Cartesian Co-ordinate system (x,y,z):

- $\hat{a}_x \times \hat{a}_y = \hat{a}_z \quad \hat{a}_x \cdot \hat{a}_x = \hat{a}_y \cdot \hat{a}_y = \hat{a}_z \cdot \hat{a}_z = 1$
- $\hat{a}_y \times \hat{a}_z = \hat{a}_x \quad \hat{a}_x \cdot \hat{a}_y = \hat{a}_y \cdot \hat{a}_z = \hat{a}_z \cdot \hat{a}_x = 0$
- $\hat{a}_z \times \hat{a}_x = \hat{a}_y \quad \hat{a}_x \cdot \hat{a}_x = \hat{a}_y \cdot \hat{a}_y = \hat{a}_z \cdot \hat{a}_z = 0$

Spherical co-ordinate system (r, ϕ , θ)

$$d\vec{l} = \begin{cases} \pm dx \hat{a}_x \\ \pm dy \hat{a}_y \\ \pm dz \hat{a}_z \end{cases} \quad \text{To find elementary surface in } z\text{-}$$

direction, take cross product of $d\vec{l}$ in x & y direction and so on.

Elementary Surface

$$d\vec{s} = \begin{cases} \pm dy dz \hat{a}_x \\ \pm dx dz \hat{a}_y \\ \pm dx dy \hat{a}_z \end{cases}$$

Elementary volume:

$$dv = dx \cdot dy \cdot dz$$

Cylindrical co-ordination System (ρ, ϕ, z)

$$\text{Elementary length: } d\vec{l} = \begin{cases} \pm d\rho \hat{a}_\rho \\ \pm \rho d\phi \hat{a}_\phi \\ \pm dz \hat{a}_z \end{cases}$$

Elementary Surface : Elementary volume

$$d\vec{s} = \begin{cases} \pm \rho d\phi dz \hat{a}_\rho \\ \pm \rho d\rho dz \hat{a}_\phi \\ \pm \rho d\rho d\phi \hat{a}_z \end{cases} \quad dv = \rho d\rho d\phi dz$$

$$\text{General limits} \quad \begin{cases} 0 \leq \rho \leq \infty \\ 0 \leq \phi \leq 2\pi \\ -\infty \leq z \leq \infty \end{cases}$$

Elementary length

$$d\vec{l} = \begin{cases} \pm dr \hat{a}_r \\ \pm r d\theta \hat{a}_\theta \\ \pm r \sin\theta d\phi \hat{a}_\phi \end{cases}$$

Elementary Surface

$$d\vec{s} = \begin{cases} \pm r^2 \sin\theta d\theta d\phi \hat{a}_r \\ \pm r \sin\theta dr d\phi \hat{a}_\theta \\ \pm r dr d\theta \hat{a}_\phi \end{cases}$$

Elementary Volume

$$dv = r^2 \sin\theta dr d\theta d\phi$$

General limits

$$\begin{cases} 0 \leq r \leq \infty \\ 0 \leq \theta \leq \pi \\ 0 \leq \phi \leq 2\pi \end{cases}$$

Conversion of points from one to other

Cartesian to cylindrical

Cartesian to Spherical

$$\begin{cases} \rho = \sqrt{x^2 + y^2} \\ \phi = \tan^{-1}\left(\frac{y}{x}\right) \\ z = z \end{cases}$$

$$\begin{cases} r = \sqrt{x^2 + y^2 + z^2} \\ \theta = \cos^{-1}\left(\frac{z}{\sqrt{x^2 + y^2 + z^2}}\right) \text{ or } \tan^{-1}\left(\frac{\sqrt{x^2 + y^2}}{z}\right) \\ \phi = \tan^{-1}\left(\frac{y}{x}\right) \end{cases}$$

Cylindrical to cartesian

$$\begin{cases} x = \rho \cos\phi \\ y = \rho \sin\phi \\ z = z \end{cases}$$

Spherical to cartesian

$$\begin{cases} x = r \sin\theta \cos\phi \\ y = r \sin\theta \sin\phi \\ z = r \cos\theta \end{cases}$$

Cylindrical to Spherical

$$\begin{cases} r = \sqrt{\rho^2 + z^2} \\ \theta = \cos^{-1}\left(\frac{z}{\sqrt{\rho^2 + z^2}}\right) \\ \phi = \phi \end{cases}$$

Spherical to Cylindrical

$$\begin{cases} \rho = r \sin\theta \\ \phi = \phi \\ z = r \cos\theta \end{cases}$$

To find component of any vector in a particular direction, we just have to take dot product of that vector with the unit vector of desired direction. Any vector $\vec{A}$ is represented in different co-ordinate system as.

$$\vec{A} = A_x \hat{a}_x + A_y \hat{a}_y + A_z \hat{a}_z \text{ (Cartesian)}$$

$$\vec{A} = A_\rho \hat{a}_\rho + A_\phi \hat{a}_\phi + A_z \hat{a}_z \text{ (Cylindrical)}$$

$$\vec{A} = A_r \hat{a}_r + A_\theta \hat{a}_\theta + A_\phi \hat{a}_\phi \text{ (Spherical)}$$

Dot product tables of unit vectors to convert vector from one system to other

	$\hat{a}_\rho$	$\hat{a}_\phi$	$\hat{a}_z$
$\hat{a}_x \cdot$	$\cos \phi$	$-\sin \phi$	0
$\hat{a}_y \cdot$	$\sin \phi$	$\cos \phi$	0
$\hat{a}_z \cdot$	0	0	1

	$\hat{a}_r$	$\hat{a}_\theta$	$\hat{a}_\phi$
$\hat{a}_\rho \cdot$	$\sin \theta$	$\cos \theta$	0
$\hat{a}_\phi \cdot$	0	0	1
$\hat{a}_z \cdot$	$\cos \theta$	$-\sin \theta$	0

	$\hat{a}_r$	$\hat{a}_\theta$	$\hat{a}_\phi$
$\hat{a}_x \cdot$	$\sin \phi \cos \phi$	$\cos \theta \cos \phi$	$-\sin \phi$
$\hat{a}_y \cdot$	$\sin \theta \sin \phi$	$\cos \theta \sin \phi$	$\cos \phi$
$\hat{a}_z \cdot$	$\cos \theta$	$-\sin \theta$	0

Concept of general Curvilinear Co-ordinate System is helpful to remember different formulae in terms of u, v, w, h_1, h_2, h_3 and by taking appropriate values, we can convert formula to any co-ordinate system.

General Curvilinear Co-ordinate System (u, v, w)

$$\text{Differential length: } = \begin{cases} \pm h_1 du \hat{a}_u \\ \pm h_2 dv \hat{a}_v \\ \pm h_3 dw \hat{a}_w \end{cases}$$

$$\hat{a}_u \times \hat{a}_v = \hat{a}_w$$

$$\hat{a}_v \times \hat{a}_w = \hat{a}_u$$

$$\hat{a}_w \times \hat{a}_u = \hat{a}_v$$

$$h_1, h_2, h_3 = f(u, v, w)$$

Co-ordinate system	u	v	w	h_1	h_2	h_3
Cartesian	x	y	z	1	1	1
Cylindrical	ρ	ϕ	z	1	ρ	1
Spherical	r	θ	ϕ	1	r	$r \sin \theta$

Different operators

- Del operator

∇ ---- 3 dimensional differential vector operator.

$$\nabla = \hat{a}_u \frac{1}{h_1} \frac{\partial}{\partial u} + \hat{a}_v \frac{1}{h_2} \frac{\partial}{\partial v} + \hat{a}_w \frac{1}{h_3} \frac{\partial}{\partial w}$$

Unit of ∇ operator is $\frac{1}{m}$

- Gradient of a scalar

$$\nabla = \frac{1}{h_1} \frac{\partial V}{\partial u} \hat{a}_u + \frac{1}{h_2} \frac{\partial V}{\partial v} \hat{a}_v + \frac{1}{h_3} \frac{\partial V}{\partial w} \hat{a}_w$$

Directional derivative of v in the

direction of $\vec{A} = (\nabla v) \cdot \hat{a}_A$

- Divergence of a vector

$$\nabla \cdot \vec{A} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial u} (h_2 h_3 A_u) + \frac{\partial}{\partial v} (h_3 h_1 A_v) + \frac{\partial}{\partial w} (h_1 h_2 A_w) \right]$$

Where $\vec{A} = A_u \hat{a}_u + A_v \hat{a}_v + A_w \hat{a}_w$

- Curl of vector

$$\nabla \times \vec{A} = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1 \hat{a}_u & h_2 \hat{a}_v & h_3 \hat{a}_w \\ \frac{\partial}{\partial u} & \frac{\partial}{\partial v} & \frac{\partial}{\partial w} \\ h_1 A_u & h_2 A_v & h_3 A_w \end{vmatrix}$$

- Laplacian of scalar

$$\nabla^2 V = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial u} \left(\frac{h_2 h_3}{h_1} \frac{\partial V}{\partial u} \right) + \frac{\partial}{\partial v} \left(\frac{h_3 h_1}{h_2} \frac{\partial V}{\partial v} \right) + \frac{\partial}{\partial w} \left(\frac{h_1 h_2}{h_3} \frac{\partial V}{\partial w} \right) \right]$$

Laplacian of a vector

$$= (\nabla^2 A_x) \hat{a}_x + (\nabla^2 A_y) \hat{a}_y + (\nabla^2 A_z) \hat{a}_z$$

- Stokes' theorem

$$\oint \vec{A} \cdot d\vec{l} = \iint_S (\nabla \times \vec{A}) \cdot d\vec{s}$$

- Gauss's divergence theorem

$$\oint \vec{A} \cdot d\vec{s} = \iiint_V (\nabla \cdot \vec{A}) \cdot dv$$

- Poisson's equation

$$\nabla^2 V = -\rho / \epsilon$$

Laplace's equation

$$\nabla^2 V = 0$$

- Transmission line.

Primary constants of the line

R..... Resistance per unit length (Ω/m)

L..... Inductance per unit length (H/m)

C..... Capacitance per unit length (F/m)

G..... Conductance per unit length (S/m)

Z_0 = Characteristic impedance {Secondary constants of the line}

$$\gamma = \alpha + j\beta = \infty \text{ constant}$$

{Secondary constants of the line}

$$V = V^+ e^{-\gamma z} + V^- e^{+\gamma z}$$

$$I = \frac{1}{Z_0} [V^+ e^{-\gamma z} - V^- e^{+\gamma z}]$$

$$V = \sqrt{(R + j\omega L)(G + j\omega C)}$$

$$Z_0 = \sqrt{\frac{(R + j\omega L)}{(G + j\omega C)}}$$

$$\beta = \frac{\omega}{v_p}$$

$$v_p = \frac{\omega}{\beta} = \frac{1}{\sqrt{LC}}$$

$$= \frac{c}{\sqrt{\epsilon_r}} = \frac{1}{\sqrt{\mu\epsilon}}$$

For loss-less line

$$R = 0 \quad G = 0 \quad \alpha = 0$$

$$\beta = \omega\sqrt{LC} \quad \gamma = j\beta$$

$$v_p = \frac{1}{\sqrt{LC}} \quad Z_0 = \sqrt{\frac{L}{C}}$$

For terminated transmission line

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{\bar{Z}_L - 1}{\bar{Z}_L + 1} \quad \Gamma = \rho e^{j\theta}$$

$$T = 1 + \Gamma$$

$$T = \frac{2Z_L}{Z_L + Z_0} = \frac{2\bar{Z}_L}{\bar{Z}_L + 1}$$

$$Z_L = Z_0 \left(\frac{1 + \Gamma}{1 - \Gamma} \right)$$

VSWR

$$S = \frac{1 + \rho}{1 - \rho}$$

$$0 \leq \rho \leq 1$$

$$1 \leq S \leq \infty$$

$$Z_{in} = Z_0 \frac{Z_L + jZ_0 \tan \beta l}{Z_0 + jZ_L \tan \beta l} \quad \text{for lossless line}$$

For Short Circuited line

$$Z_{SC} = +j Z_0 \tan \beta l$$

$$\Gamma = -1, \rho = 1, \theta = 180^\circ, S = \infty$$

For Open Circuited line

$$Z_L = \infty$$

$$Z_{OC} = -j Z_0 \cot \beta l$$

$$\Gamma = 1, \rho = 1, \theta = 0^\circ, S = \infty$$

For perfectly Matched line

$$Z_L = Z_0$$

$$Z_{in} = Z_0 = Z_L$$

$$\Gamma = 0, \rho = 0, \theta = 0^\circ (\text{min}), S = 1$$

For purely reactive load ($Z_L = jX$)

$$Z_{in} = j \left[Z_0 \frac{X + Z_0 \tan \beta l}{Z_0 - X \tan \beta l} \right]$$

$$\Gamma = \frac{jX - Z_0}{jX + Z_0}, \rho = 1, \theta = -2 \tan^{-1} \left(\frac{X}{Z_0} \right), S = \infty$$

For length of line $l = \lambda$ or $l = \lambda/2$

$$Z_{in} = Z_L$$

For Quarter Wave Transformer

$$l = \lambda/4 \quad Z_{in} = \frac{Z_0^2}{Z_L}$$

$$Z_0 = \sqrt{Z_{in} Z_L}$$

To find $Z_{\max}$ and $Z_{\min}$ along line

$$Z_{\max} = Z_0 \cdot S \quad Z_{\min} = \frac{Z_0}{S}$$

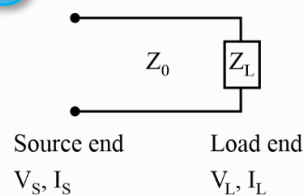
$$\bar{Z}_{\max} = S \quad \bar{Z}_{\min} = \frac{1}{S}$$

$$\frac{1}{S} \leq \bar{Z} \leq S$$

- $Z_0 = \sqrt{Z_{SC} \cdot Z_{OC}}$ for any length of line

Transmission Matrices

$$\begin{bmatrix} V_S \\ I_S \end{bmatrix} \leftrightarrow \begin{bmatrix} V_L \\ I_L \end{bmatrix}$$



For lossless line

$$\begin{bmatrix} V_S \\ I_S \end{bmatrix} = \begin{bmatrix} \cos \beta l & jZ_0 \sin \beta l \\ (j/Z_0) \sin \beta l & \cos \beta l \end{bmatrix} \begin{bmatrix} V_L \\ I_L \end{bmatrix}$$

$$\begin{bmatrix} V_L \\ I_L \end{bmatrix} = \begin{bmatrix} \cos \beta l & -jZ_0 \sin \beta l \\ -(j/Z_0) \sin \beta l & \cos \beta l \end{bmatrix} \begin{bmatrix} V_S \\ I_S \end{bmatrix}$$

Condition for distortion less line

$$RC = LG$$

$$\alpha \neq f(\Omega)$$

$$\beta = k \cdot \Omega$$

- Input impedance of stub is purely reactive
- Distance of stub: $d = \frac{\lambda}{2\pi} \tan^{-1} \sqrt{\frac{Z_L}{Z_0}}$
- Susceptance, that should be added by the stub

$$b_0 = (Y_0 - Y_L) \sqrt{\frac{Y_0}{Y_L}}$$

- Length of stab $l = \frac{\lambda}{2\pi} \tan^{-1} \left(\frac{Z_L Z_0}{Z_L - Z_0} \right)$

For coaxial transmission line

- $C = \frac{2\pi\epsilon}{\ln(b/a)}$ F/m
- $L = \frac{\mu}{2\pi} \ln(b/a)$ H/m
- $Z_0 = \sqrt{\frac{L}{C}}$ For lossless line
- $Z_0 = \frac{1}{2\pi} \sqrt{\frac{\mu}{\epsilon}} \ln(b/a) \sqrt{\frac{\mu}{\epsilon}} = \eta$

For free space,

- $\eta = \eta_0 = \sqrt{\frac{\mu_0}{\epsilon_0}} = 120\pi = 377\Omega$
- $Z_0 = 60 \ln(b/a)\Omega$

continuity equation of current

$$\nabla \cdot \vec{J} = -\frac{\partial \rho}{\partial t}$$

relaxation time,

$$T_r = \frac{\epsilon}{\sigma}$$

MAXWELL'S EQUATION IN GENETAL TIME RANGING FIELDS

Point/differential form	Integral form
1. $\nabla \cdot \vec{D} = \rho$	$\oiint \vec{D} \cdot d\vec{s} = \iiint \rho \cdot dv$
2. $\nabla \cdot \vec{B} = 0$	$\oiint \vec{B} \cdot d\vec{s} = 0$
3. $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$	$\oint \vec{E} \cdot d\vec{l} = -\iint \frac{\partial \vec{B}}{\partial t} \cdot d\vec{s}$
4. $\nabla \times \vec{H} = \sigma \vec{E} + \frac{\partial \vec{D}}{\partial t}$	$\oint \vec{H} \cdot d\vec{l} = \iint \left(\sigma \vec{E} + \frac{\partial \vec{D}}{\partial t} \right) \cdot d\vec{s}$

$\sigma \vec{E} = \vec{J}_c$ conduction current density

$$\frac{\partial \vec{D}}{\partial t} = \vec{J}_d = \text{Displacement current density.}$$

Case-I For good conductor,

$$\sigma \text{ is high, } |\vec{J}_c| \gg |\vec{J}_d| \Rightarrow \text{neglect } \frac{\partial \vec{D}}{\partial t}$$

Case-II, For good dielectric (free space)

$$\rho = 0, \sigma \approx 0 \Rightarrow \text{Neglect } \sigma \vec{E}$$

Case-III For static fields

$$\frac{\partial \vec{B}}{\partial t} = 0, \frac{\partial \vec{D}}{\partial t} = 0$$

Case-IV for sinusoidal varying fields

$$\frac{d}{dt} \approx j\omega; \frac{d^2}{dt^2} \approx (j\omega)^2$$

$$\nabla \cdot \vec{D} = \rho$$

$$\nabla \cdot \vec{B} = 0$$

$$\nabla \times \vec{E} = \frac{-\partial \vec{B}}{\partial t} = -j\omega \mu \vec{H}$$

$$\nabla \times \vec{H} = \sigma \vec{E} + \frac{\partial \vec{D}}{\partial t} = (\sigma + j\omega\epsilon) \vec{E}$$

Loss tangent

$$\tan \delta = \frac{|\vec{J}_c|}{|\vec{J}_d|} = \frac{|\sigma \vec{E}|}{|j\omega\epsilon \vec{E}|} = \frac{\sigma}{\omega\epsilon}$$

If $\frac{\sigma}{\omega\epsilon} \gg 1 \Rightarrow$ Medium behaves as good conductor

$\frac{\sigma}{\omega\epsilon} \ll 1 \Rightarrow$ Medium behaves as good dielectric.

Wave equations in $\vec{E}$ & $\vec{H}$:-

$$\nabla^2 \vec{E} = \mu\sigma \frac{\partial \vec{E}}{\partial t} + \mu\epsilon \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\nabla^2 \vec{H} = \mu\sigma \frac{\partial \vec{H}}{\partial t} + \mu\epsilon \frac{\partial^2 \vec{H}}{\partial t^2}$$

For lossless or perfectly dielectric medium ($\sigma = 0$)

$$\nabla^2 \vec{E} = \mu\epsilon \frac{\partial^2 \vec{E}}{\partial t^2} = \frac{1}{v_p^2} \frac{\partial^2 \vec{E}}{\partial t^2}$$

For free space, $v_p = c$.

$$\nabla^2 \vec{E} = \frac{1}{c^2} \frac{\partial^2 \vec{E}}{\partial t^2}$$

For harmonically varying fields

$$\nabla^2 \vec{E} = \mu\sigma j\omega \vec{E} + \mu\epsilon (j\omega)^2 \vec{E}$$

$$= (j\omega\mu\sigma - \omega^2\mu\epsilon) \vec{E}$$

$$\boxed{\nabla^2 \vec{E} = \gamma^2 \vec{E}} \text{ where } \gamma = \sqrt{j\omega\mu\sigma - \omega^2\mu\epsilon}$$

$$= \alpha + j\beta$$

For harmonically varying fields in lossless medium:-

$$\boxed{\nabla^2 \vec{E} = -\beta^2 \vec{E}}$$

Poynting vector

$$\vec{P} = \vec{E} \times \vec{H} \Rightarrow \text{watt / m}^2 \Rightarrow \text{power per unit area}$$

$$|\vec{P}| = EH \sin \theta = P$$

$$\text{Power} = \oint_S \vec{P} \cdot d\vec{s}$$

$\frac{E}{H} = \text{constant} = \eta = \text{intrinsic impedance of medium.}$

$$(1) \eta = \frac{E}{H} = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\epsilon}} \text{ for general medium.}$$

$$(2) \eta = \sqrt{\frac{\mu}{\epsilon}} \text{ for lossless medium } \sigma = 0.$$

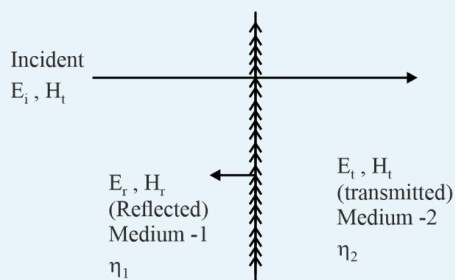
$$(3) \eta = \sqrt{\frac{\mu_0}{\epsilon_0}} \text{ for free space } \eta = \eta_0 = 120\pi\Omega$$

$$\boxed{P = EH}$$

$$= \eta H^2$$

$$= E^2 / \eta$$

Reflection & refraction of EM waves:



Normal Incidence

$$\eta_1 = \sqrt{\frac{j\omega\mu_1}{\sigma_1 + j\omega\epsilon_1}} \quad \eta_2 = \sqrt{\frac{j\omega\mu_2}{\sigma_2 + j\omega\epsilon_2}}$$

$$\frac{E_r}{E_i} = \rho = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{\sqrt{\epsilon_1} - \sqrt{\epsilon_2}}{\sqrt{\epsilon_1} + \sqrt{\epsilon_2}}$$

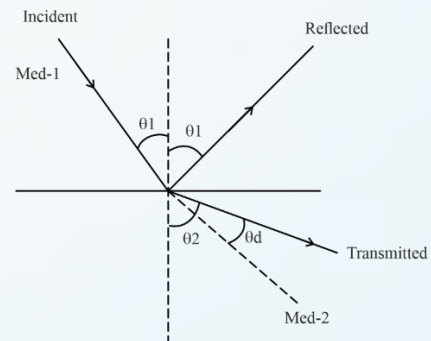
$$\frac{E_t}{E_i} = T = \frac{2\eta_2}{\eta_2 + \eta_1} = \frac{2\sqrt{\epsilon_1}}{\sqrt{\epsilon_1} + \sqrt{\epsilon_2}}$$

$$\frac{H_r}{H_i} = \rho' = \frac{\eta_1 - \eta_2}{\eta_1 + \eta_2} = \frac{\sqrt{\epsilon_2} - \sqrt{\epsilon_1}}{\sqrt{\epsilon_2} + \sqrt{\epsilon_1}}$$

$$\frac{H_t}{H_i} = T' = \frac{2\eta_1}{\eta_1 + \eta_2} = \frac{2\sqrt{\epsilon_2}}{\sqrt{\epsilon_2} + \sqrt{\epsilon_1}}$$

for perfect dielectric

Oblique incidence



Angle of deviation $\theta_d = \theta_2 - \theta_1$

Snell's law

$$\frac{\sin \theta_2}{\sin \theta_1} = \frac{v_2}{v_1} \quad v = \frac{1}{\sqrt{\mu\epsilon}}$$

$\mu = \mu_0$ for perfect dielectric

$$\boxed{\frac{\sin \theta_2}{\sin \theta_1} = \frac{\sqrt{\epsilon_1}}{\sqrt{\epsilon_2}}}$$

$\theta_1 = \theta_c$, $\theta_2 = 90^\circ$, $\theta_1 > \theta_c \Rightarrow \text{TIR.}$

$$\frac{1}{\sin \theta_c} = \sqrt{\frac{\epsilon_1}{\epsilon_2}} \quad \boxed{\sin \theta_c = \sqrt{\frac{\epsilon_2}{\epsilon_1}}}$$

$\epsilon_2 < \epsilon_1$ For total internal reflection

$$\boxed{\frac{\sin \theta_2}{\sin \theta_1} = \frac{\eta_1}{\eta_2}}$$

Brewster's angle

$$\tan \theta_1 = \tan \theta_B = \sqrt{\frac{\epsilon_2}{\epsilon_1}}$$

Propagation of EM wave in good conductor

$$\frac{\sigma}{\omega\epsilon} \gg 1$$

$$\alpha = \sqrt{\frac{\omega\mu\sigma}{2}} \quad \beta = \sqrt{\frac{\omega\mu\sigma}{2}}$$

$$\delta = \frac{1}{\alpha} = \frac{1}{\sqrt{\pi f \mu \sigma}} \dots \dots \text{Depth of penetration}$$

$$V_p = \frac{\omega}{\beta} \quad \lambda = \frac{2\pi}{\beta}$$

$$\eta = \sqrt{\frac{\omega\mu}{2\sigma}} (1+j)$$

Propagation of EM wave in good dielectric

$$\frac{\sigma}{\omega\epsilon} \ll 1$$

$$\alpha = \frac{\sigma}{2} \sqrt{\frac{\mu}{\epsilon}}$$

$$\beta = \beta_0 \left(1 + \frac{\sigma^2}{8\omega^2 \epsilon^2} \right); \quad \beta_0 = \omega \sqrt{\mu\epsilon} = \frac{\omega}{V_0}$$

$$V_p = \frac{\omega}{\beta} \quad \lambda = \frac{2\pi}{\beta}$$

$$\delta = \frac{1}{\alpha} \quad \eta = \eta_0 \left(1 + \frac{j\sigma}{2\omega\epsilon} \right); \quad \eta_0 = \sqrt{\frac{\mu}{\epsilon}}$$

Wave polarization

$$E_x = E_1 \sin(\omega t - \beta z)$$

$$E_y = E_2 \sin(\omega t - \beta z + \delta)$$

$$\vec{E} = E_x \hat{a}_x + E_y \hat{a}_y$$

$$\text{If } z = 0, E_x = E_1 \sin \omega t, E_y = E_2 \sin(\omega t + \delta)$$

$$\delta = \text{Time phase angle between } E_x \text{ \& } E_y.$$

(i) Linear polarization

$$\delta = 0^\circ \text{ or } 180^\circ, \quad \frac{E_y}{E_x} = \frac{E_2}{E_1} = m$$

$$E_y = m \cdot E_x$$

Case-I: $E_1 = 0 \Rightarrow m = \infty \Rightarrow$ Wave is vertically polarized.

Case-II: $E_2 = 0 \Rightarrow m = 0 \Rightarrow$ Wave is horizontally polarized.

Case-III: $E_1 = E_2 \Rightarrow m = 1 \Rightarrow$ Wave is linearly polarized at 45°

(2) Circular polarization

$$\boxed{E_1 = E_2 = E_0}$$

$$\delta = \pm 90^\circ$$

Let $\delta = +90^\circ \Rightarrow E_x^2 + E_y^2 = E_0^2 \Rightarrow$ equation of circle.

If $\delta = +90^\circ \Rightarrow$ Left circularly polarized (LCP)

$\delta = -90^\circ \Rightarrow$ Right circularly polarized (RCP)

(3) Elliptical polarization

$$E_1 \neq E_2, \quad \delta = \pm 90^\circ$$

$$\left(\frac{E_x}{E_1} \right)^2 + \left(\frac{E_y}{E_2} \right)^2 = 1 \Rightarrow \text{equation of ellipse}$$

$\delta = +90^\circ \Rightarrow$ LEP

$\delta = -90^\circ \Rightarrow$ REP

Important points

- RCP waves incidents on perfect conductor.
Result:- Reflected wave is LCP wave.
- LCP wave incidents on perfect conductor.
Result:- Reflected wave is RCP wave.
- RCP wave incidents on polystyrene.
Result:- Reflected wave is LEP transmitted wave is REP
- Elliptically polarized or circularly polarized wave incidents at Brewster's angle.
Results:- Reflected and transmitted waves are linearly polarized.

Rectangular waveguide

Dominant mode $\rightarrow TE_{10}, m = 1, n = 0$

$$f_c = \frac{V_0}{2\pi} \sqrt{\left(\frac{m\pi}{a} \right)^2 + \left(\frac{n\pi}{b} \right)^2}$$

$$\beta_g = \beta_0 \sqrt{1 - \left(\frac{f_c}{f} \right)^2}$$

$$V_g = \frac{V_0}{\sqrt{1 - \left(\frac{f_c}{f} \right)^2}} \quad (\text{Phase velocity})$$

$$V_{g_r} = V_0 \sqrt{1 - \left(\frac{f_c}{f} \right)^2} \quad (\text{Group velocity})$$

$$V_g \cdot V_{g_r} = V_0^2$$

Rectangular waveguide

$$\lambda_g = \frac{2\pi}{\beta_g} = \frac{\lambda_0}{1 - \left(\frac{f_c}{f}\right)^2} = \frac{\lambda_0}{\sqrt{1 - \left(\frac{\lambda_0}{\lambda_c}\right)^2}}$$

$$\lambda_0 = \frac{\lambda_g \cdot \lambda_c}{\sqrt{\lambda_g^2 + \lambda_c^2}}$$

$$Z_{TE} = \frac{\eta_0}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}}$$

$$Z_{TM} = \eta_0 \sqrt{1 - \left(\frac{f_c}{f}\right)^2}$$

$$Z_{TE} \cdot Z_{TM} = \eta_0^2$$

$$V_0 = \frac{1}{\sqrt{\mu\epsilon}}$$

$$\beta_0 = \frac{\omega}{V_0} = \omega\sqrt{\mu\epsilon}$$

$$\lambda_0 = \frac{V_0}{f} = \frac{2\pi}{\beta_0}$$

$$\eta_0 = \sqrt{\frac{\mu}{\epsilon}}$$

Dominant mode

$$TE_{mm}, \quad TM_{mm}$$

$$m = 1$$

$$n = 0 \quad \boxed{TE_{10}}$$

$$f_c = \frac{V_0}{2a} = \frac{V_0}{\lambda_c}$$

$$\lambda_c = 2a$$

$$\lambda_e = \frac{2a}{m} \text{ [For other than dominant mode]}$$

Circular waveguide : TE_{nm}

Dominant: TE_{11} , $n = 1$, $m = 1$

$$f_e = \frac{1.841V_0}{2\pi a}$$

$$TM_{nm} \Rightarrow \text{Dominant} \Rightarrow TM_{01}$$

$$n = 0, m = 1$$

$$f_e = \frac{2.405V_0}{2\pi a}$$

TE_{11} is overall dominant mode

Cavity resonators (TE_{mnp} , TM_{mnp})

$$f_0 = \frac{V_0}{2\pi} \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 + \left(\frac{p\pi}{d}\right)^2} \text{ Frequency of oscillation}$$

Dominant mode TE_{101}

Cylindrical Cavity Resonator (TE_{nmp} , TM_{nmp}):

Overall dominant mode: TE_{111}

$$f_0 = \frac{V_0}{2\pi} \sqrt{\left(\frac{1.841}{a}\right)^2 + \left(\frac{p\pi}{d}\right)^2} \text{ --- } TE_{111}$$

$$f_0 = \frac{V_0}{2\pi} \sqrt{\left(\frac{2.405}{a}\right)^2 + \left(\frac{p\pi}{d}\right)^2} \text{ --- } TM_{011}$$

Antenna	Radiation resistance	Installation
Short monopole	$R_{rad} = 40\pi^2 \left(\frac{l}{\lambda}\right)^2$	Vertical
Short dipole	$R_{rad} = 80\pi^2 \left(\frac{l}{\lambda}\right)^2$	Horizontal
Quarter wave monopole	$R_{rad} = 36.5 \Omega$	Vertical
Half wave dipole	$R_{rad} = 73 \Omega$	Horizontal

Antenna array

α = Progressive phase shift of feed

ψ = Total phase shift between E fields

$$\psi = \beta d \cos \phi + \alpha$$

$$E_n = \frac{1}{n} = \frac{\sin\left(\frac{n\psi}{2}\right)}{\sin\left(\frac{\psi}{2}\right)}$$

Normalized E -field due to array of n elements.

Broadside array:-

$$\psi = 0^\circ \text{ at } \phi = 90^\circ$$

$$E_{\max} \text{ occurs at } \phi = 90^\circ$$

$$\psi = \beta d \cos \phi + \alpha, \alpha = 0, \psi = \beta d \cos \phi$$

Endfire array:

$$\psi = 0^\circ \text{ at } \phi = 0^\circ$$

$$E_{\max} \text{ occurs at } \phi = 0^\circ$$

$$\alpha = -\beta d$$

$$\psi = \beta d (\cos \phi - 1)$$

Parabolic reflector:-

$$\text{BEFN} \approx \frac{140^\circ \cdot \lambda}{D^1}$$

D = Diameter

$$\text{HPBW} \approx \frac{58^\circ \cdot \lambda}{D^1}$$

$$D = 9.81 \left(\frac{D^-}{\lambda}\right)^2$$

Directivity

$$G_p \approx 6 \left(\frac{D^-}{\lambda} \right)^2 \quad \text{Power gain}$$

$$G_p = \eta \pi^2 \left(\frac{D^-}{\lambda} \right)^2 \quad \eta = \text{efficiency}$$

$$D_{dB} = 10 \log_{10} D$$

$$D = G_{P(\max)}$$

$$D = \left(\frac{4\pi}{\lambda^2} \right) A_e \quad A_e = \frac{\omega}{P}$$

ω = Total power

P = Power density

- For concentric spherical shell carrying equal and opposite charges $\pm Q$:-

$$E = \frac{Q}{4\pi\epsilon r^2} \hat{a}_r, \quad a < r < b$$

$$V_{ab} = \frac{Q(b-a)}{4\pi\epsilon ab}$$

$$C = \frac{4\pi\epsilon ab}{b-a}$$

- Electrical energy density

$$w_e = \frac{1}{2} \vec{D} \cdot \vec{E} = \frac{1}{2} \epsilon E^2$$

- For long coaxial cable carrying equal and opposite charge densities $\pm \rho_L$:-

$$\vec{E} = \frac{\rho_L}{2\pi\epsilon r} \hat{a}_r, \quad a < r < b$$

$$w_e = \frac{\rho_L^2}{8\pi^2 \epsilon r^2} \quad a < r < b$$

$$C = \frac{2\pi\epsilon}{\ln(b/a)} \quad \text{F/m}$$

- Magnetic dipole moment

$$\vec{m}_m = IA \hat{a}_n$$

- Magnetic field intensity due to long solid cylindrical conductor carrying current I

$$H = \begin{cases} \frac{Ir}{2\pi a^2} \hat{a}_\phi & r < a \\ \frac{I}{2\pi r} \hat{a}_\phi & r > a \\ \frac{I}{2\pi a} \hat{a}_\phi & r = a \end{cases}$$

- Magnetic energy density

$$w_m = \frac{1}{2} \vec{B} \cdot \vec{H} = \frac{1}{2} \text{mH}^2$$

- For long coaxial cable with inner & outer radius a & b carrying equal and opposite currents

$$H = \begin{cases} 0, & r < a \\ \frac{I}{2\pi r} \hat{a}_\phi, & a < r < b \\ 0 & r > b \\ \frac{I}{2\pi a} \hat{a}_\phi, & r = a \\ \frac{I}{2\pi b} \hat{a}_\phi & r = b \end{cases}$$

- Vector magnetic potential vector ($\vec{A}$)

$$\vec{B} = \nabla \times \vec{A}$$





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ELECTRICAL MACHINE

TRANSFORMERS

- Induced EMF $E = 4.44\pi f \phi N_{ph}$
- Equivalent impedance referred to primary is $Z_{eq} = (R_1 + jX_1) + a^2(R_2 + jX_2)$, Where, $a = \frac{N_1}{N_2}$.
- Induced EMT $E = 4.44 \times T N_{ph}$ Equivalent impedance referred to primary is $Z_{eq} = (R_1 + jX_1) + a^2(R_2 + jX_2)$
- Where, $a = \frac{N_1}{N_2}$
- Core loss component of current is
- $I_C = I_0 \cos \phi$
- Magnetizing component of current is $I_m = I_0 \sin \phi$
- Hysteresis loss $P_h = K_h f B_m^{1.6} \text{ W/m}^3$
- Eddy current loss $P_e = K_e f^2 B_m^2 t^2 \text{ W/m}^3$
- In open circuit test, $R_c = \frac{(V_{oc})^2}{P_{oc}}, X_m = \frac{V_{oc}}{I_m}$
- $Z_{p.u.(new)} = Z_{p.u.(old)} \times \left[\frac{M.V.A_b(\text{new})}{M.V.A_b(\text{old})} \right] \times \left[\frac{K \cdot V_b(\text{old})}{K \cdot V_b(\text{new})} \right]^2$
= p.u. voltage drop across resistance
- $X_{p.u} = \text{p.u. voltage drop across reactance} = \frac{X_{eq}^n}{Z_{Base}}$
- Voltage Régulation $= R_{pu} \cos \phi_2 + X_{p.u.} \sin \phi_2$
- $\cos \phi_2 = \frac{X_{p.u.}}{Z_{p.u.}}$ leading for zero voltage regulation
- $\cos \phi_2 = \frac{R_{p.u.}}{Z_{p.u.}}$ lagging for maximum voltage regulation
- $\eta = \left[\frac{x \times S \times \cos \phi_2}{x \times S \times \cos \phi_2 + Pi + x^2 P_{cu(f.I)}} \right],$

x = Loading factor,

S = rated K.V.A.

- For maximum efficiency,

$$S_{\eta_{max}} = S_{fl} \sqrt{\frac{P_i}{P_{cu(fl)}}}$$

$$\eta_{max} = \frac{S_{\eta_{max}} \cos \phi_2}{S_{\eta_{max}} \cos \phi_2 + 2P_i}$$

$$\eta_{All\ day} = \frac{\text{Energy o/p in kWh in 24 hrs}}{\text{Energy i/P in kWh in 24 hrs.}}$$

- In auto transformer,

$$\frac{(Cu\ weight)_{autot/f}}{(Cu\ weight)_{2\ winding}} = \left[\frac{a_{auto} - 1}{a_{auto}} \right]$$

- Inductive power transfer $= \left[1 - \frac{1}{a_{auto}} \right] \times S_{auto}$

- Conductive power transfer $= \left[\frac{1}{a_{auto}} \right] \times S_{auto}$

$$S_{auto} = \left[\frac{a_{auto}}{a_{auto} - 1} \right] S_{2-winding}$$

- When two transformers operate in parallel then,

$$\vec{I}_A = \vec{I}_L \times \left[\frac{\vec{Z}_B}{\vec{Z}_A + \vec{Z}_B} \right], \vec{S}_A^* = \left[\frac{\vec{Z}_B}{\vec{Z}_A + \vec{Z}_B} \right] \times \vec{S}_L^*$$

$$\text{Maximum kVA shared is } |S_L|_{max} = S_A + S_B \left(\frac{Z_A}{Z_B} \right)$$

where, $Z_A < Z_B$.

- Stacking factor $S_K = \frac{\text{Gross Cross-Sectional Area}}{\text{Net Cross-Sectional Area}}$

INDUCTION MACHINE

$$\text{Synchronous speed } N_s = \frac{120f}{P}$$

$$\text{Slip } s = \frac{N_s - N}{N_s}$$

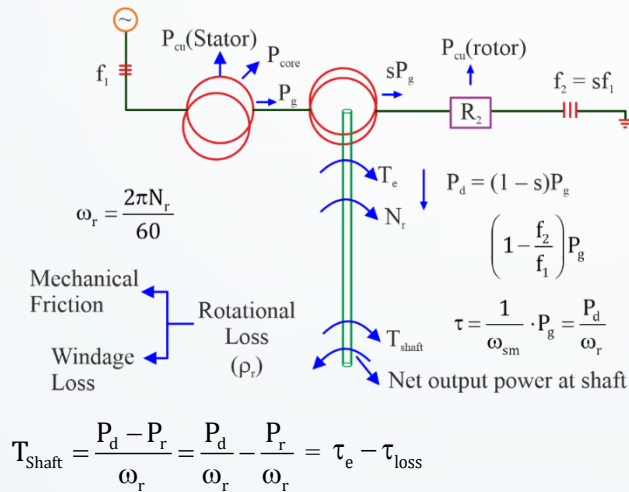
Induction motor rotates below the synchronous speed.

Three-phase induction motor is self starting.

When supply is given to stator:

Relative speed of	With Respect to			
	Stator	Rotor	Stator field	Rotor field
(i) Stator	0	$-N_s(1-s)$	$-N_s$	$-N_s$
(ii) Stator field	N_s	sN_s	0	0
(iii) Rotor	$N_r = (N_s(1-s))$	0	$-sN_s$	$-sN_s$
(iv) Rotor field	N_s	sN_s	0	0

Power flow in Induction Motor



Torque Equation

$$|I'_2| = \frac{V_1}{\sqrt{\left(\frac{R'_2}{s}\right)^2 + (X'_2)^2}}$$

$$P = |I'_2|^2 \cdot \frac{R'_2}{s}$$

$$\tau = \frac{3}{\omega_{sm}} \cdot \left[\frac{V_1^2}{(R'_2/s)^2 + (X'_2)^2} \right] \cdot \frac{R'_2}{s}$$

Low Slip Zone	High Slip Zone
$\tau = \frac{3}{\omega_{sm}} \cdot \left[\frac{V_1^2}{(R'_2)^2} \right] \times s$ $\tau \propto s$	$\tau = \frac{3}{\omega_{sm}} \times \left[\frac{V_1^2}{(X'_2)^2} \right] \times \frac{R'_2}{s}$ $\tau \propto \frac{1}{s}$

$$s_{T(\max)} = \frac{R'_2}{X_2} = \frac{R_2}{X_2}, \quad s_{T(\max)} \propto R_2$$

$$T_{(\max)} = \frac{3}{2\omega_{sm}} \cdot \left[\frac{V_1^2}{(X'_2)^2} \right]$$

$$\frac{T_{st}}{T_{(\max)}} = \frac{2}{\left(\frac{R'_2}{X'_2}\right) + \left(\frac{X'_2}{R'_2}\right)} = \frac{2}{\frac{s_{T(\max)}}{1} + \frac{1}{s_{T(\max)}}}$$

No Load Test	Blocked Rotor Test
$P_{OC} = \frac{V_{OC}^2}{R_c}$ $R_c = \frac{V_{OC}^2}{P_{OC}}$ $I_c = \frac{V_{OC}}{R_c}$ $I_m = \sqrt{I_0^2 - I_c^2}$ $X_m = \frac{V_{OC}}{I_c}$	$R_{B.R} \cong R_1 + R'_2 = \frac{P_{B.R}}{(I_{B.R})^2}$ $Z_{B.R} = V_{B.R} / I_{B.R}$ $X_{B.R} = \sqrt{Z_{B.R}^2 - R_{B.R}^2}$ $X_{B.R} \cong (X_1 + X'_2)$

Starting of Motor

DOL starting : $\frac{T_{st}}{T_{fl}} = \left[\left(\frac{I_{SC}}{I_{fl}} \right)^2 s_{fl} \right]$

Impedance starting : $\frac{T_{st}}{T_{fl}} = x^2 \left(\frac{I_{SC}}{I_{fl}} \right)^2 \times s_{fl}$

Auto transformer starting : $\frac{I_{st}}{I_{fl}} = x^2 \cdot \left(\frac{I_{SC}}{I_{fl}} \right)^2 \cdot s_{fl}$

Star-Delta starting : $\frac{T_{st(Y)}}{T_{fl(\Delta)}} = \left(\frac{1}{\sqrt{3}} \right)^2 \times \left[\frac{I_{sc(\Delta)}}{I_{fl(\Delta)}} \right] \times s_{fl}$

Double Cage Rotor

$$\frac{T_{\text{outer}}}{T_{\text{inner}}} = \left[\frac{I'_{2(\text{outer})}}{I'_{2(\text{inner})}} \right]^2 \times \frac{R'_{2(\text{outer})}}{R'_{2(\text{inner})}}$$

SYNCHRONOUS MACHINE

$$\text{Induced Emf } E = \sqrt{2} \pi f \times N_{ph} K_c K_d$$

$$K_c = \cos \frac{\alpha}{2}, \quad K_d = \frac{\sin \left(\frac{m\beta}{2} \right)}{m \sin \left(\frac{\beta}{2} \right)}$$

Armature reaction in Synchronous Generator:

Condition	Armature reaction effect
- At UPF - At Zero PF lagging - At zero PF leading - At 0.8 PF lagging - At 0.8 PF leading	- Cross magnetising effect, Normally excited. - Directly demagnetising effect, Over excited. - Directly magnetising effect, Under excited. - Both demagnetising and cross magnetising effects, Over excited. - Both cross magnetising and magnetising effects, Under excited.

Power angle equation in Round Rotor Synchronous Generator:

$$P_{dev} = \frac{E_f^2}{Z_s} \cos \theta_s - \frac{E_f V}{Z_s} \cos(\theta_s + \delta)$$

$$Q_{dev} = \frac{E_f^2}{Z_s} \sin \theta_s - \frac{E_f V}{Z_s} \sin(\theta_s + \delta)$$

$$P_{out} = \frac{VE_f}{Z_s} \cos(\theta_s - \delta) - \frac{V^2}{Z_s} \cos \theta_s$$

$$P_{out(max)} \text{ occurs at } \delta = \theta_s$$

$$Q_{out} = \frac{VE_f}{Z_s} \sin(\theta_s - \delta) - \frac{V^2}{Z_s} \sin \theta_s$$

Armature reaction in Synchronous Motor:-

Condition	Armature reaction effect
- At UPF - At Zero PF lagging - At zero PF leading - At 0.8 PF lagging	- Cross magnetising effect, Normally excited. - Directly magnetising effect, Under excited. - Directly demagnetising effect, Over excited.

- | | |
|---|---|
| <ul style="list-style-type: none"> At 0.8 PF leading | <ul style="list-style-type: none"> Both demagnetising and cross magnetising effects, Under excited. Both cross magnetising and magnetising effects, Over excited. |
|---|---|

Power angle equation in Round Rotor Synchronous Motor:

$$P_{dev} = \frac{E_f V}{Z_s} \cos(\theta_s - \delta) - \frac{E_f^2}{Z_s} \cos \theta_s$$

$$Q_{dev} = \frac{E_f V}{Z_s} \sin(\theta_s - \delta) - \frac{E_f^2}{Z_s} \sin \theta_s$$

$$P_{in} = \frac{V^2}{Z_s} \cos \theta_s - \frac{VE_f}{Z_s} \cos(\theta_s + \delta)$$

$$Q_{in} = \frac{V^2}{Z_s} \sin \theta_s - \frac{VE_f}{Z_s} \sin(\theta_s + \delta)$$

Salient pole Synchronous Generator:

$$E'_f = \sqrt{(V \cos \phi + I_a R_a)^2 + (V \sin \phi \pm I_a X_q)^2}$$

$$\psi = \tan^{-1} \left(\frac{V \sin \phi \pm I_a X_q}{V \cos \phi + I_a R_a} \right)$$

$$P_{out} = \frac{VE_f}{X_d} \sin \delta + \frac{V^2}{2} \left(\frac{1}{X_q} - \frac{1}{X_d} \right) \sin 2\delta$$

Salient Pole Synchronous Motor:

$$E'_f = \sqrt{(V \cos \Phi - I_a R_a)^2 + (V \sin \Phi \pm I_a X_q)^2}$$

$$\Psi = \tan^{-1} \left(\frac{V \sin \Phi \pm I_a X_q}{V \cos \Phi - I_a R_a} \right)$$

Synchronizing power coefficient

For round rotor machine,

$$S_p = \frac{VE_f}{X_s} \cos \delta$$

For salient pole machine,

$$S_p = \frac{VE_f}{X_s} \cos \delta + V^2 \left[\frac{1}{X_q} - \frac{1}{X_d} \right] \cos 2\delta$$

DC MACHINE

$$\text{Induced Emf } E_a = \frac{\phi ZN}{60} \times \frac{P}{A}$$

$$\text{In motor, developed power } P = E_b I_a$$

$$\text{Torque } T = \frac{P}{\omega_m} = \frac{E_b I_a}{2\pi N/60}$$

Separately Excited DC Machine

Generator	Motor
$E_g = V + I_a R_a$	$E_f = V - I_a R_a$

DC Shunt Machine

Generator	Motor
$I_a = I_L + I_{sh}$	$I_a = I_L - I_{sh}$
$I_{sh} = \frac{V}{R_{sh}}, E_g = V + I_a R_a$	$I_{sh} = \frac{V}{R_{sh}}, E_b = V - I_a R_a$

DC Series Machine

Generator	Motor
$I_a = I_L$	$I_a = I_L$
$E_g = V + I_a(R_a + R_{se})$	$E_b = V - I_a(R_a + R_{se})$

DC Compound Machine

Short Shunt Compound Generator	Short shunt Compound Motor
$I_a = I_{sh} + I_L$	$I_a = I_L - I_{sh}$
$E_g = V + I_{se} R_{se} + I_a R_a$	$E_f = V - I_{se} R_{se} - I_a R_a$
Long Shunt Compound Generator	Lung Shunt Compound Motor
$I_a = I_{sh} + I_L$	$I_a = I_L + I_{sh}$
$E_g = V + I_a(R_a + R_{se})$	$E_g = V - I_a(R_a + R_{se})$

DC Generator characteristics

- Magnetization characteristics $\rightarrow V_{OC}$ vs I_f
- Internal characteristics $\rightarrow E_g$ vs I_L
- External characteristics $\rightarrow V$ vs I_L

Demagnetizing and cross magnetizing ampere turns.

$$F_{ar} (\text{demagnetising}) = \left[\frac{Z/2}{P} \times \frac{2\beta_{electrical}}{180} \times \frac{I_a}{A} \right], \quad F_{ar} (\text{Cross-magnetising}) = \frac{Z/2}{P} \cdot \left[1 - \frac{2\beta_{electrical}}{180} \right] \times \frac{I_a}{A}$$

Speed control of DC motor

$$N \times \frac{E_f}{\phi}, \quad \frac{N_2}{N_1} = \frac{E_{f2}}{E_{f1}} \times \frac{\phi_1}{\phi_2}$$



POWER SYSTEM

Transmission Line Parameter

Inductance

Inductance of conductor due to flux between point 1 & 2

$$L_{12} = \frac{\Psi_{12}}{I} = \frac{\mu}{2\pi} \ln \frac{D_2}{D_1} \text{ or } 2 \times 10^{-7} \ln \frac{D_2}{D_1}$$

Inductance of a single conductor due to flux up to distance D

$$L = 2 \times 10^{-7} \ln \frac{D}{r'} \text{ H/m or } L = 0.4605 \log \frac{D}{r'} \text{ mH/km}$$

mH/km

Skin Effect

δ = skin depth.

$$\delta = \sqrt{\frac{2}{\omega \mu \sigma}} = \sqrt{\frac{1}{\pi f \mu \sigma}}$$

$$\frac{R_{ac}}{R_{dc}} = \frac{A_{act}}{A_{eff}}$$

Capacitance Between 1 ϕ , 2 wire line

$$c_{ab} = \frac{q}{v_{ab}} = \frac{\pi \epsilon}{\ln D/r} \text{ F/m}$$

Charging current between two conductor $I_C = j\omega c_{ab} v_{ab}$

Capacitance to each line to neutral is $c_n = 2c_{ab} = \frac{2\pi \epsilon}{\ln D/r} \text{ F/m}$

Performance of Transmission Line

Efficiency

$$\eta = \frac{P_r}{P_r + \text{losses}} = \frac{P_r}{P_r + 3I^2 R}$$

Voltage Regulation

$$V.R. = \frac{|V_{RO}| - |V_R|}{|V_R|} \times 100\%$$

$$V_{RO} = \frac{v_s}{A}$$

Charging current

$$I_C = c v_{ro} = \frac{c v_s}{A}$$

Short Transmission Line

$$v_s^2 = (v_r \cos \phi_r + I_r R)^2 + (v_r \sin \phi_r + I_r X)^2$$

$$\tan \phi_s = \frac{v_r \sin \phi_r + I_r X}{v_r \cos \phi_r + I_r R}$$

Condition of zero V.R. $\cos(\theta - \phi_r) = \frac{-I_r z}{2v_r}$

Condition for maximum V.R. $\theta = \phi_r$

Approx :

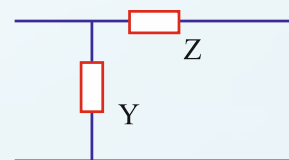
$$V.R. = \frac{v_s - v_r}{v_r} = \frac{I_r R \cos \phi_r + I_r X \sin \phi_r}{V_r}$$

Condition for zero voltage regulation $\theta - \phi_r = \frac{\pi}{2}$ it occur at

leading p.f.

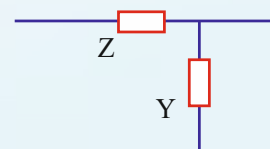
Medium Transmission Line

1. Sending end C Model



$$= \begin{bmatrix} 1 & Z \\ Y & 1 + YZ \end{bmatrix}$$

2. Receiving end C model.

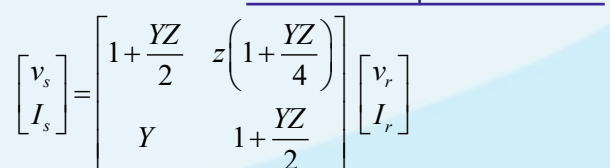


$$= \begin{bmatrix} 1 + YZ & Z \\ Y & 1 \end{bmatrix}$$

Nominal T Model :

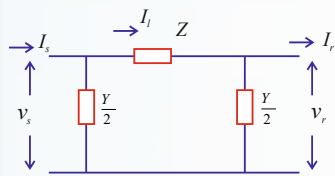
$$Z = R + j\omega L$$

$$Y = \frac{1}{j\omega c}$$



$$\begin{bmatrix} v_s \\ I_s \end{bmatrix} = \begin{bmatrix} 1 + \frac{YZ}{2} & z \left(1 + \frac{YZ}{4} \right) \\ Y & 1 + \frac{YZ}{2} \end{bmatrix} \begin{bmatrix} v_r \\ I_r \end{bmatrix}$$

Nominal π Model :



$$\begin{bmatrix} v_s \\ I_s \end{bmatrix} = \begin{bmatrix} 1 + \frac{YZ}{2} & Z \\ Y\left(1 + \frac{YZ}{4}\right) & 1 + \frac{YZ}{2} \end{bmatrix} \begin{bmatrix} v_r \\ I_r \end{bmatrix}$$

Long Transmission Line

$$\begin{bmatrix} v_s \\ I_s \end{bmatrix} = \begin{bmatrix} \cosh rl & z_c \sinh rl \\ \frac{1}{z_c} \sinh rl & \cosh rl \end{bmatrix} \begin{bmatrix} v_r \\ I_r \end{bmatrix}$$

$r = \sqrt{zy}$ is called propagation constant

$z_c = \sqrt{\frac{z}{y}}$ is called characteristic Impedance.

$$r = \alpha + j\beta$$

α = attenuation constant

β = phase constant

Approx value of

$$A = 1 + \frac{YZ}{2}$$

$$B = Z \left[1 + \frac{YZ}{6} \right]$$

$$C = Y \left[1 + \frac{YZ}{6} \right]$$

$$D = A$$

Power Formula

$$P_r = \frac{|v_s||v_r|}{|B|} \cos(\beta - \delta) - \frac{|A||v_r|^2}{|B|} \cos(\beta - \alpha)$$

$$Q_r = \frac{|v_s||v_r|}{|B|} \sin(\beta - \delta) - \frac{|A||v_r|^2}{|B|} \sin(\beta - \alpha)$$

$$\delta = B$$

For short T.L $P_r = \frac{|v_s||v_r|}{|x|} \sin \delta$

In general if $v_s = |v_s| < \delta_1$ and $v_r = |v_r| < \delta_2$

$$P_r = \frac{|v_s||v_r|}{|x|} \sin(\delta_1 - \delta_2)$$

Sending end Power Formula

$$P_s = \frac{|A||v_s|^2}{|B|} \cos(\beta - \alpha) - \frac{|v_s||v_r|}{|B|} \cos(\beta + \delta)$$

$$Q_s = \frac{|A||v_s|^2}{|B|} \sin(\beta - \alpha) - \frac{|v_s||v_r|}{|B|} \sin(\beta + \delta)$$

For short transmission line

$$P_s = \frac{|v_s||v_r|}{|x|} \sin \delta$$

$$Q_s = \frac{|v_s|^2}{|x|} - \frac{|v_s||v_r|}{|x|} \cos \delta$$

Characteristic Impedance: $Z_c = \sqrt{\frac{z}{y}} = \sqrt{\frac{Z}{Y}} = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$

Surge Impedance :

It is characteristic Impedance of a loss less line $Z_s = \sqrt{\frac{L}{C}}$

Wave Nature

wave length $X = \frac{2\pi}{\beta}$, time period $T = \frac{2\pi}{\omega}$

velocity $v = \frac{\omega}{\beta} = \frac{1}{\sqrt{L \cdot C}} = \frac{1}{\sqrt{\mu \epsilon}}$

Surge Impedance loading / Natural Loading:

$$P = \frac{v_r^2}{z_s} \quad z_s = \sqrt{\frac{L}{C}}$$

Corona

Critical Disruptive Voltage

$v_d = 2rg_o \delta m_o \ln \frac{D}{r}$ (For 1 ϕ 2 wire line)

$v_d = rg_o \delta m_o \ln \frac{D}{r}$ (For 3 ϕ line)

$g_o = \frac{3 \times 10^6}{\sqrt{2}} v / m$ (RMS)

$\delta = \frac{3.92P}{273 + Q}$

Corona Loss:

As per peak formula

$$P = 241 \times 10^{-5} \frac{(f + 25)}{\delta} \sqrt{\frac{r}{D}} (v_p - v_d)^2 \text{ kw/km/ph.}$$

this formula give more accurate result if $\frac{v_p}{v_D} > 1.8$

if $\frac{v_p}{v_D} < 1.8$ Peterson formula give more accurate result.

$$P_c = \frac{21 \times 10^{-6} f v^2}{(\log_{10} D/r)^2} \times F$$

Factor Affecting Corona:

1. Increase with freq. $P_c \propto (f \uparrow + 25)$
2. Increase with voltage $P_c \propto (V_p \uparrow - V_D)^2$
3. At hills $P \uparrow \Rightarrow \delta \downarrow \Rightarrow P_c \uparrow$ corona loss is more

Corona

Voltage Control

$$v_2 = \left[v_1 - \frac{QX}{v_1} \right] - j \left[\frac{PX}{v_1} \right]$$

$$v_2 \approx v_1 - \frac{QX}{V_1}$$

Shunt Capacitor:

$$v_r' = v_r + \frac{Q_c X}{v_s}$$

$$\text{Increment of voltage } \Delta v = \frac{Q_c X}{v_s}$$

$$Q_c = \frac{v_s \Delta v}{X}$$

$$Q_c = P_1 [\tan \phi_1 - \tan \phi_2]$$

$$\frac{3v_r^2}{X_c} = P_1 [\tan \phi_1 - \tan \phi_2]$$

$$c_{pn} = \frac{P_1 [\tan \phi_1 - \tan \phi_2]}{3\omega v_{r(ph)}^2}$$

Shunt Reactor

It reduce the voltage

$$v_r' = v_r - \frac{Q_c X}{v_s}$$

$$\text{Reduction in voltage } \Delta v = \frac{Q_c X}{v_s}$$

$$Q_c = \frac{\Delta v v_s}{X}$$

$$X_L = \frac{|B|}{1 - |A|}$$

Series Capacitor :

$$\Delta v = I X_C \sin \phi$$

$$Q_c = \frac{3\Delta v}{\sin \phi} \cdot I$$

Synchronous Condenser/Synchronous Coil

$$\phi_{sm} = p_1 \tan \phi_1 - (p_1 + p_{sm}) \tan \phi_2$$

Synchronous Phase Modifier

$$Q_{sm} = p_1 \tan \phi_1 - (p_1 + p_{sm}) \tan \phi_2$$

Cables and Insulator

CABLES

Min electrical stress occur at inner surface of sheath at $x = R$

$$E_{mn} = \frac{v}{R \ln \frac{R}{r}}$$

Maximum electrical stress occur at surface of conductor at $x = r$

$$E_{(mn)} = \frac{v}{r \ln \frac{R}{r}}$$

$$\frac{r}{R} = \frac{1}{e}$$

$$r = 0.365R$$

G_1, G_2, G_3 is dielectric strength of insulating material

$$\text{Total operating voltage} = g_{1mx} r \ln \frac{r_1}{r} + g_{2nx} r_1 \ln \frac{r_2}{r_1} + g_{3nx} r_2 \ln \frac{R}{r_2}$$

All insulation is subjected to same maximum stress

$$\epsilon_1 r = \epsilon_2 r_2 = \epsilon_3 r_3$$

$$r < r_1 < r_2$$

$$\epsilon_1 > \epsilon_2 > \epsilon_3$$

Capacitance of single core cable

$$c = \frac{2\pi \epsilon}{\ln R/r}$$

Capacitance of 3 core cable

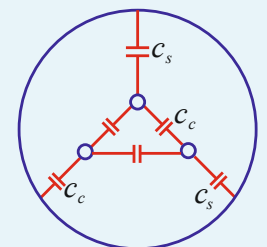
$$\text{Per phase capacitance } C_{ph} = 3c_c + c_s$$

Capacitance between two conductor = $1.5 c_c + 0.5 c_s$

Bunch all conductors the capacitance between bunched conductor & sheath

$$c_x = 3c_s$$

Connect any two conductor with sheath, the capacitance between remaining conductor & sheath is



$$c_y = 2c_c + c_s$$

$$c_{ph} = \frac{3}{2}c_y - \frac{c_x}{6}$$

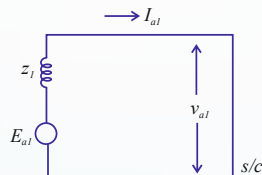
String Efficiency:

$$\eta = \frac{\text{voltage across whole string}}{\text{No. of Discs} \times \text{voltage across lower most unit}}$$

Symmetrical Component & Fault Analysis

Symmetrical Fault Analysis:

$$I_f - I_a = I_{a1} = \frac{E_{a1}}{z_1} = \frac{E}{z_1}$$



If fault impedance is given $I_f = \frac{E}{z_1 + z_f}$

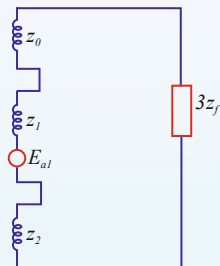
If fault occur at no. load condition $I_f = \frac{1 \angle 0^\circ}{z_1 + z_f}$

Unsymmetrical fault

L-G Fault

$$I_{a0} = I_{a1} = I_{a2} = \frac{1 \angle 0}{z_1 + z_2 + (z_0 + 3z_f)}$$

$$I_f = \frac{3 \angle 0}{z_1 + z_2 + (z_0 + 3z_f)}$$



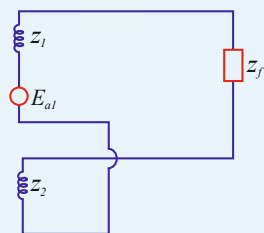
$$\text{Actual fault MVA} = \frac{3 \times \text{base MVA}}{z_1 + z_2 + (z_0 + 3z_b)}$$

$$\text{Fault MVA} = \sqrt{3} \text{ pre fault voltage} \times \text{post fault current}$$

L-L Fault

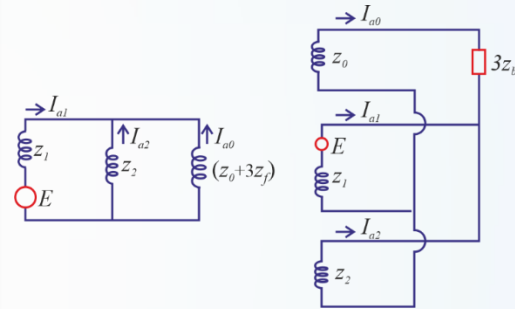
$$I_{a1} = \frac{E_a}{z_1 + z_2 + z_f}$$

$$I_f = \frac{-j\sqrt{3}}{z_1 + z_2 + z_f} \quad \begin{cases} I_b = -j\sqrt{3}I_{a1} \\ I_c = -I_b \\ I_a = 0 \end{cases}$$



$$\text{fault MVA} = I_f(p.u.)$$

L-L-G Fault :



$$I_{a1} = \frac{E}{z_1 + \frac{z_2(z_0 + 3z_f)}{z_2 + z_0 + 3z_f}}$$

$$I_{a2} = -\frac{(z_0 + 3z_b)}{z_2 + z_0 + 3z_f} \cdot I_{a1}$$

$$I_{a0} = -\frac{z_2}{(z_2 + z_0 + 3z_f)} \cdot I_{a1}$$

$$I_f = I_b + I_c$$

POWER SYSTEM STABILITY

Steady State Stability

If rate of change of load < rate of change of excitation

$$p_{mx} = \text{SSSL} = \frac{v_1 v_2}{x}$$

If R is present $p_{nx} = \frac{|v_s| |v_r|}{|z|} - \frac{|v_r|^2}{|z|} \cdot \frac{R}{|z|}$

For Maximum power $X = \sqrt{3}R$

So for $\theta = 60$ (line impedance angle)

Dynamic of Synchronous Machine

$$KE = \frac{1}{2} M \omega_s^2$$

$$M = \text{Inertia constant} = J \left(\frac{2}{p} \right)^2 \omega_s \frac{\text{MJ} - \text{sec}}{\text{elect rad.}}$$

$$H = \text{Store energy} = \frac{K.E}{\text{MVA} \cdot (G)} \frac{\text{MJ}}{\text{MVA}} \rightarrow \frac{M_0 - \text{sec}}{\text{MVA}}$$

$$GH = \frac{1}{2} M \omega_s^2$$

$$M = \frac{GH}{\pi f} \quad \text{MJ-sec/elect rad.}$$

$$M = \frac{GH}{180f} \quad \text{MJ-sec/elect degree.}$$

Swing Equation:

$$\frac{Md^2\delta}{dt^2} = p_s - p_e$$

$$\frac{GH}{180f} \cdot \frac{d^2\delta}{dt^2} = p_s - p_e$$

$$\frac{H}{180f} \cdot \frac{d^2\delta}{dt^2} = (p_s - p_e)$$

(I) Two machine swinging coherently

$$M_{eq} = M_1 + M_2$$

$$H_{eq} = \frac{G_1 H_1 + G_2 H_2}{G_{eq}}$$

(II) Two machine are swinging together.

$$M_{eq} = \frac{d^2\delta}{dt^2} = p_s - p_e$$

$$M_{eq} = \frac{M_1 M_2}{M_1 + M_2}$$

$$P_s = \frac{P_{s1} M_2 - P_{s2} M_1}{M_1 + M_2}$$

$$P_e = \frac{P_{e1} M_2 - P_{e2} M_1}{M_1 + M_2}$$

Load Flow Studies

Load Flow Equations:

$$P_i = \sum_{k=1}^n |V_i| |V_k| |Y_{ik}| \cos(\delta_i - \delta_k - \theta_{ik})$$

$$\theta_i = \sum_{k=1}^n |V_i| |V_k| |Y_{ik}| \sin(\delta_i - \delta_k - \theta_{ik})$$

Where $V_i = |V_i| \angle \delta_i$

$$Y_{ik} = |Y_{ik}| \angle \theta_{ik}$$

1. Load Bus or PQ Bus. $\begin{cases} \text{Specified quantity} = p_i, \theta_i \\ \text{Unknown quantity} = |V|, \delta \end{cases}$

2. Generator Bus/PV Bus. $\begin{cases} \text{Specified quantity} = p_i, |V| \\ \text{Unknown quantity} = \theta_i, \delta_i \end{cases}$

Slack Bus:

Specified quantity = $|V_i|, \delta_i$

$$|V_i| = 1 \quad \delta_i = 0$$

Unknown quantity = p_i, θ_i

GS Method : It PV bus is absent-

$$V_j = \frac{1}{Y_{ji}} \left[\frac{p_i - jQ_i}{V_i^*} - \sum_{k=1, k \neq i}^n Y_{ik} V_k \right]$$

The $(r+1)^{th}$ iteration of V_i

$$V_j^{r+1} = \frac{1}{Y_{ji}} \left[\frac{p_i - jQ_i}{(V_i^r)^*} - \sum_{k=1}^{k=i-1} Y_{ik} V_k^{r+1} - \sum_{k=i+1}^n Y_{ik} V_k^r \right]$$

Fast Decoupled Method

$$\begin{bmatrix} \Delta P_i / |V_i| \\ \Delta Q_i / |V_i| \end{bmatrix} = \begin{bmatrix} B \\ C \end{bmatrix} \begin{bmatrix} \Delta \delta_i \\ \Delta V_i \end{bmatrix}$$

Economic Load Dispatch

Input- Output Curve:

Economic load dispatch neglecting loss -

For optimal operation

$$\frac{df_1}{dP_1} = \frac{df_2}{dP_2} = \frac{df_3}{dP_3} \dots = \frac{df_n}{dP_n} = \lambda$$

Loss Equations :

$$P_L = \sum_n \sum_m P_m B_{mn} P_n$$

matrix form $P_L = P^T B P$

In N bus system we have N^2 coefficient

Lagrangian Method (Including Losses):

$$\frac{df_1}{dP_1} L_1 = \frac{df_2}{dP_2} L_2 \dots = \frac{df_n}{dP_n} L_n = \lambda$$

$$L_n = \text{Penalty factor} = \frac{1}{\left(1 - \frac{\partial P_L}{\partial P_n}\right)}$$

Switch Gear and Protection

Attracted Armature type Relay:

$$I > \frac{\sqrt{k_2}}{\sqrt{k_1}}$$

Balance Beam Relay :

$$\frac{I_1}{I_2} > \sqrt{\frac{k_2}{k_1}}$$

Induction Disc Relay :

1. Shaded Pole Relay :

$$T \propto \phi_{m1} \phi_{m2} \sin \theta$$

2. Watthour meter type

$$\text{PSM} = \frac{\text{Secondary current in CT}}{\text{Relay current setting}}$$

Induction Cup Type Relay:

- It has two, four or more electromagnet

PMMC Relay:

- $T \propto I$
- Rectifier Relay
- Thermal Relay
- Under Frequency/Over Frequency Relay
- Under Voltage/Over Voltage Relay
- Directional Relay

Over Current Relays:

Very inverse Relay $t = \frac{K}{I^n - n}$

$$N = (1.02-2)$$

Extreme inverse relay $I = \frac{K}{I^n} \quad (n = 2 - 8)$

Torque developed in directional unit.

$$T = KVI \cos(\theta - \tau)$$

Differential Protection :

Restraining torque $\propto \left(\frac{I_1 + I_2}{2} \right) N$

$$\% \text{ pickup value} = \frac{(I_1 + I_2)}{N \frac{(I_1 + I_2)}{2}} \times 100\%$$

Distance Relay :

$$T = k_1 I^2 + k_2 V^2 + k_3 VI \cos(\theta - \tau) + k_4$$

Impedance Relay:

$$k_1 I^2 > k_2 V^2 \text{ for Relay operation}$$

Reactance Relay:

$$T = k_1 I^2 - k_2 VI \cos(\theta - \tau)$$

Mho Relay :

$$T = k_3 VI \cos(\theta - \tau) - k_2 V^2$$



POWER ELECTRONICS

Basic expressions

Average expression

$$X_{avg} = \frac{1}{T} \int_0^T x(t) dt \quad \dots (1)$$

RMS expression

$$X_{rms} = \sqrt{\frac{1}{T} \int_0^T x^2(t) dt} \quad \dots (2)$$

$$X_{rms} = \sqrt{\sum_{n=1}^N X_{n,rms}^2} \quad \dots (3)$$

Fourier series

$$f(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)] \quad \dots (4)$$

where,

$$a_0 = \frac{1}{T} \int_{-T/2}^{T/2} f(t) dt \quad \dots (5)$$

$$a_n = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \cos(n\omega_0 t) dt \quad \dots (6)$$

$$b_n = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \sin(n\omega_0 t) dt \quad \dots (7)$$

Root-Mean-Square Values of waveforms, $i(t)$

	$I_{rms} = \frac{I_p}{\sqrt{2}}$
	$I_{rms} = I_p \sqrt{\frac{k}{2}}, k = \frac{T_0}{T}$
	$I_{rms} = I_p \sqrt{k}, k = \frac{T_0}{T}$
	$I_{rms} = I_p \sqrt{\frac{k}{3}}, k = \frac{T_0}{T}$
	$I_{rms} = \frac{\Delta I}{2\sqrt{3}}$

Inductor

$$\text{Voltage } \Rightarrow v_L = L \frac{di}{dt} \quad \dots (1)$$

$$\text{Current } i_L = \frac{1}{L} \int v_L(dt) + I_0 \quad \dots (2)$$

$$\text{Energy } w(t) = \frac{1}{2} Li^2(t) \quad \dots (3)$$

$$\tau = \frac{L}{R} \quad \dots (4)$$

Capacitor

$$\text{Current } i_C = C \frac{dv}{dt} \quad \dots (1)$$

$$\text{Voltage } v_C = \frac{1}{C} \int i_C(dt) + V_0 \quad \dots (2)$$

$$\text{Energy } w(t) = \frac{1}{2} Cv^2(t) \quad \dots (3)$$

$$\tau = \frac{1}{RC} \quad \dots (4)$$

Phasors

$$\text{Impedance } (Z) = \sqrt{R^2 + (\omega L)^2} \quad \dots (1)$$

$$\theta = \tan^{-1} \left(\frac{\omega L}{R} \right) \quad \dots (2)$$

Power & Energy

Instantaneous Power

$$p(t) = v(t)i(t) \quad \dots (1)$$

Average power

$$P = \frac{1}{T} \int_{t_0}^{t_0+T} p(t) dt = \frac{W}{T} \quad \dots (2)$$

Average power $P_{dc} = V_{dc} I_{avg}$

Apparent Power $S = V_{rms} I_{rms}$

$$S = \sqrt{3} V_{L-L,rms} I_{s,rms} \quad \dots (3)$$

Reactive power $Q = V_{rms} I_{rms} \sin(\theta - \phi)$

$$\text{Energy } (W) = \int_{t_1}^{t_2} p(t) dt \quad \dots (4)$$

$$W = \int_0^T p(t) dt \quad \dots (5)$$

$$\text{Power factor } (pf) = \frac{P}{S} \quad \dots (6)$$

Diode Rectifiers (Uncontrolled Rectifiers)

Performance Parameters for Un-controlled and Controlled Rectifiers

Form Factor	$FF = \frac{V_{o,rms}}{V_{o,dc}}$
Ripple Factor	$RF = \frac{\sqrt{V_{o,rms}^2 - V_{o,dc}^2}}{V_{o,dc}} = \sqrt{FF^2 - 1}$
Input Power Factor, PF (I_{s1} is the fundamental component of the input current I_{ss})	$PF = \frac{V_{s,rms} I_{s1,rms} \cos \phi_1}{V_{s,rms} I_{s,rms}} = \frac{I_{s1,rms}}{I_{s,rms}} \cos \phi_1$

Power Factor of Load	$PF = \frac{\text{effective output power}}{V_{s,rms} I_{s,rms}} = \frac{V_{o,rms} I_{o,rms}}{V_{s,rms} I_{s,rms}}$
Transformer Utilization Factor	$TUF = \frac{P_{o,dc}}{V_s I_s} = \frac{V_{o,dc} I_{o,dc}}{V_{s,rms} I_{s,rms}}$
Crest Factor	$CF = \frac{I_{s(peak)}}{I_{s(rms)}}$
Efficiency (Practical definition)	$\eta = \frac{P_{o,dc}}{P_{in,ac}} = \frac{V_{o,dc} I_{o,dc}}{V_{s,rms} I_{s,rms}}$ Or $\eta = \frac{P_{o,dc}}{P_{in,ac}} = \frac{P_{o,dc}}{P_{o,dc} + P_{loss}}$ Notes : include the power loss.
Efficiency (Text Book definition)	$\eta = \frac{P_{o,dc}}{P_{o,ac}} = \frac{V_{o,dc} I_{o,dc}}{V_{o,rms} I_{o,rms}}$ Notes : measuring the quality of the output de in relation to the output rms value.

Basic Characteristics of DC Motors

Armature Voltage	$V_a = i_a R_a + E_g$ i_a = armature current R_a = armature resistance
Field Voltage	$V_f = i_f R_f$
Motor back emf (speed voltage), E_g	$E_g = K_v \omega$ or i_f K_v = Voltage constant ω = Motor angular speed or rotor angular frequency i_f = Field current
Developed torque, T_d	$T_d = K_t i_f i_a$ $= K_t \phi i_a = B \omega + T_L$ Where Torque constant, $K_t = K_v$, T_L = Load torque, and B = Viscous friction constant
Developed Power	$P_d = T_d \omega$
1 st Quadrant Criteria's	$+V_a, +E_a, +I_a, +\omega, +T$ and $+P$
2 nd Quadrant Criteria's	$+V_a, +E_a, -I_a, +\omega, -T$ and $-P$
3 rd Quadrant Criteria's	$-V_a, -E_a, -I_a, -\omega, +T$ and $+P$
4 th Quadrant Criteria's	$-V_a, -E_a, +I_a, -\omega, +T$ and $-P$

Half Wave Rectifier with RL Load

$$i_{(\omega t)} = \begin{cases} \frac{V_m}{Z} [\sin(\omega t - \theta) + \sin(\theta) e^{-\omega t / \omega L}] & \text{for } 0 \leq \omega t \leq \beta \\ 0 & \beta \leq \omega t \leq 2\pi \end{cases} \quad \dots (1)$$

For controlled rectifier

$$i_{(\omega t)} = \begin{cases} \frac{V_m}{Z} [\sin(\omega t - \theta) - \sin(\alpha - \theta) e^{(\alpha - \omega t) / \omega L}] & \text{for } \alpha \leq \omega t \leq \beta \\ 0 & \text{otherwise} \end{cases} \quad \dots (2)$$

$$V_0 = \frac{1}{2\pi} \int_{\alpha}^{\beta} V_m \sin(\omega t) d(\omega t) = \frac{V_m}{2\pi} (\cos \alpha - \cos \beta) \quad \dots (3)$$

With free wheeling diode

$$v(t) = \frac{V_m}{\pi} + \frac{V_m}{2} \sin(\omega_0 t) - \sum_{n=2,4,6,\dots}^{\infty} \frac{2V_m}{(n^2-1)\pi} \cos(n\omega_0 t) \quad \dots (4)$$

With source inductance

$$V_0 = \frac{V_m}{2\pi} (1 + \cos u) = \frac{V_m}{\pi} \left(1 - \frac{I_L X_s}{2V_m} \right) \quad \dots (5)$$

Full Wave Rectifier

Full wave rectified waveform Fourier is

$$v_0(t) = \frac{2V_m}{\pi} + \sum_{n=2,4,\dots}^{\infty} \left(\frac{2V_m}{\pi} \left(\frac{1}{n-1} - \frac{1}{n+1} \right) \right) \cos(n\omega_0 t + \pi) \quad \dots (1)$$

Ripple voltage for C filter

$$\Delta V_0 \approx \frac{V_m \pi}{\pi R C} = \frac{V_m}{2fRC} \quad \dots (2)$$

With LC filter

In CCM

$$V_0 = \frac{2V_m}{\pi} \quad \dots (3)$$

$$I_L = \frac{2V_m}{\pi R} \quad \dots (4)$$

$$\frac{3\omega L}{R} > 1 \quad \dots (5)$$

In DCM average inductor current.

$$i_L = \frac{1}{\pi} \int_{\alpha}^{\beta} \frac{1}{\omega L} [V_m (\cos \alpha - \cos \omega t) - V_0 (\omega t - \alpha)] d(\omega t) \quad \dots (6)$$

RLE load CCM

$$I_0 = \frac{V_0 - E}{R} = \frac{\frac{2V_m}{\pi} - E}{R} \quad \dots (7)$$

$$v_0(t) = V_0 + \sum_{n=2,4,\dots}^{\infty} V_n \cos(n\omega_0 t + \pi) \quad \dots (8)$$

$$\text{where, } VC_0 = \frac{2V_m}{\pi} \text{ and } V_n = \frac{2V_m}{\pi} \left(\frac{1}{n-1} - \frac{1}{n+1} \right)$$

Full controlled with RL load

DCM

$$i_0(\omega t) = \frac{V_m}{Z} [\sin(\omega t - \theta) - \sin(\alpha - \theta) e^{-(\omega t - \alpha) / \omega L}] \quad \dots (9)$$

for $\alpha \leq \omega t \leq \beta$

$$\beta < \alpha + \pi \Rightarrow \text{DCM} \quad \dots (10)$$

For CCM

$$\alpha \leq \tan^{-1} \left(\frac{\omega L}{R} \right) \quad \dots (11)$$

$$v_0(\omega t) = V_0 + \sum_{n=1}^{\infty} V_n \cos(n\omega_0 t + \theta_n) \quad \dots (12)$$

$$V_0 = \frac{1}{\pi} \int_{\alpha}^{\alpha+\pi} V_m \sin(\omega t) d(\omega t) = \frac{2V_m}{\pi} \cos \alpha \quad \dots (13)$$

The harmonics are

$$V_n = \sqrt{a_n^2 + b_n^2} \quad \dots (14)$$

where,

$$a_n = \frac{2V_m}{\pi} \left[\frac{\cos(n+1)\alpha}{n+1} - \frac{\cos(n-1)\alpha}{n-1} \right] \quad \dots (15)$$

$$b_n = \frac{2V_m}{\pi} \left[\frac{\sin(n+1)\alpha}{n+1} - \frac{\sin(n-1)\alpha}{n-1} \right] \quad \dots (16)$$

$$V_0 = \frac{2V_m}{\pi} \cos \alpha \quad \dots (17)$$

Three Phase Full Wave Rectifier

Uncontrolled with R load

$$V_0(t) = V_0 + \sum_{n=6,12,18,\dots}^{\infty} V_n \cos(n\omega_0 t + \pi) \quad \dots (1)$$

where,

$$V_0 = \frac{1}{\pi/3} \int_{\pi/3}^{2\pi/3} V_{m,L-L} \sin(\omega t) d(\omega t) = \frac{3V_{m,L-L}}{\pi} \quad \dots (2)$$

The amplitudes of the ac voltage terms are

$$V_n = \frac{6V_{m,L-L}}{\pi(n^2 - 1)} \quad \dots (3)$$

for $n = 6, 12, 18, \dots$

DC-DC Converters - CCM

Buck Converter

$$D = \frac{t_{on}}{t_{on} + t_{off}} = \frac{t_{on}}{T} = t_{on} f_a \quad \dots (1)$$

$$I_{\max} = I_L + \frac{\Delta I_L}{2} \text{ and } I_{\min} = I_L - \frac{\Delta I_L}{2}$$

$$V_0 = DV_s \quad \dots (1)$$

Minimum inductance $L_{\min}$

$$L_{\min} = \frac{(1-D)R}{2f} \quad \dots (2)$$

$$L = \frac{V_0(1-D)}{\Delta I_L f_s} \quad \dots (3)$$

Output capacitor C

$$C = \frac{1-D}{8L(\Delta V_0 / V_0)f^2} \quad \dots (4)$$

$$\Delta I_L = \left(\frac{V_s - V_0}{L} \right) DT$$

Boost Converter

$$V_0 = \frac{V_s}{1-D} \quad \dots (1)$$

Minimum inductance $L_{\min}$

$$L_{\min} = \frac{D(1-D)^2 R}{2f} \quad \dots (2)$$

$$L = \frac{V_s D}{\Delta I_L f} \quad \dots (3)$$

Output capacitor C

$$C = \frac{D}{R(\Delta V_0 / V_0)f} \quad \dots (4)$$

$$\Delta I_L = \left(\frac{V_s}{L} \right) DT$$

Buck-Boost Converter

$$V_0 = -V_s \left(\frac{D}{1-D} \right) \quad \dots (1)$$

Minimum inductance $L_{\min}$

$$L_{\min} = \frac{(1-D)^2 R}{2f} \quad \dots (2)$$

$$L = \frac{V_s D}{f \Delta I_L} \quad \dots (3)$$

Output capacitor C

$$C = \frac{D}{R(\Delta V_0 / V_0)f} \quad \dots (4)$$

$$\Delta I_L = \frac{V_s DT}{L}$$

Flyback Converter

$$V_0 = V_s \left(\frac{D}{1-D} \right) \left(\frac{N_1}{N_2} \right) \quad \dots (1)$$

Minimum inductance $L_{\min}$

$$(L_m)_{\min} = \frac{(1-D)^2 R}{2f} \left(\frac{N_2}{N_1} \right)^2 \quad \dots (2)$$

$$L = \frac{V_s D}{f \Delta I_{Lm}} \quad \dots (3)$$

Output capacitor C

$$\frac{\Delta V_0}{V_0} = \frac{D}{RCf} \quad \dots (4)$$

$$\Delta I_L = \frac{V_s DT}{L_m}$$

DC-AC Converters

Square wave inverter (RL series load)

$$i_0(t) = \begin{cases} \frac{V_{dc}}{R} + \left(I_{\min} - \frac{V_{dc}}{R} \right) e^{-t/\tau} & \text{for } 0 \leq t \leq \frac{T}{2} \\ -\frac{V_{dc}}{R} + \left(I_{\min} + \frac{V_{dc}}{R} \right) e^{-(t-T/2)/\tau} & \text{for } \frac{T}{2} \leq t \leq T \end{cases} \quad \dots (1)$$

$$I_{\max} = -I_{\min} = \frac{V_{dc}}{R} \left(\frac{1 - e^{-(T/2\tau)}}{1 + e^{-(T/2\tau)}} \right) \quad \dots (2)$$

$$I_{rms} = \sqrt{\frac{1}{T} \int_0^T i^2(t) dt}$$

$$I_{rms} = \sqrt{\frac{2}{T} \int_0^{T/2} \left[\frac{V_{dc}}{R} + \left(I_{\min} - \frac{V_{dc}}{R} \right) e^{-t/\tau} \right]^2 dt} \quad \dots (3)$$

$$v_0(t) = \sum_{1,3,5,\dots}^{\infty} \frac{4V_{dc}}{n\pi} \sin n\omega_0 t \quad \dots (4)$$

$$THD = \frac{\sqrt{\sum_{n=2}^{\infty} (V_{n,rms})^2}}{V_{1,rms}} = \frac{\sqrt{V_{rms}^2 - V_{1,rms}^2}}{V_{1,rms}} \quad \dots (5)$$

Quasi Square wave inverter (RL series load)

$$V_{rms} = \sqrt{\frac{1}{\pi} \int_{\alpha}^{\pi-\alpha} V_{dc}^2 d(\omega t)} = V_{dc} \sqrt{1 - \frac{2\alpha}{\pi}} \quad \dots (1)$$

$$v_n = \frac{2}{\pi} \int_{\alpha}^{\pi-\alpha} V_{dc} \sin(n\omega_0 t) d(\omega_0 t) = \frac{4V_{dc}}{n\pi} \cos(n\alpha) \quad \dots (2)$$

$$V_1 = \left(\frac{4V_{dc}}{\pi} \right) \cos \alpha \quad \dots (3)$$

Bipolar PWM Inverter

$$v_0(t) = \sum_{n=1}^{\infty} V_n \sin(n\omega_0 t) \quad \dots (1)$$

6 Step Inverter

$$V_{n,L-L} = \left| \frac{2V_{dc}}{3n\pi} \left[2 + \cos\left(\frac{n\pi}{3}\right) - \cos\left(\frac{2n\pi}{3}\right) \right] \right| \quad \dots (1)$$

for $n = 1, 5, 7, 11, 13, \dots$

3 phase PWM Inverter

$$V_{n3} = \sqrt{A_{n3}^2 + B_{n3}^2} \quad \dots (1)$$

Where

$$\begin{cases} A_{n3} = V_n \sin\left(\frac{n\pi}{2}\right) \sin\left(\frac{n\pi}{3}\right) \\ B_{n3} = V_n \cos\left(\frac{n\pi}{2}\right) \sin\left(\frac{n\pi}{3}\right) \end{cases} \quad \dots (2)$$

Harmonics — Performance Parameters

Harmonic factor of n^{th} harmonic, HF_n	$HF_n = \frac{V_{on}}{V_{ol}}$; V_{ol} = RMS value of the fundamental component V_{on} = RMS value of the n^{th} harmonic component
Total Harmonic Distortion, THD	$THD = \frac{1}{V_{ol}} \sqrt{\sum_{n=2,3,\dots}^{\infty} V_{on}^2}$
Distortion factor, DF	$DF = \frac{1}{V_{ol}} \sqrt{\sum_{n=2,3,\dots}^{\infty} \left(\frac{V_{on}}{n^2}\right)^2}$
Distortion factor of n^{th} harmonic, DF_n	$DF_n = \frac{V_{on}}{V_{ol} n^2}$
Lowest order harmonic, LOH	Harmonic component with frequency $\cong$ fundamental frequency and amplitude $\geq 3\% V_{ol}$

Voltage control of single-phase full-bridge inverters

Amplitude modulation index	$M_a = \frac{A_r}{A_c}$ A_r = Reference signal amplitude A_c = Carrier signal amplitude
Frequency modulation ratio	$M_f = \frac{f_c}{f_r}$ f_r = Reference signal frequency f_c = Carrier signal frequency

3-Phase Inverters-180-Degree Conduction with Y-connected load

Total RMS line-to-line output voltage	$V_{LL} = \sqrt{\frac{2}{3}} V_s = 0.8165 V_s$
RMS n^{th} component of the line voltage	$V_{LL,n} = \frac{4V_s}{n\pi\sqrt{2}} \cos\left(\frac{n\pi}{6}\right)$
Instantaneous line-to-line output voltage	$V_{ab} = \sum_{n=1,2,\dots}^{\infty} \frac{4V_s}{n\pi} \cos\left(\frac{n\pi}{6}\right) \sin n\theta$ $\theta = \omega t$ V_{bc} and V_{ca} is phase shifted by $\frac{2\pi}{3}$, and $\frac{4\pi}{3}$, and $\frac{4\pi}{3}$ $V_{bc} = \sum_{n=1,2,\dots}^{\infty} \frac{4V_s}{n\pi} \cos\left(\frac{n\pi}{6}\right) \sin n\left(\omega t - \frac{2\pi}{3}\right)$ $V_{ba} = \sum_{n=1,2,\dots}^{\infty} \frac{4V_s}{n\pi} \cos\left(\frac{n\pi}{6}\right) \sin n\left(\omega t - \frac{4\pi}{3}\right)$
Instantaneous phase output voltage	$V_{an} = \sum_{n=1,2,\dots}^{\infty} \frac{4V_s}{n\pi\sqrt{3}} \cos\left(\frac{n\pi}{6}\right) \sin n(\omega t)$ V_{bn} and V_{cn} is phase shifted by $\frac{2\pi}{3}$, and $\frac{4\pi}{3}$ $V_{bn} = \sum_{n=1,2,\dots}^{\infty} \frac{4V_s}{n\pi\sqrt{3}} \cos\left(\frac{n\pi}{6}\right) \sin n\left(\omega t - \frac{2\pi}{3}\right)$ $V_{cn} = \sum_{n=1,2,\dots}^{\infty} \frac{4V_s}{n\pi\sqrt{3}} \cos\left(\frac{n\pi}{6}\right) \sin n\left(\omega t - \frac{4\pi}{3}\right)$
The instantaneous line (phase) currents for Resistive Load	$I_a = \sum_{n=1,2,\dots}^{\infty} \frac{4V_s}{n\pi R\sqrt{3}} \cos\left(\frac{n\pi}{6}\right) \sin n(\omega t)$ $I_b = \sum_{n=1,2,\dots}^{\infty} \frac{4V_s}{n\pi R\sqrt{3}} \cos\left(\frac{n\pi}{6}\right) \sin n\left(\omega t - \frac{2\pi}{3}\right)$ $I_c = \sum_{n=1,2,\dots}^{\infty} \frac{4V_s}{n\pi R\sqrt{3}} \cos\left(\frac{n\pi}{6}\right) \sin n\left(\omega t - \frac{4\pi}{3}\right)$
The instantaneous line (phase) currents for RL Load	$I_a = \sum_{n=1,2,\dots}^{\infty} \frac{4V_s}{n\pi Z\sqrt{3}} \cos\left(\frac{n\pi}{6}\right) \sin n(\omega t - \theta_n)$ $I_b = \sum_{n=1,2,\dots}^{\infty} \frac{4V_s}{n\pi Z\sqrt{3}} \cos\left(\frac{n\pi}{6}\right) \sin n\left(\omega t - \frac{2\pi}{3} - \theta_n\right)$ $I_c = \sum_{n=1,2,\dots}^{\infty} \frac{4V_s}{n\pi Z\sqrt{3}} \cos\left(\frac{n\pi}{6}\right) \sin n\left(\omega t - \frac{4\pi}{3} - \theta_n\right)$ where $Z = r + j\omega L$; $\theta_n = \tan^{-1}\left(\frac{\omega L}{R}\right)$

MOST AFFORDABLE
QUALITY CONTENT, INDIA'S BEST FACULTY TEAM

• GATE 2025 RESULTS •

RESULTS HON TOH KAISE – PW JAISE!

RANK 1 Branch ECE  SWARNA BISWAS BATCH – PARAKRAM	RANK 1 Branch IN  KAILASH GOYAL BATCH – SHRESHTH	RANK 2 Branch EE  KAILASH GOYAL BATCH – SHRESHTH	RANK 2 Branch PI  PRAKHAR ADARSH BATCH – VIJAY
RANK 3 Branch ME  SANKET TUPKAR BATCH – SHRESHTH	RANK 3 Branch XE  ROHAN KUMAR BISWAL BATCH – SUPER 1500	RANK 4 Branch CE  HARSHIL MAHESHWARI BATCH – PARAKRAM & VIJAY GATE TEST SERIES	RANK 5 Branch ME  SATYAKI DATTA BATCH – PARAKRAM
RANK 5 Branch XE  SOUMAJIT NATH BATCH – VIJAY	RANK 7 Branch XE  SANKET TUPKAR BATCH – SHRESHTH	RANK 7 Branch DA  HARSHIT BISHT GATE TEST SERIES	RANK 7 Branch CH  YUG NISHCHAY BATCH – PARAKRAM
RANK 8 Branch XE  ROHIT KUMAR BATCH – PARAKRAM	RANK 8 Branch CSE  RISHI SHARMA BATCH – SUPER 1500	RANK 10 Branch ME  ASHUTOSH BATCH – GATE TEST SERIES	

& many more.....

ELECTRICAL & ELECTRONICS MEASUREMENT

Basic Instruments

Arithmetic Mean

$$\bar{x} = \frac{x_1 + x_2 + x_3 + x_4 + \dots + x_n}{n} = \frac{\sum x_i}{n}$$

Average Deviation

$$D = \frac{|d_1| + |d_2| + |d_3| + \dots + |d_n|}{n} = \frac{\sum |d_i|}{n}$$

Standard Deviation

$$\sigma = \sqrt{\frac{d_1^2 + d_2^2 + d_3^2 + \dots + d_n^2}{n-1}} = \sqrt{\frac{\sum d_i^2}{n-1}}$$

Relative Static Error

$$\epsilon_r = \frac{\text{absolute error}}{\text{true value}} = \frac{\delta A}{A_t} = \frac{\epsilon_0}{A_t}$$

Percentage Limiting Error

$$= \frac{A_m - A_t}{A_t} \times 100$$

Basic Instruments

PMMC Instrument

It works on D.C. only

$$T_d = BINA, T_c = k_c \theta$$

$$\text{Frictional error} \propto \frac{\text{Weight}}{\text{Torque}}$$

For extension of range of PMMC Ammeter,

$$\text{Shunt Resistance, } R_{sh} = \frac{R_m}{m-1}, m = \frac{I_{ext}}{I}$$

For extension of range at PMMC voltmeter,

$$\text{Series Resistance } R_{se} = R_m(m-1), m = \frac{V_{ext}}{V}$$

Internal resistance of ideal ammeter is zero and that of ideal voltmeter is infinity.

Moving Iron Type Instrument

$$T_d = \frac{1}{2} I^2 \frac{dL}{d\theta}, \theta = \frac{1}{2k_c} I^2 \frac{dL}{d\theta}$$

$$T_d \propto I^2$$

It works for both A.C. and D.C.

Electromagnet Moving Coil Type Instrument

It works on the change in mutual inductance

$$\tau_d = |I_{FC}| \cdot |I_{MC}| \cdot \cos \angle I_{FC} \text{ \& } I_{MC} \cdot \frac{dM}{d\theta}$$

Electrostatic Voltmeter

It works on change in capacitance.

1. Linear motion type

$$F_d = \frac{1}{2} V^2 \frac{dC}{dx} \quad x = \frac{1}{2} \frac{V^2}{K_c} \cdot \frac{dC}{dx}$$

2. Circular motion type

$$\tau_d = \frac{1}{2} V^2 \frac{dC}{d\theta} \quad \tau_c = K_c \theta$$

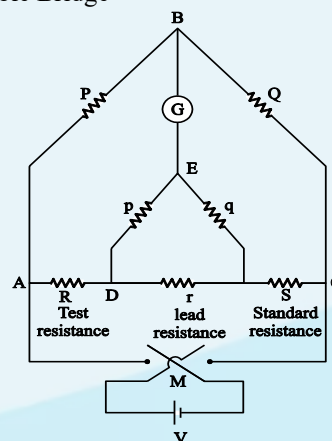
Measurement of Resistance and AC Bridges

Measurement of Resistance

Low resistance can be measured by:

1. Potentiometer method.
2. Ammeter voltage method.
3. Kelvin's Double bridge method.

Kelvin's Double Bridge

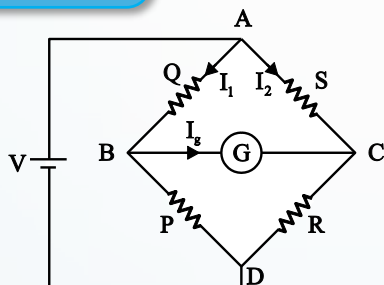


$$R = \frac{PS}{Q} + \frac{qr}{p+q+r} \left[\frac{P}{Q} - \frac{P}{q} \right]$$

Medium Resistance can be measured by

1. Wheatstone bridge method
2. Substitution method
3. Voltmeter-Ammeter method
4. Ohmmeter method

Wheatstone Bridge



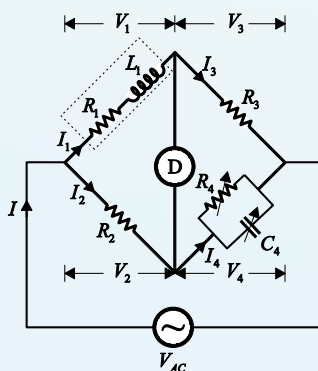
$$R = \frac{PS}{Q}$$

High resistance can be measure by

1. Direct deflection method
2. Los of charge method
3. Megger
4. Megohm bridge method

AC Bridges

1. Maxwell (L-C) bridge

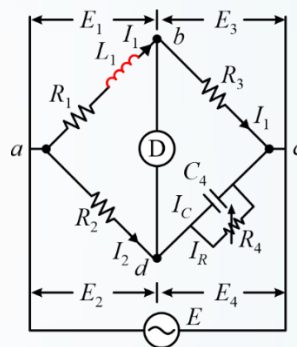


$$R_1 = \frac{R_2 R_3}{R_4}$$

$$L_1 = R_2 R_3 C_4$$

$$Q = \omega C_4 R_4$$

2. Maxwell Inductance Capacitance Bridge:

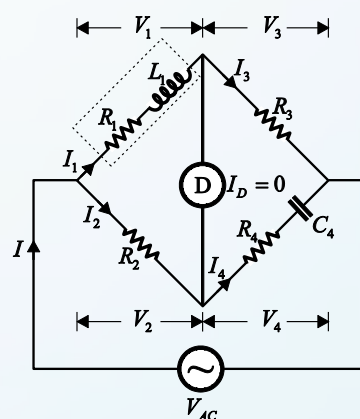


Maxwell's Inductance Capacitance Bridge

$$R_1 = \frac{R_2 R_3}{R_4}, Q = \frac{\omega L_1}{R_1} = \omega C_4 R_4$$

$$L_1 = R_2 R_3 C_4$$

3. Hay's bridge

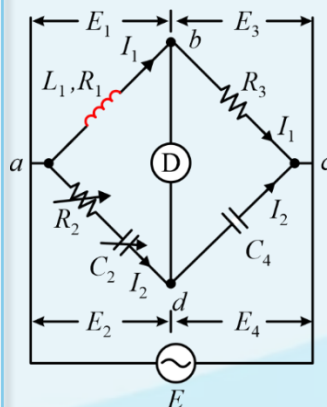


$$R_1 = \frac{\omega^2 R_2 R_3 R_4 C_4^2}{1 + \omega^2 R_4^2 C_4^2}$$

$$L_1 = \frac{R_2 R_3 C_4}{1 + \omega^2 R_4^2 C_4^2}$$

$$Q = \frac{1}{\omega C_4 R_4}$$

4. Owen's bridge

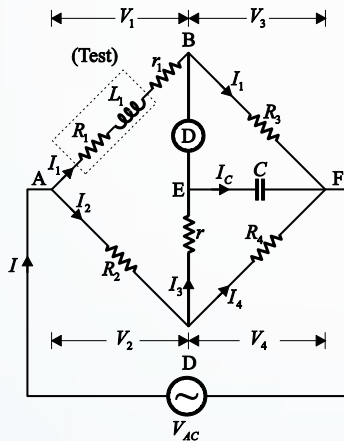


$$\Rightarrow L_1 = R_2 R_3 C_4$$

$$\frac{-R_1}{\omega C_4} = \frac{-R_3}{\omega C_2}$$

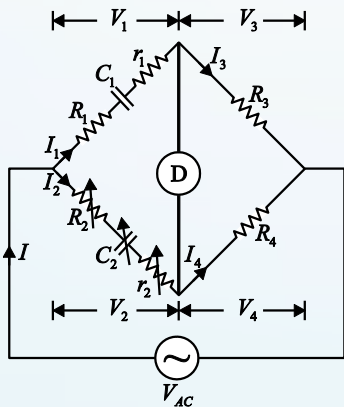
$$\Rightarrow R_1 = R_3 \frac{C_4}{C_2}$$

5. Anderson Bridge



$$L_1 = R_2 R_3 C_4, R_1 = \frac{R_3 C_4}{C_2}$$

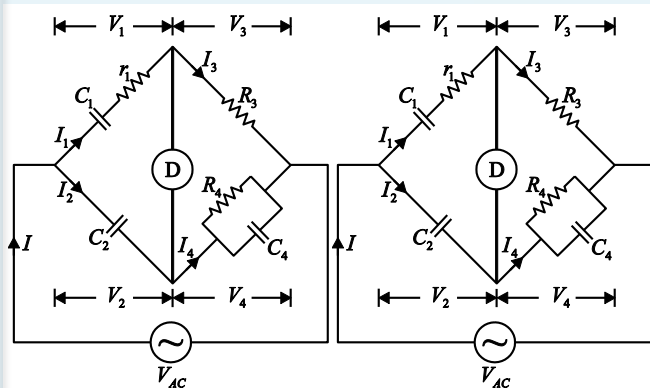
6. De-Sauty's Bridge



$$r_1 = \frac{R_3(R_2 + r_2) - R_1}{R_4}$$

$$C_1 = \frac{R_4 C_2}{R_3}$$

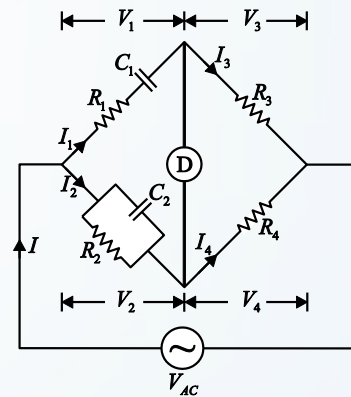
7. Schering Bridge



$$C_1 = \frac{R_4 C_2}{R_3}$$

$$r_1 = \frac{R_3 C_4}{C_2}$$

8. Wein's Bridge



$$\omega = \frac{1}{\sqrt{R_1 R_2 C_1 C_2}}$$

$$f = \frac{1}{2\pi \sqrt{R_1 R_2 C_1 C_2}}$$

$$\frac{R_1}{R_2} + \frac{C_2}{C_1} = \frac{R_3}{C_4}$$

Measurement of Energy and Power

Single-Phase Energy Meter

- (i) Energy meter constant

$$(K) = \frac{\text{Number of revolutions made by disc}}{\text{Energy recorded in KWhr}} = \frac{N}{Pt}$$

- (ii) Measured energy (W_m) = $\frac{\text{Total number of revolution}}{K}$

- (iii) True energy (W_T) = $\frac{VI \cos \phi}{1000} \times \frac{t}{3600}$ KWhr

Where, V = Voltage and I = Current.

- (iv) % Error

$$(\varepsilon_r) = \frac{\text{Measured energy} - \text{True energy}}{\text{True energy}} \times 100\%$$

$\varepsilon_r = +ve$ means meter is running fast, $\varepsilon_r = -ve$ means meter is running slow.

Measurement of DC Power:

$$V-A \text{ method} - \% \text{Error} = \frac{P_m - P_T}{P_T} \times 100$$

$$= \frac{I_L^2 R_a}{V_L I_L} \times 100 = \frac{R_a}{R_L} \times 100$$

$$A-V \text{ method} - \% \text{Error} = \frac{V_L^2 / R_V}{V_L \times X_L} \times 100 = \frac{R_L}{R_V} \times 100$$

$$\text{Measurement of AC Power } P_{avg.} = V_{pc} I_{cc} \cos \phi$$

For signals with harmonics,

$$P_{avg} = V_0 I_0 + \frac{1}{2} \left[\sum_{k=1}^n V_k I_k \cos(\phi_k - \phi'_k) \right]$$

Two Wattmeter Method

$$P_1 = \sqrt{3} V_{ph} I_{ph} \cos(30^\circ \mp \phi); P_2 = \sqrt{3} V_{ph} I_{ph} \cos(30^\circ \pm \phi)$$

$$P_{3\phi} = P_1 + P_2 = \sqrt{3} V_L I_L \cos \phi = 3 V_{ph} I_{ph} \cos \phi$$

$$P_1 - P_2 = \sqrt{3} V_{ph} I_{ph} \sin \phi = Q_{3\phi} / \sqrt{3} \quad \mp \pm \text{lag load}$$

$$P.F. = \cos \left[\tan^{-1} \left(\frac{\sqrt{3}(P_1 - P_2)}{P_1 + P_2} \right) \right]$$

CRO & Electronic Measurement

Single-Phase Energy Meter

CRO is used for display, measurement and analysis of waveform in electrical and electronic circuit.

Cathode Ray Tube (CRT) is the heart of CRO.

Lissajous pattern is used for measurement of phase between two sinusoidal signals on CRO screen.

Phase Angle		Lissajous Pattern	
1.	0°, 180° 360°	1.	Straight line
2.	90°, 270°	2.	Circle
3.	0° < φ 180° but ≠ 90°	3.	Ellipse
4.	180° ≤ φ < 360° but ≠ 270°	4.	Ellipse

$$\frac{f_y}{f_x} = \frac{\text{Maximum number of horizontal tangencies}}{\text{Maximum number of vertical tangencies}}$$

Digital Voltmeter

Resolution $R = R = \frac{V_{\max}}{10^N}$ or $R = \frac{I_{\max}}{10^N}$ where, N is the number of full digits.

Instrument Transformers

Instrument transformers are used for high voltage and high current measurement.

Current Transformer

$$\text{Turns ratios} = (n) = \frac{N_2}{N_1}, \text{ nominal ratio} = \frac{I_P}{I_S}$$

$$\text{Actual/Transformation ratio} = (R) = \frac{I_P \text{ actual}}{I_S \text{ actual}}$$

$$\text{Secondary burden angle} \Rightarrow (\delta) = \tan^{-1} \left(\frac{X_e + X_s}{R_e + R_s} \right)$$

$$R = \frac{I_w \cos \delta + I_m \sin \delta}{I_S} + n$$

not valid for $\delta = 0$ because some approximation is made

$$\text{Ratio correlation factor} = \frac{R}{K}$$

$$\text{Ratio error} = \frac{K - R}{R} \times 100\%$$

$$\text{Phase angle error} = (\theta) = \frac{I_m \cos S - I_w \sin S}{n I_S} \times \frac{180}{\pi} \text{ degree}$$

Potential Transformer

$$\text{Turns ratio} = (n) = \frac{N_1}{N_2}$$

$$\text{Nominal ratio} (k) = \frac{V_P}{V_S} = \frac{V_1}{V_2}$$

$$\text{Actual transformation ratio} = (R) = \frac{V_P}{V_S} = \frac{V_1}{V_2} \rightarrow \text{T.V.}$$

$$\text{Secondary burden angle} = (\Delta) = \tan^{-1} \left(\frac{X_1}{R_1} \right)$$

$$R = \frac{\overline{V_P}}{V_S} = n + \frac{V I_S (R_{2e} \cos \Delta + X_{2e} \sin \Delta) + I_w R_p + I_u \times p}{V_S}$$

$$R_{2e} = R_S + \frac{R_P}{n^2}$$

$$Z_{2e} = X_S + \frac{X_P}{n^2}$$

Phase angle error

$$(\theta) = \frac{I_S}{V_S} [X_{2e} \cos \Delta - R_{2e} \sin \Delta] + \frac{I_w p r \times p}{n V_S} \text{ (rad)}$$



COMMUNICATION SYSTEM

Unit - I Amplitude Modulation

Unmodulated carrier $v_c(t) = A_C \cos \omega_c t$

Modulating/Baseband signal $= x(t)$

Expression of AM wave in time & frequency domains:

$$\begin{aligned} x_{AM}(t) &= [A_C + x(t)] \cos \omega_c t \\ &= A_C \cos \omega_c t + x(t) \cos \omega_c t \\ x_{AM}(\omega) &= A_C \pi [\delta(\omega - \omega_c) + \delta(\omega + \omega_c)] \\ &\quad + \frac{1}{2} [X(\omega - \omega_c) + X(\omega + \omega_c)] \end{aligned}$$

Sinusoidal AM

$x(t) = A_m \cos \omega_m t$, $\omega_m \ll \omega_c$

$$\begin{aligned} x_{AM}(t) &= [A_C + A_m \cos \omega_m t] \cos \omega_c t \\ &= A_C [1 + m \cos \omega_m t] \cos \omega_c t \end{aligned}$$

Where m = Modulation index $= \frac{A_m}{A_C} = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}}$

$$\Rightarrow x_{AM}(t) = A_C \cos \omega_c t + \frac{mA_C}{2} [\cos(\omega_c - \omega_m)t + \cos(\omega_c + \omega_m)t]$$

$\Rightarrow$ Upper side band frequency or maximum frequency component $= f_c + f_m$

$\Rightarrow$ Lower side band frequency or minimum frequency component $= f_c - f_m$

$$\text{Amplitude of side bands} = \frac{mA_C}{2}$$

$$\text{Bandwidth} = (f_c + f_m) - (f_c - f_m) = 2f_m$$

Power Relations in AM

Total transmitted power

$$P_T = \frac{A_C^2}{2} + \frac{\left(\frac{mA_C}{2}\right)^2}{2} + \frac{\left(\frac{mA_C}{2}\right)^2}{2} = \frac{A_C^2}{2} \left[1 + \frac{m^2}{2}\right]$$

$$\Rightarrow P_T = P_C \left[1 + \frac{m^2}{2}\right]$$

Total power in both sidebands

$$P_{SB} = P_C \cdot \frac{m^2}{2}$$

Carrier power if antenna is represented by a resistive load of $R\Omega$

$$P_C = \frac{A_C^2}{2R}$$

Transmission Efficiency

$$\eta = \frac{\text{useful transmitted power}}{\text{total transmitted power}} = \frac{P_{SB}}{P_T}$$

$$\text{For full AM } \% \eta = \frac{m^2}{2 + m^2} \times 100\%$$

Current relation in AM

$$P_T = P_C \left(1 + \frac{m^2}{2}\right)$$

$$I_T^2 \cdot R = I_C^2 \cdot R \left(1 + \frac{m^2}{2}\right)$$

$$\frac{I_T}{I_C} = \sqrt{1 + \frac{m^2}{2}}$$

Modulation by Several sine Wave Overall Modulation Index

$$P_T = P_C \left[1 + \frac{m_1^2}{2} + \frac{m_2^2}{2} + \dots\right]$$

$$m_T = \sqrt{m_1^2 + m_2^2 + m_3^2 + \dots}$$

Square law modulator output

$$V_o(t) = A_1 V_i(t) + A_1 V_i^2(t), V_i(t) = x(t) + V_c(t)$$

$$\text{Desired part of } V_o(t) = A_1 V_c(t) [1 + K_a x(t)] \cos \omega_c t$$

$$\text{Where } K_a = \frac{2A_2}{A_1} = \text{Modulation Sensitivity}$$

Double Side band Suppressed Carrier (DSB-SC)

Carrier $= V_c(t) = V_c \cos \omega_c t$, Modulating signal

$$= x(t) = A_m \cos \omega_m t$$

$$x_{DSB}(t) = v_c \cdot x(t) \cos \omega_c t$$

$$x_{DSB}(\omega) = \frac{v_c}{2} [x(\omega - \omega_c) + x(\omega + \omega_c)]$$

$$\text{Bandwidth} = 2f_m$$

$$\text{Total transmitted power} = P_{DSB-SC} = P_{LSB} + P_{USB}$$

$$P_{LSB} = P_{USB} = \frac{m^2 V_c^2}{8R}$$

$$\% \eta = \frac{2}{2 + m^2} \times 100\%$$

Single Side band with Suppressed Carrier (SSB-SC) Modulation

Hilbert Transform $\hat{x}(t) = \frac{1}{\pi t} * x(t)$

$$\hat{x}(\omega) = -j \operatorname{sgn}(\omega) X(\omega)$$

$$x_{SSB}(t) = x(t) C \cos \omega_c t \pm \hat{x}(t) \sin \omega_c t$$

+ve sign $\Rightarrow$ For LSB

-ve sign $\Rightarrow$ For USB

$$\boxed{\text{Bandwidth} = f_m} \quad \boxed{\% \eta = \frac{4+m^2}{4+2m^2} \times 100}$$

- For synchronous demodulation of AM wave, a locally generated carrier signal, which is having its phase exactly equal to the received carrier signal phase is multiplied with the received signal.
- Envelope detector / diode detector is used for asynchronous demodulation. It is used only for $m < 1$.
- Condition to avoid diagonal clipping

$$\frac{1}{f_m} \gg R.C \gg \frac{1}{f_c} \quad R.C \leq \frac{\sqrt{1-m^2}}{m \omega_m}$$

AM Receivers

- Tuned radio frequency (TRF) receivers suffer from the disadvantages of variable bandwidth and instability.
- Superheterodyne receivers use frequency mixing to convert RF to IF.
- Frequency of image signal

$$f_i = f_{LO} + I.F = f_s + I.F + I.F$$

$$\boxed{f_i = f_s + 2I.F.}$$

- The rejection of image frequency by a single tuned circuit is given by

$$\boxed{\alpha = \sqrt{1+Q^2 \rho^2}}$$

- Where Q = Quality factor of tuned circuit

$$\boxed{\rho = \frac{f_{si}}{f_s} - \frac{f_s}{f_{si}}}$$

- In a SHD receiver, the image channel selectivity is determined by the preselectors and RF stages only. Image rejection should be achieved before I.F. stage.

Unit - II Angle Modulation

Unmodulated Carrier $V_c(t) = A \cos \omega_c t = A \cos \theta(t)$

$\theta(t)$ = phase of unmodulated carrier.

$$\omega_c = \frac{d}{dt} \theta(t) = \text{frequency of unmodulated carrier}$$

Frequency Modulation:-

Modulating signal $= m(t)$

Instantaneous frequency of unmodulated carrier

$$\boxed{\omega_i = \omega_c + K_f m(t)}$$

K_f = Frequency sensitivity (rad/sec/volt)

$$x_{FM}(t) = A \cos \theta_i(t) \quad \text{where } \theta_i(t) = \int_0^t \omega_i dt$$

$$\therefore \boxed{x_{FM}(t) = A \cos \left[\omega_c t + K_f \int_0^t m(t) dt \right]}$$

Instantaneous phase of modulated carrier

$$\boxed{\theta_i(t) = \omega_c t + K_f \int_0^t m(t) dt}$$

For sinusoidal FM, $m(t) = A_m \cos \omega_m t$

$$\omega_i(t) = \omega_c + K_f A_m \cos \omega_m t$$

$$\boxed{\Delta \omega = K_f A_m} \Rightarrow \text{maximum frequency deviation}$$

$$\theta_i(t) = \int_0^t \omega_i(t) dt = \omega_c t + \frac{\Delta \omega}{\omega_m} \sin \omega_m t$$

$$\boxed{\beta = \frac{\Delta \omega}{\omega_m} = \text{Modulation Index}}$$

$$\therefore \boxed{x_{FM}(t) = A \cos \theta_i(t) = A \cos [\omega_c t + \beta \sin \omega_m t]} \quad \dots (i)$$

Narrow band Frequency modulation (NBFM)

- β is very small.

From equation (i)

$$x_{FM}(t) = A \cos \omega_c t \cos(\beta \sin \omega t)$$

$$-A \sin \omega_c t \sin(\beta \sin \omega t)$$

As $\beta \ll 1$, hence

$$x_{NBFM}(t) = A \cos \omega_c t - \beta A \sin \omega_c t \sin \omega_m t$$

$$\boxed{x_{NBFM}(t) = A \cos \omega_c t - \frac{A\beta}{2} \cos(\omega_c - \omega_m)t + \frac{A\beta}{2} \cos(\omega_c + \omega_m)t}$$

- B.W. requirement $= 2\omega_m \text{ rad/s} = 2f_m \text{ Hz}$
- In NBFM, carrier and USB are in phase and LSB is out of phase.

Wide Band Frequency Modulation (WBFM)

- $\beta \gg 1$, $\cos(\beta \sin \omega_m t)$ & $\sin(\beta \sin \omega_m t)$ are now periodic signals and given in terms of Bessel's functions of 1st kind as:

$$\cos(\beta \sin \omega_m t) = J_0(\beta) + 2J_2(\beta) \cos^2 \omega_m t + 2J_4(\beta) \cos^4 \omega_m t + \dots + 2J_{2n}(\beta) \cos^{2n} \omega_m t + \dots$$

$$\sin(\beta \sin \omega_m t) = 2J_1(\beta) \sin \omega_m t + 2J_3(\beta) \sin^3 \omega_m t + \dots + 2J_{2n-1}(\beta) \sin(2n-1) \omega_m t + \dots$$

$$x_{WBFM}(t) = A_c \left[J_0(\beta) \cos \omega_c t - J_1(\beta) \{ \cos(\omega_c - \omega_m)t - \cos(\omega_c + \omega_m)t \} \right. \\ \left. + J_2(\beta) \{ \cos(\omega_c - 2\omega_m)t + \cos(\omega_c + 2\omega_m)t \} \right. \\ \left. - J_3(\beta) \{ \cos(\omega_c - 3\omega_m)t - \cos(\omega_c + 3\omega_m)t \} + \dots \right]$$

- For $\beta = 2.4, 5.5, 8.6, 11.8 \dots$, $J_0(\beta) = 0$ hence carrier can be suppressed.
- β determines how many sidebands have significant amplitudes, hence B.W. of WBFM depends on β .
- Carson's Bandwidth Formula:

$$B.W. = 2(\beta + 1)\omega_m = 2(\Delta\omega + \omega_m)$$

Phase Modulation

- Unmodulated carrier

$$V_c(t) = A \cos \omega_c t = A \cos \theta(t)$$

Modulating signal = $m(t)$

- Instantaneous phase of modulated carrier

$$\theta_i(t) = \omega_c t + K_p m(t)$$

K_p = phase constant (rad/volt)

$$x_{PM}(t) = A \cos [\omega_c t + K_p m(t)]$$

Inter-relation between $x_{FM}(t)$ & $x_{PM}(t)$

- $m(t) \rightarrow \frac{d}{dt} \rightarrow \text{F.M.} \rightarrow \text{Phase modulation Wave}$
- $m(t) \rightarrow \int \rightarrow \text{P.M.} \rightarrow \text{Frequency Modulated Wave}$

Sinusoidal PM

- $m(t) = A_m \cos \omega_m t$
- $\theta_i(t) = \omega_c t + K_p A_m \cos \omega_m t$
Phase deviation = $\phi_p = K_p A_m$
- $x_{PM}(t) = A \cos [\omega_c t + \phi_p \cos \omega_m t]$
- In phase modulation, bandwidth increases with modulating frequency.
- At the minimum of the modulating signal, the phase of the carrier is delayed by maximum amount.
- Pre-emphasis improves SNR for high frequency signals before transmission.
- De-emphasis restores the original signal at receiver.
- Armstrong method is the indirect method of generation of FM wave.
- Phase detector, low pass filter, DC amplifiers and voltage controlled oscillators are basic parts of phase-locked-loop.
- FM is more immune to noise compared to AM.

Unit - III Random Variables & Noise

- A random variable is a function whose domain is set S of all experimental outcomes and the range is subset of R (set of real numbers).

Cumulative Distribution Function (CDF):

CDF of R.V. X is a function

$$F_X(x) = P\{X \leq x\}$$

defined for every 'x' from $-\infty$ to $+\infty$.

Properties of CDF:

- $0 \leq F_X(x) \leq 1$
 - $F_X(\infty) = 1$ $F_X(-\infty) = 0$
 - If $x_2 > x_1$ then $F_X(x_2) \geq F_X(x_1)$
 $\Rightarrow$ CDF is a non-decreasing function.
 - If $F_X(x_0) = 0$
then $F_X(x) = 0$ for all $x \leq x_0$.
 - If $F_X(x^+) = P[X \leq x + \epsilon]$, $0 < \epsilon \rightarrow 0$
 $F_X(x^-) = P[X \leq x - \epsilon]$
CDF is continuous from right, i.e.
 $F_X(x^+) = F_X(x)$
 - $P\{x_1 < x \leq x_2\} = F_X(x_2) - F_X(x_1)$
 - $P\{X = x\} = F_X(x) - F_X(x^-)$
 - $P\{x_1 \leq x \leq x_2\} = F_X(x_2) - F_X(x_1^-)$
- If $F_X(x_0) - F_X(x_0^-) > 0$
then R.V. has discontinuity.
 - $P(X > x) = 1 - F_X(x)$

Probability Density Function (PDF):-

PDF is defined in terms of CDF as

$$f_X(x) \triangleq \frac{d}{dx} F_X(x)$$

$$f_X(x) = \lim_{\Delta x \rightarrow 0} \frac{F_X(x + \Delta x) - F_X(x)}{\Delta x} \geq 0$$

- $F_X(x) = \int_{-\infty}^x f_X(\tau) d\tau$
- PDF is a continuous function of x for continuous R.V. and PDF of a discrete R.V. can have the general expression

$$f_X(x) = \sum_{i=1}^N P_i \delta(x - x_i)$$

Properties of PDF:

- $f_X(x) \triangleq \frac{d}{dx} F_X(x)$
- $f_X(x) \geq 0$

$$(iii) F_X(x) = \int_{-\infty}^x f_X(\tau) d\tau$$

$$(iv) F_X(\infty) = \int_{-\infty}^{\infty} f_X(\tau) d\tau = \text{Area under PDF} = 1$$

$$(v) P\{x_1 < X \leq x_2\} = \int_{x_1}^{x_2} f_X(\tau) d\tau [= F_X(x_2) - F_X(x_1)]$$

(vi) If R.V. is continuous, then

$$P\{x_1 < X \leq x_2\} = P\{x_1 \leq X \leq x_2\} = \int_{x_1}^{x_2} f_X(\tau) d\tau$$

For discrete R.V.

$$P\{x_1 \leq X \leq x_2\} = F_X(x_2) - F_X(x_1)$$

Statistical Averages:-

(i) First moment or mean value:

$$m_X = E[X] = \int_{-\infty}^{\infty} x \cdot f_X(x) dx$$

(ii) Second moment or mean square value:

$$E[X^2] = \int_{-\infty}^{\infty} x^2 \cdot f_X(x) dx$$

$$(iii) n^{\text{th}} \text{ moment } E[X^n] = \int_{-\infty}^{\infty} x^n \cdot f_X(x) dx$$

$$(iv) \text{ Variance } \sigma_X^2 = E[(X - m_X)^2] = E[X^2] - m_X^2$$

$$\bullet \text{ Standard deviation } \sigma_X = \sqrt{\text{var}(X)}$$

Multiple Random Variables:-

$$\bullet F_{X,Y}(x, y) = P[X \leq x, Y \leq y]$$

$$\bullet f_{X,Y}(x, y) = \frac{\partial^2}{\partial x \partial y} F_{X,Y}(x, y)$$

$$\bullet \text{Cov}(X, Y) = E[XY] - E[X] \cdot E[Y]$$

$$\bullet \text{Correlation coefficient } \rho_{X,Y} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

$\bullet$ If $\rho_{X,Y} = 0$ then X & Y are uncorrelated.

Random Processes:-

$\bullet$ A collection of time functions $\{X(t, \omega)\}$, indexed by t , defined over sample space ω .

$\bullet$ **Strictly stationary R.P.:** If $X(t)$ & $X(t - \alpha)$ have the same statistical properties, i.e. mean, PDF, variance, auto correlation functions etc. for all α .

$\bullet$ **Wide sense stationary R.P.:** If only mean and correlation are stationary, i.e. mean is constant & ACF is independent of time.

EX - Noise

Auto Correlation Function (ACF):

$$R_X(t_1, t_2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_1 \cdot x_2 f_{X_1, X_2}(x_1, x_2) dx_1 dx_2$$

For a stationary R.P.

$$R_X(t_1, t_2) = R_X(t_2 - t_1) = R_X(\tau)$$

$$R_X(\tau) = E[X(t) \cdot X(t + \tau)]$$

$$R_X(\tau) = \int_{-\infty}^{\infty} [X(t) \cdot X(t + \tau)] f_X(x) dx$$

Cross Correlation:

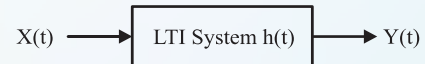
$$R_{XY}(\tau) = E[X(t) Y(t + \tau)]$$

$$\bullet R_X(0) \geq |R_X(\tau)| \quad R_X(0) = \text{Power}$$

$$\bullet R_X(-\tau) = R_X(\tau)$$

Linear Systems & Random Processes:

Wide sense stationary R.P.



$$\bullet \text{Output mean: } \mu_Y = \mu_X \cdot \int_{-\infty}^{\infty} h(t) dt$$

$$\bullet \text{Output ACF: } R_Y(\tau) = R_X(\tau) * h(\tau) * h(-\tau)$$

$$\bullet \text{Output PSD: } S_Y(f) = |H(f)|^2 \cdot S_X(f)$$

Relation between Power Spectral Density (PSD) & Autocorrelation Function (ACF):

$$S_X(f) = \text{Fourier Transform of } [R_X(\tau)]$$

$$\bullet S_X(0) = \int_{-\infty}^{\infty} R_X(\tau) d\tau$$

$$\bullet R_X(0) = \int_{-\infty}^{\infty} S_X(f) df$$

$\bullet$ PSD is always non-negative and an even function of frequency.

Different types of distributions:

Distribution	PDF	Mean	Variance
Uniform	$f_X(x) = \frac{1}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Normal (Gaussian)	$f_X(x) = \frac{1}{\sqrt{2\pi\sigma_X^2}} e^{-\frac{(x-m_X)^2}{2\sigma_X^2}}$	m_X	σ_X^2
Poisson	$f_X(x) = P(X=k) = \frac{e^{-\lambda} \cdot \lambda^k}{k!}$	λ	λ
Binomial	$f_X(x) = P(X=k) = {}^n C_k p^k (1-p)^{n-k}$	p	$p(1-p)$

Noise Performance:

$$\text{Figure merit } FOM = \frac{(SNR)_{o/p}}{(SNR)_{i/p}}$$

MODULATION SCHEME	FOM
DSB – SC	2
SSB – SC	1
FULL AM	$\frac{2k_a^2 \overline{m^2(t)}}{1 + k_a^2 \overline{m^2(t)}}$
FM	$\frac{3}{2} \beta^2$

THERMAL NOISE:

$$P_n = KTB$$

K = Boltzman Constant

T = Temperature in Kelvin

B = Bandwidth in Hz

$$V_n = \sqrt{4KTB R}$$

Equivalent Noise Temperature:

$$T_{eq} = T_0 (f - 1)$$

$T_0 = 290 \text{ K}$ (if not given)

f = Noise Figure

Effective Noise Temperature:

$$T_e = T_{e1} + \frac{T_{e2}}{G_1} + \frac{T_{e3}}{G_1 \cdot G_2}$$

Unit – IV Baseband Pulse Modulation

Pulse Code Modulation (PCM):

$$\text{Step size } s = \frac{V_{\max} - V_{\min}}{m}$$

Where m = no. of quantization levels = 2^n

$$\text{For } n\text{-bit PCM system } 2^n \geq m$$

$$\text{Bit rate } R_b = n \cdot f_s$$

$$\text{Quantization noise is uniformly distributed between } \pm \frac{s}{2}$$

$$\text{Quantization noise power } P_{nq} = \frac{s^2}{12}$$

Signal to quantization noise power ratio

$$SN_{qR} = \frac{3}{2} m^2 = \frac{3}{2} (4)^n = \frac{3}{2} (4)^{B/f_m}$$

$$(SN_{qR})_{dB} = 1.761 + 6.02^n$$

SN_{qR} gets improved by 6dB for every 1 bit increment.

$$\text{Bandwidth } B = \frac{\text{Bit rate}}{2} = \frac{n f_s}{2}$$

Delta modulation:

Also known as one bit PCM system

To avoid slope overload

$$s f_s \geq 2\pi f_m A_m$$

$$A_m \leq \frac{s f_s}{2\pi f_m}$$

$$m(t) = A_m \sin \omega_m t = \text{modulating signal}$$

Modulation	SNR	Bandwidth	Complexity
PAM	Minimum	$\sim 2m f_m$ (min)	Minimum
PWM	Moderate	$\sim 10m f_m$ (Moderate)	Moderate
PPM	Maximum	$\sim \frac{1}{t_r}$ (Maximum)	Maximum

Bandwidth required in case of raised cosine pulse having roll off factor ' α ' is given as

$$B.W. = \frac{R_b}{2} (1 + \alpha)$$

Unit – V Digital Bandpass Transmission

- $\operatorname{erfc}(z) = \frac{2}{\sqrt{\pi}} \int_z^{\infty} e^{-x^2} dx$
- $\frac{e^{-x^2}}{\sqrt{\pi} \cdot x} \left(1 - \frac{1}{2x^2} \right) \leq \operatorname{erfc}(x) \leq \frac{e^{-x^2}}{\sqrt{\pi} \cdot x}$
- $\operatorname{erfc}(x) \approx \frac{e^{-x^2}}{\sqrt{\pi} \cdot x}$
- $Q(x) = \frac{1}{2} \operatorname{erfc} \left(\frac{x}{\sqrt{2}} \right)$

Scheme	Transmitted signals	B.ω.	P _e
ASK	$S_1(t) = A \cos \omega_c t$ $S_2(t) = 0$	$2R_b$	$\frac{1}{2} \operatorname{erfc} \left(\sqrt{\frac{E}{4\eta}} \right)$
BPSK	$S_1(t) = A \cos \omega_c t$ $S_2(t) = -A \cos \omega_c t$	$2R_b$	$\frac{1}{2} \operatorname{erfc} \left(\sqrt{\frac{E}{\eta}} \right)$
BFSK	$S_1(t) = \sqrt{\frac{2E}{T_b}} \cos 2\pi f_1 t$ $S_2(t) = \sqrt{\frac{2E}{T_b}} \cos 2\pi f_2 t$	$2R_b + (f_2 - f_1)$	$\frac{1}{2} \operatorname{erfc} \sqrt{\frac{E}{2\eta}}$
QPSK	$S_1(t) = \sqrt{\frac{2E_s}{T_s}} \cos \left[2\pi f_c t + \frac{\pi}{4} \right]$ $S_2(t) = \sqrt{\frac{2E_s}{T_s}} \cos \left[2\pi f_c t + \frac{3\pi}{4} \right]$ $S_3(t) = \sqrt{\frac{2E_s}{T_s}} \cos \left[2\pi f_c t + \frac{5\pi}{4} \right]$ $S_4(t) = \sqrt{\frac{2E_s}{T_s}} \cos \left[2\pi f_c t + \frac{7\pi}{4} \right]$	R_b	$\frac{1}{2} \operatorname{erfc} \sqrt{\frac{E_b}{2\eta}}$

- Bandwidth efficiency

$$\rho = \frac{r_b}{B} \text{ bits/sec/Hz}$$

- For M-ary PSK signals

$$B = \frac{2}{T_S} \text{ where } T_S = T_b \log_2 M$$

For BPSK, $M = 2$, $T_S = T_b$

For QPSK, $M = 4$, $T_S = 2T_b$

$$\rho = \frac{r_b}{B} = \frac{r_b}{\frac{2}{T_S}} = \frac{r_b \cdot T_b \cdot \log_2 M}{2}$$

$$\therefore \rho = \frac{\log_2 M}{2}$$

- For M-ary FSK signal

$$B = \frac{M}{2T_S} \quad T_S = T_b \log_2 M$$

$$\rho = \frac{r_b}{\left(\frac{M}{2T_S}\right)} = \frac{2r_b \cdot T_b \cdot \log_2 M}{M}$$

$$\rho = \frac{2 \log_2 M}{2}$$

- $\text{Baud rate} = \frac{\text{Bit rate}}{\log_2 M}$

Matched filter

$$\text{SNR} = \frac{|S_o(T)|^2}{P_{n_o}} = \frac{2}{\eta} \int_{-\infty}^{\infty} S_i^2(t) dt$$

$$h(t) = S_i^*(T-t)$$

$$P_e = \frac{1}{2} \text{erfc} \sqrt{\frac{E_s}{\eta}}$$

Maximum amplitude of filter output

$$= \frac{A^2 T}{2}$$

Unit-VI Information theory & Coding

$$\text{Information} \propto \frac{1}{\text{Probability}}$$

$$I = \log_2 \frac{1}{P} \text{ bits}$$

$$\text{Entropy } H(X) = \sum_{i=1}^M P(X_i) \log_2 \frac{1}{P(X_i)} \text{ bits / symbol}$$

$$H(X)_{\max} = \log_2 M \text{ bits/symbol}$$

When all messages are equiprobable

$$\text{Information rate } R_B = r \cdot H \text{ bits/sec}$$

Average code word length

$$\bar{L} = \sum_{i=1}^M \eta_i P(\eta_i) = \text{Average no. of bits per message}$$

$$\text{Coding efficiency } \eta = \frac{H(X)}{\bar{L}} \times 100\%$$

Variance of average code word length

$$\sigma^2 = \sum_{k=0}^{m-1} P_k (l_k - \bar{L})^2$$

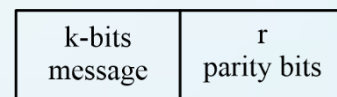
Capacity of a Gaussian channel

$$C = B \log_2 \left(1 + \frac{S}{N} \right) = B \log_2 \left(1 + \frac{S}{\eta_B} \right)$$

Bandwidth – SNR trade off: channel capacity with ∞ bandwidth

$$C = 1.44 \frac{S}{\eta}$$

Linear block codes



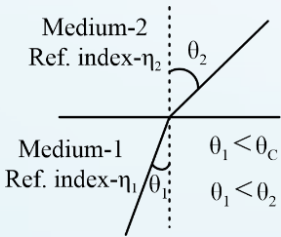
$$\eta = k + r$$

Minimum hamming distance	Error control capacity
1	Error can't be detected
2	Single error detection
3	Single error correction
4	Single error correction + double error detection
5	Double error correction
6	Double error correction + Triple error detection

- $\frac{d-1}{2}$ errors can be corrected if d is odd.

- $\frac{d-2}{2}$ errors can be corrected and $d/2$ errors can be detected if d is even where d = minimum Hamming distance.
- Discrete memoryless channel
 $P(Y) = P(X) \cdot P(Y/X)$
 Marginal probability $\sum_{i=1}^m P(x_i y_i) = P(y_i)$
 Joint probability $P(X, Y) = [P(X)_d [P(Y/X)]]$
- Mutual information
 $I(X; Y) = H(X) - H(X/Y)$
 $= H(Y) - H(Y/X)$
 $H(XY) = H(Y/X) + H(X)$
 $= H(X/Y) + H(Y)$

Unit-VII. Optical communication

- Snell's law
 $\eta_1 \sin \theta_1 = \eta_2 \sin \theta_2$
 Critical angle:
 Angle θ_1 for which $\theta_2 = 90^\circ$
 $\eta_1 \sin \theta_C = \eta_2 \sin 90^\circ$
 $\Rightarrow \theta_C = \sin^{-1} \left(\frac{\eta_2}{\eta_1} \right)$
- 
- Medium-2
Ref. index- η_2
- Medium-1
Ref. index- η_1
- If $\theta_1 = \theta_C$
 $\theta_2 = 90^\circ$
 If $\theta_1 > \theta_C \Rightarrow$ total internal reflection takes place
- Numerical aperture
 $NA = \sqrt{\eta_1^2 - \eta_2^2}$
 $NA = \eta_1 \sqrt{2\Delta}$
 $NA = \sin \theta_a$
 $\theta_a = \sin^{-1}(NA) = \text{Acceptance angle}$
 $\Delta = \text{Relative refractive index}$
 $\Delta = \frac{\eta_1 - \eta_2}{\eta_1}$
 $\eta_1 = \text{refractive index of core}$
 $\eta_2 = \text{refractive index of cladding}$
 $\eta_1 > \eta_2$

- Normalized frequency parameter (V-number)
 For single mode propagation, $V < 2.405$

$$V = \frac{2\pi a}{\lambda} (NA)$$

- For multimode fiber, $V > 2.405$
- No. of modes of propagation:

$$\text{For step index fiber } Mg = \frac{V^2}{2}$$

- Attenuation $\alpha_d \text{ (dB/km)} = \frac{10}{L} \log \left(\frac{P_{in}}{P_{out}} \right)$

- Dispersion $D = \frac{\Delta L}{\Delta \tau}$

ΔL = Fiber length,
 $\Delta \tau$ = pulse spreading

- $B.W. = \frac{0.35}{t_r}$

Photodiodes:

- Responsivity $\Rightarrow R = \frac{I_P}{P_O} = \frac{\eta \cdot e}{E}$

- $R = \frac{\eta \cdot e}{h \cdot f} = \frac{\eta \cdot e \cdot \lambda}{h \cdot c}$

Planck's constant



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ELECTRONIC DEVICES AND CIRCUITS

Energy Band and Charge Carriers in Semiconductors

Fermi Level

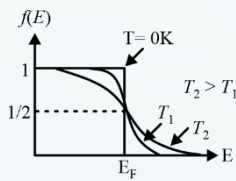
Fermi Dirac distribution function $f(E)$ gives the probability that an available energy state E will be occupied by an electron at absolute temperature T , under conditions of thermal equilibrium

$$f(E) = \frac{1}{1 + \exp[(E - E_F) / kT]}$$

where, E_F Fermi energy level; $K \rightarrow$ Boltzmann's constant

$T \rightarrow$ Absolute temperature in Kelvin

The Fermi Dirac distribution function is given as:



Note

$[1 - f(E)]$ gives the probability that energy state E will be occupied by a hole.

Concentration of Carriers

Concentration of electrons in conduction band is given by

$$n_0 = N_C e^{-(E_C - E_F) / kT}$$

where, $n_0 \rightarrow$ Concentration e^- in conduction band

$E_F \rightarrow$ Fermi energy level

$k \rightarrow$ Boltzmann's constant

$T \rightarrow$ Absolute temperature

$N_C \rightarrow$ Effective density of states in conduction band

Effective density of states N_C is given by

$$N_C = 2 \left[\frac{2\pi m_n^* kT}{h^2} \right]^{3/2}$$

Concentration of holes in valence band is given

$$p_0 = N_V e^{-(E_F - E_V) / kT}$$

where, $p_0 \rightarrow$ Concentration of holes

$N_V \rightarrow$ Effective density of states in valence band

$E_F \rightarrow$ Fermi energy level

Effective density of states in valence band is given by

$$N_V = 2 \left[\frac{2\pi m_p^* kT}{h^2} \right]^{3/2}$$

Mass Action Law

It states that at thermal equilibrium product of concentration of free electrons and holes is equal to the square of intrinsic concentration at that temperature i.e.

$$n_0 p_0 = n_i^2$$

Where, $n_0 \rightarrow$ Concentration of electron in conduction band

$p_0 \rightarrow$ Concentration of holes in valence band

$n_i \rightarrow$ Intrinsic concentration at given temperature

Intrinsic concentration is given by

$$n_i = \sqrt{N_C N_V} e^{-E_g / 2kT} \text{ where, } E_g \rightarrow \text{band gap}$$

n_i can also be given as

$$n_i^2 = A_0 T^3 e^{-E_g / kT} \text{ where, } A_0 \rightarrow \text{is a constant}$$

Concentration of electrons in conduction band can also be given as

$$n_0 = n_i e^{(E_F - E_i) / kT}$$

Where, $E_i \rightarrow$ intrinsic level lies near the middle of bandgap

Concentration of holes in valence band can also be given as

$$p_0 = n_i e^{(E_i - E_F) / kT}$$

Space Charge Neutrality

If the material is to remain electrostatically neutral, the sum of positive charges must balance the sum of negative charges i.e.

$$p_0 + N_d^+ = n_0 + N_a^-$$

where, $N_d^+ \rightarrow$ concentration of donor atoms

$N_a^- \rightarrow$ concentration of acceptor atoms

$p_0 \rightarrow$ concentration of holes

$n_0 \rightarrow$ concentration of electrons

Doping in Semiconductors

For N-type semiconductors	For p-type semiconductors
$n \cong N_d^+$	$p \cong N_a^-$
$p = \frac{n_i^2}{N_d^+}$	$n = \frac{n_i^2}{N_a^-}$

Note

Expression for p and n in above equations are approximate under the assumption that $N_d^+ \gg n_i, N_a^- \gg n_i$

Exact expressions are:

$$n = \frac{N_d^+ + \sqrt{(N_d^+)^2 + 4n_i^2}}{2} \quad p = \frac{N_a^- + \sqrt{(N_a^-)^2 + 4n_i^2}}{2}$$

Mobility of Charge Carriers

It is drift velocity per unit electric field. $\left(\text{Unit} \rightarrow \frac{m^2}{V \cdot s} \right)$.

It defines how fast the charge carrier travels from one place to other and is given by

$$\mu = \frac{V_d}{E}$$

where,

$V_d \rightarrow$ Drift velocity ; $E \rightarrow$ Electric field

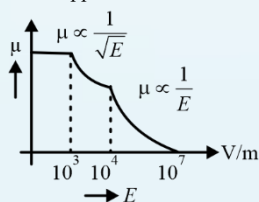
Ge

Si

e^- mobility 3800 cm² / V sec 1300 cm² / V sec

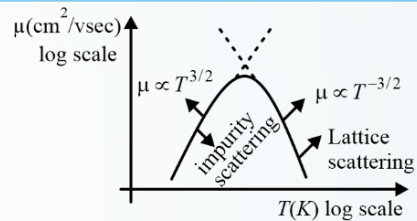
hole mobility 1800 cm² / V sec 500 cm² / V sec

- Mobility of charge carriers decreases with temperature and varies as $\mu \propto T^{-m}$ Where, m is a constant
- In Ge, $m = 1.66$ for e^- and 2.33 for hole
- In Si, $m = 2.5$ for e^- and 2.7 for hole
- Mobility also varies with applied electric field



Note

- At smaller electric field mobility is constant.
- At very high electric field product of mobility and electric field becomes constant and is equal to saturation value of drift velocity.
- The variation of mobility in accordance with scattering is shown as



- Overall mobility is given by

$$\frac{1}{\mu} = \frac{1}{\mu_1} + \frac{1}{\mu_2} + \frac{1}{\mu_3} + \dots$$

Where, $\mu \rightarrow$ overall mobility

$\mu_1, \mu_2, \mu_3 \rightarrow$ Mobility corresponding to different scattering mechanism.

Hall Effect

- Hall Voltage (V_H) is given by

$$V_H = Ed \quad \text{or} \quad V_H = \frac{BI}{\rho W}$$

where,

$E \rightarrow$ Electric field induced

$d \rightarrow$ Separation between upper and lower surface of specimen or height of specimen

$W \rightarrow$ Width of specimen

$B \rightarrow$ Applied magnetic field

$I \rightarrow$ Current flowing in specimen

$\rho \rightarrow$ Charge density

- $1/\rho$ is called **Hall coefficient** R_H i.e. $R_H = \frac{1}{\rho}$
- V_H can also be given as $V_H = R_H \frac{BI}{W}$
- By Hall experiment. mobility is given by $\mu = \frac{8}{3\pi} \sigma R_H$

Where, $\mu \rightarrow$ Mobility of charge carriers ; $\sigma \rightarrow$ Conductivity of material.

Note

- Hall effect is utilized in determining whether a given material is metal, n-type SC or p-type SC.
- For metal, value of V_H is lesser compared to SC.
- Hall voltage is -ve for metal or n-type SC.
- Hall voltage is +ve for p-type SC.
- Hall voltage is zero for intrinsic SC.
- It can be used in finding mobility of charge carriers, concentration of charge carriers, and type of semiconductor.

Conductivity (σ)

$$\sigma = q \cdot n \mu_n + q p \mu_p$$

where, $n \rightarrow$ Concentration e_s^- in conduction band

$p \rightarrow$ Concentration of holes in valence band

$\mu_n \rightarrow$ Mobility of e_s^-

$\mu_p \rightarrow$ Mobility of holes

- Conductivity of pure SC is given by

$$\sigma = q n_i [\mu_n + \mu_p]$$

where, $n_i \rightarrow$ Intrinsic concentration

Note

- Conductivity of pure SC increases with temperature.
- Conductivity of pure SC at 0K is zero.

- Conductivity of extrinsic SC is given by

$$\sigma \cong q N_d \mu_n \text{ for } n\text{-type where, } N_d \rightarrow \text{donor concentration}$$

$$\sigma \cong q N_a \mu_p \text{ for } p\text{-type } N_a \rightarrow \text{acceptor concentration}$$

Note

- Conductivity of extrinsic SC decreases with increase in temperature above normal temperature.
- At Curie temperature, conductivity becomes equal to intrinsic conductivity.
- Conductivity increases with Increase in doping concentration.

Fermi Level in Intrinsic and Extrinsic SC

Fermi Level in Intrinsic and Extrinsic SC

- Fermi level E_F in intrinsic SC is given by

$$E_F = \frac{E_C + E_V}{2} - \frac{kT}{2} \ln \left(\frac{N_C}{N_V} \right)$$

Note

- If $N_V \cong N_C$ then fermi level lies in the middle of energy gap.
- At 0 K Fermi level lies in the middle of energy gap.
- Fermi level in pure SC depends on temperature.
- As temperature increases, fermi level moves away from the middle of bandgap.

Fermi level in n-type SC is given by

$$E_F = E_C - kT \ln \left(\frac{N_C}{N_D} \right)$$

Where,

$N_C \rightarrow$ density of states in conduction band

$N_D \rightarrow$ donor concentration

- Fermi level in p -type SC is given by

$$E_F = E_V + kT \ln \left(\frac{N_V}{N_A} \right)$$

where,

$N_V \rightarrow$ density of states in valence band

$N_A \rightarrow$ concentration of acceptors

- Shift in the position of fermi level with respect to intrinsic fermi level in n -type SC due to doping is given by

$$\text{Shift} = kT \ln \left(\frac{n}{n_i} \right) \cong kT \ln \left(\frac{N_D}{n_i} \right) \text{ upwards}$$

- Shift in position of fermi level in p -type SC with respect intrinsic fermi level due to doping is given by

$$\text{Shift} = kT \ln \frac{p}{n_i} \cong kT \ln \frac{N_A}{n_i} \text{ downwards}$$

In a semiconductor minimum conductivity occurs at:

$$n = n_i \sqrt{\frac{\mu_p}{\mu_n}}; p = n_i \sqrt{\frac{\mu_n}{\mu_p}}$$

$$\sigma_{\min} = 2 n_i q \sqrt{\mu_n \mu_p}$$

Excess Carriers in Semiconductor

Diffusion Process

- Diffusion is the natural result of the random motion of the individual molecules.
- Diffusion happens due to concentration gradient.
- Diffusion current density is given by

$$J_n(\text{diff}) = q D_n \frac{dn(x)}{dx}; J_p(\text{diff}) = -q D_p \frac{dp(x)}{dx}$$

where, $J_n(\text{diff}) \rightarrow$ diffusion current density due to electrons

$J_p(\text{diff}) \rightarrow$ diffusion current density due to holes

$D_n \rightarrow$ electron diffusion coefficient

$D_p \rightarrow$ hole diffusion coefficient

$n(x) \rightarrow$ profile of e_s^- with respect to x

$p(x) \rightarrow$ profile of holes with respect to x

Drift Process

- Transport of carriers under the influence of applied electric field.
- $J_n(\text{drift}) = nq\mu_n E$; $J_p(\text{drift}) = nq\mu_p E$
where, $J_n(\text{drift}) \rightarrow$ Drift current density due to electrons
 $J_p(\text{drift}) \rightarrow$ Drift current density due to holes
 $n, p \rightarrow$ Concentration of electron and holes respectively
 $E \rightarrow$ Applied Electric Field

Total Current Density

- Overall current density is given by

$$\begin{aligned} J_n(x) &= q\mu_n E(x)n(x) + qD_n \frac{dn(x)}{dx} \\ J_p(x) &= q\mu_p E(x)p(x) - qD_p \frac{dp(x)}{dx} \end{aligned} ; \quad J(x) = J_n(x) + J_p(x)$$

where, $J(x) \rightarrow$ Overall current density
 $J_n(x) \rightarrow$ Current density due to electrons
 $J_p(x) \rightarrow$ Current density due to holes
 $E(x) \rightarrow$ Electric field applied

Einstein Relation

- This gives the relation between diffusion coefficient (D), mobility (μ) and temperature and given as

$$\frac{D_n}{\mu_n} = \frac{D_p}{\mu_p} = \frac{kT}{q}$$

where, $\frac{kT}{q} = V_T \rightarrow$ Thermal voltage

Continuity Equation

- This equation is given by

$$\frac{\partial \delta p}{\partial t} = -\frac{1}{q} \frac{\partial J_p}{\partial x} - \frac{\delta p}{\tau_p}; \quad \frac{\partial \delta n}{\partial t} = \frac{1}{q} \frac{\partial J_n}{\partial x} - \frac{\delta n}{\tau_n}$$

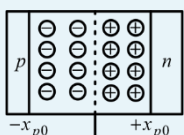
where, $\delta p, \delta n \rightarrow$ Excess carrier concentration
 $\tau_n, \tau_p \rightarrow$ Carrier life time for e^- s and holes respectively

- Another form of this equation is given by

$$\frac{\partial \delta n}{\partial t} = D_n \frac{\partial^2 \delta n}{\partial x^2} - \frac{\delta n}{\tau_n}; \quad \frac{\partial \delta p}{\partial t} = D_p \frac{\partial^2 \delta p}{\partial x^2} - \frac{\delta p}{\tau_p}$$

Junction Diode

Contact Potential or Build-in Potential



$$V_0 = \frac{kT}{q} \ln \left[\frac{N_a N_d}{n_i^2} \right]$$

where, $V_0 \rightarrow$ Contact potential
 $k \rightarrow$ Boltzmann's constant (1.38×10^{-28} J/K)
 $q \rightarrow$ Electron charge (1.6×10^{-19} coulomb)
 $T \rightarrow$ Temperature in Kelvin of p-n junction

Maximum Electric Field

- Maximum electric field is given by

$$E_0 = -\frac{q}{\epsilon} \cdot N_d x_{n0} = -\frac{q}{\epsilon} N_a x_{p0}$$

where, $x_{n0} \rightarrow$ Width of depletion region on n-side
 $x_{p0} \rightarrow$ Width of depletion region on p-side
 $\epsilon \rightarrow$ Permittivity of material from which diode is made

Note

- Contact potential in the terms of maximum electric field is given by

$$V_0 = -\frac{1}{2} E_0 W$$

where, $V_0 \rightarrow$ Contact potential
 $E_0 \rightarrow$ Maximum electric field at junction
 $W \rightarrow$ Width of depletion region

- Negative sign is due to the fact that this electric field is from n to p.
- In case of unbiased p-n junction diode electric field is maximum at the junction and decreases on the two sides of junction and is zero outside the space charge region.

Width of Depletion Region

- Width of depletion region in unbiased condition is given by

$$W = \left[\frac{2\epsilon V_0}{q} \left(\frac{N_a + N_d}{N_a N_d} \right) \right]^{1/2}$$

Note

- If we reverse bias the diode by voltage V then in formula of depletion width V_0 is replaced by $|V_0 + V|$.
- Depletion width increases with reverse and decreases with forward biased.
- Depletion approximation is assumed in calculating the above equations.

Charge Neutrality Equation

$$x_{n0} N_d = x_{p0} N_a$$

also $x_{n0} + x_{p0} = W$

Current Equation

$$I_f = I_0 \left[e^{V_d / \eta V_T} - 1 \right]$$

where, $I_0 \rightarrow$ Reverse saturation or leakage current

$V_d \rightarrow$ Forward biased voltage

$\eta \rightarrow$ Recombination factor (material constant)

= 1 for Ge

= 2 for Si

$V_T \rightarrow$ thermal voltage

$$V_T = \frac{kT}{q} = 0.0259 \text{ V at room temperature}$$

Note

- $I_f \rightarrow$ is majority carrier current and diffusion current.
- $I_0 \rightarrow$ drift current
- Reverse saturation current (I_0) doubles for every 10°C rise in temperature.

$$I_{o(T_2)} = I_{o(T_1)} \times 2^{(T_2 - T_1)/10}$$

where, $I_{o(T_2)} \rightarrow$ Reverse saturation current at temperature T_2

$I_{o(T_1)} \rightarrow$ Reverse saturation current at temperature T_1

Here, $T_2 > T_1$

- For 1°C rise in temperature it decreases by 2.5 mV

$$\frac{dV}{dT} = -2.5 \text{ mV}/^\circ\text{C}$$

$$I_0 = Aqn_i^2 \left[\frac{D_p}{L_p N_D} + \frac{D_n}{L_n N_A} \right]$$

where, $A \rightarrow$ Area of cross-section

$D_p, D_n \rightarrow$ Diffusion constant of holes and electrons respectively

$L_p, L_n \rightarrow$ Diffusion length of holes and electrons respectively

Junction Capacitance

- Junction capacitance (C_j)

$$C_j = \frac{A\epsilon}{W}$$

where, $A \rightarrow$ Area of cross-section of diode

$W \rightarrow$ Width of depletion region

$\epsilon \rightarrow$ Permittivity of material

Semiconductor Devices

Photo Conductors

- Photo conductive equation is given by

$$E_g = h\nu = \frac{hc}{\lambda}$$

where, $h \rightarrow$ Plank's constant,

$c \rightarrow$ Speed of light in free space,

$\lambda \rightarrow$ Wavelength of light incident on SC

- Wave length of light incident can be given as

$$\lambda = \frac{1.24}{E_g (eV)} \mu\text{m}$$

LED (Light Emitting Diode)

- It works on the principle of Electro-luminescence.
- It is fabricated using direct band gap material.

Zener Diode

- Operates on the principle of tunneling effect.
- Always operated under reverse bias.
- Mainly used in voltage regulation application.
- Maintains constant voltage across a load if properly biased.

Tunnel Diode

- It is fastest switch.
- Its response time is of the order of psec.
- It is a p^+n^+ diode having doping level of $1:10^3$.
- Works on the principle of tunneling effect.
- It exhibits voltage controlled negative resistance

Bipolar junction Transistor

BJT Characteristics

- Bipolar junction transistor was invented by William Schokley, Bartain and Bardeen.
- Current flows both due to majority as well minority carries that's why called bipolar device.
- It is a **Current Controlled Current Source (CCCS)**.
- Noisy device due to presence of minority carriers.
- Thermal stability is lesser because of leakage current or reverse saturation current.

Mode of Operation

Mode	J_E	J_C	Property	Application
Saturation	Forward bias	Forward bias	Very low resistance	'ON' switch
Cut-off	Reverse bias	Reverse bias	Very high resistance	'OFF' switch
Active	Forward bias	Reverse bias	Excellent transistor action	Amplifier
Reverse Active	Reverse bias	Forward bias	Poor transistor action	Attenuator

- A transistor has excellent transistor action if:
 - Emitter is heavily doped in comparison to base.
 - Base width is kept much smaller than diffusion length of carriers.

Current Gain

- Emitter injection efficiency γ_E is given by

$$\gamma_E = \frac{i_{Ep}}{i_{Ep} + i_{En}} \text{ for } p-n-p \text{ transistor}$$

$$\gamma_E = \frac{i_{En}}{i_{Ep} + i_{En}} \text{ for } n\text{-}p\text{-}n \text{ transistor}$$

where, $i_{Ep} \rightarrow$ Emitter current due to holes
 $i_{En} \rightarrow$ Emitter current due to electrons

- Base transport factor B is given by

$$B = \frac{i_C}{i_{Ep}} \text{ for } p\text{-}n\text{-}p \text{ transistor}$$

$$B = \frac{i_C}{i_{En}} \text{ for } n\text{-}p\text{-}n \text{ transistor}$$

- Current gain α in terms of γ_E and B is given as $\alpha = B\gamma_E$
- Value of γ_E and B is very close to 1 and in ideal conditions it is 1.
- In a **diffusion transistor** base current is only due to **diffusion**.
- In a **drift transistor** base current is due to both **diffusion** and **drift** phenomenon.

Early Effect

- Increase in depletion region width due to increase in reverse biased across collector base junction is called **Early effect**.
- Due to **early effect** effective base width decreases.
- Due to **early effect** α increases by little amount but β increases by a large amount.
- The effective base width offers a spread resistance at high frequencies.
- Emitter junction breakdown voltage is smaller than the collector junction breakdown voltage.
- Emitter junction breakdown is due to Zener effect while collector junction breakdown is due to Avalanche effect.

Field Effect Transistor

FET Characteristics

- It is a voltage controlled device
- It is a high input impedance device.
- It is a majority carrier device.
- It is less noisy than BJT.
- It is thermally stable.
- It is smaller in size compared to BJT.
- Leakage currents are negligible.
- It is unipolar device.

Note

- BJT is faster than FET devices but its power consumption is large compared to FET devices.

JFET Parameters

- In saturation region current through FET is given by

$$I_{DS} = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

where, $I_{DSS} \rightarrow$ Saturation current at $V_{GS} = 0$

$V_P \rightarrow$ Pinch-off voltage

$I_{DS} \rightarrow$ Saturation current at any V_{GS} .

- I_{DS} varies parabolically with respect to V_{GS} .
- Transconductance g_m is given by

$$g_m = -\frac{2I_{DSS}}{V_P} \left[1 - \frac{V_{GS}}{V_P} \right]$$

$$\text{or } g_m = \frac{2}{|V_P|} \sqrt{I_{DS} I_{DSS}}$$

- General equation for gate to source voltage is

$$V_{GS} = V_P \left[1 - \sqrt{\frac{I_{DS}}{I_{DSS}}} \right]$$

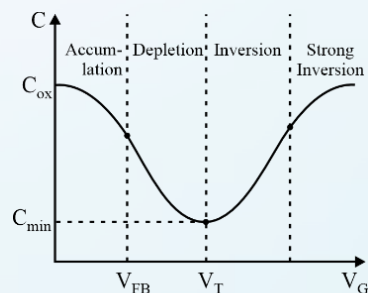
MOSFET

- Metal oxide semiconductor FETs have a very high input impedance.
- MOSFET is used as a voltage-controlled capacitor.

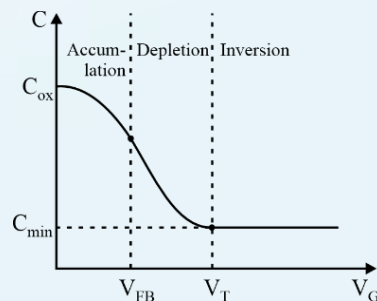
MOS Capacitor

- MOS capacitor can operate in 3 modes-Accumulation, Depletion, Inversion.

C-V Characteristics of MOS



Low Frequency Characteristics



High Frequency Characteristics

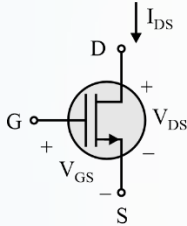
where, $V_{FB} \rightarrow$ Flat band voltage

$V_T \rightarrow$ Threshold voltage

$V_G \rightarrow$ Gate voltage

MOSFET Current Equations

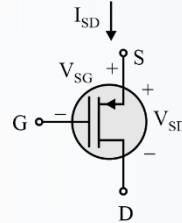
1. N-Channel MOSFET



Cut-off	$V_{GS} \leq V_{Tn}$	$I_{DS} = 0$
Linear / Triode	$V_{GS} > V_{Tn};$ $V_{DS} \leq V_{GS} - V_{Tn}$	$I_{DS} = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_{Tn})V_{DS} - \frac{V_{DS}^2}{2} \right]$
Saturation	$V_{GS} > V_{Tn};$ $V_{DS} > V_{GS} - V_{Tn}$	$I_{DS} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{Tn})^2 (1 + \lambda V_{DS})$

where, $V_{Tn} \rightarrow$ Threshold voltage
 $\mu_n \rightarrow$ Electron mobility
 $C_{ox} \rightarrow$ Oxide capacitance
 $W, L \rightarrow$ Width and length of MOSFET
 $\lambda \rightarrow$ Channel length modulation parameter

2. P-Channel MOSFET



Cut-off	$V_{SG} \leq V_{Tp} $	$I_{SD} = 0$
Linear / Triode	$V_{SG} > V_{Tp} ;$ $V_{SD} \leq V_{SG} - V_{Tp} $	$I_{SD} = \mu_p C_{ox} \frac{W}{L} \left[(V_{SG} - V_{Tp})V_{SD} - \frac{V_{SD}^2}{2} \right]$
Saturation	$V_{SG} > V_{Tp} ;$ $V_{SD} > V_{SG} - V_{Tp} $	$I_{SD} = \frac{1}{2} \mu_p C_{ox} \frac{W}{L} (V_{SG} - V_{Tp})^2 (1 + \lambda V_{SD})$

where, $V_{Tp} \rightarrow$ Threshold voltage
 $\mu_p \rightarrow$ Hole mobility

Note

- V_{Tn} is positive while V_{Tp} is negative
- For a MOSFET operating in saturation region, channel length modulation effect causes decrease in output resistance.
- In triode region FET works as voltage variable resistor (VVR).
- Transconductance $g_m = \frac{\partial I_D}{\partial V_{GS}}$.
- For depletion MOSFET channel is diffused and current flows even if V_{GS} is zero.
- It operates in two mode depletion mode and enhancement mode. For V_{GS} less than threshold voltage it works in depletion mode and for V_{GS} greater than threshold voltage it works in enhancement mode.



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